

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production Differential Cross Sections on Argon with the MicroBooNE Detector in the NuMI Beam

Analysis note for the full FHC, RHC and combined exposures

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Abstract

We measure $\nu_\mu + \bar{\nu}_\mu$ charged-current single-charged-pion ($\text{CC}\pi^\pm$) differential cross sections on argon using the full MicroBooNE NuMI exposure, in three configurations: forward horn current (FHC, 7.766×10^{20} POT), reverse horn current (RHC, 11.082×10^{20} POT) and their combination (18.848×10^{20} POT).

Events are selected by requiring a muon candidate and exactly one contained charged-pion candidate from a fiducial vertex, an electromagnetic-shower veto, and a muon–pion opening angle below 2.6 rad, giving $\approx 59\%$ purity at $\approx 17\%$ efficiency. Cross sections are extracted by Wiener-SVD unfolding in six released distributions — muon and pion momentum, muon polar angle in both parameterisations, pion polar angle and the muon–pion opening angle — together with a flux-averaged total from a one-bin extraction of the same machinery.

Each configuration carries its own integrated flux and exposure, and each result carries the nominal configured covariance: flux, interaction model, hadron re-interaction, detector response, normalisation and statistical terms. Unfolded results must be compared with a prediction through the released additional-smearing matrix A_C , which is not normalisation-preserving; A_C and the covariance are released for every extraction.

The analysis is blind. Every “data” point in this note is Poisson-thrown from the central-value Monte Carlo, so no number here is a statement about beam-on data. The four background-control regions, which contain no signal-region events, have been opened on beam-on data. They exposed a run-level exposure error — the FHC Run-1 beam-on sample was recorded without the open trigger but had been normalised with the open-trigger bookkeeping — which is corrected, and the FHC and combined results are extracted on the corrected exposure. On it the frozen global test passes and the regions do not support a class-specific background normalisation correction; the data sit 12–20% above the pre-fit prediction, inside its correlated flux and cross-section uncertainties, in every period except FHC Run 1, whose 13% offset relative to the other periods is an open item.

The implementation is validated by same-sample closure, and the size of the statistical uncertainties by an ensemble of independent Poisson realisations carried through the whole chain; the release is reproducible bit-identically from the processed ntuples. Unfolding pseudo-data thrown from a reweighted interaction model with the nominal response biases no bin by more than 0.8σ ; an independent-generator closure is not yet available, the one existing

NuWro overlay having failed validation against the standalone NuWro prediction. Sec. 2 is the one-page version of the analysis and the status of every product in it.

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1 Introduction

Charged-current single charged-pion ($\text{CC}\pi^\pm$) production,

$$\nu_\mu(\bar{\nu}_\mu) + \text{Ar} \longrightarrow \mu^\mp + \pi^\pm + X, \quad (1)$$

is among the most frequent neutrino interaction channels at energies of 1–5 GeV and constitutes a significant background for oscillation analyses at the same energy scale, most notably for ν_e appearance measurements at experiments such as NOvA [3] and T2K [4], as well as the future DUNE [5]. Despite its importance, the cross section on nuclear targets remains poorly constrained: major neutrino interaction generators (GENIE v3, NuWro, NEUT) disagree by 20–30% in absolute rate and show significant shape differences for individual kinematic variables.

MicroBooNE is a single-phase liquid argon time-projection chamber (LArTPC) [33] at Fermilab that operated from 2015–2021, providing the world’s highest-statistics ν_μ -on-argon dataset below 5 GeV during its NuMI running period. This note presents a measurement of differential cross sections in this channel over the full NuMI exposure, in the forward-horn, reverse-horn and combined configurations, using Pandora pattern recognition [6, 12] and Wiener-SVD unfolding [7].

What the measurement does

The measurement is built in five steps, and the note follows them in order.

1. **Exposure and flux** (§4). Each configuration is an independent measurement with its own exposure and its own integrated flux. The NuMI beam reaches MicroBooNE off-axis, and its direction is not the detector axis; every angle in this note is defined about the neutrino direction, which matters at the level of whole bins.
2. **Signal and selection** (§6–§8). The signal is defined by the particles leaving the nucleus, not by the interaction mechanism: one muon, one charged pion, no neutral pion, above stated momentum thresholds and inside the fiducial volume. The selection reconstructs that topology.
3. **Response and unfolding** (§10). The detector smears and loses events, so the measured spectrum is not the cross section. Wiener-SVD unfolding inverts the response with a regularisation that trades bias against variance, and reports what it did through the additional-smearing matrix A_C : a prediction must be multiplied by A_C before it is compared with these results (§2 says why the integral of a differential result is not the total).
4. **Uncertainties** (§14). Flux, interaction model, hadron re-interaction and detector response are propagated as full covariance matrices through the same unfolding as the data, and are released with the results.
5. **Validation and blinding** (§13, §15). The analysis is blind: results are extracted on fake data thrown from the central-value simulation. What the validation establishes, and what it does not, is §13; the background model is tested in four control regions that hold no signal-region event and have been opened on beam-on data.

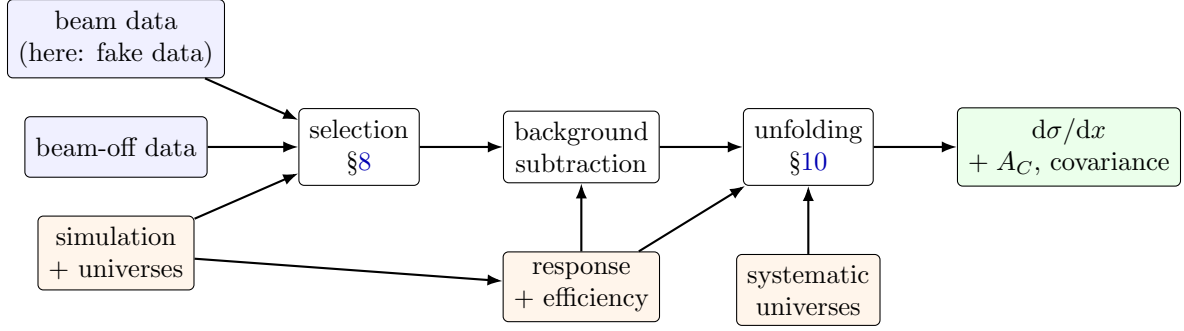


Figure 1: The extraction chain. The same selection is applied to data, beam-off data and simulation; the simulation additionally supplies the response and the systematic universes. What comes out is a cross section together with the additional-smearing matrix A_C and the covariance — all three are needed to compare it with a prediction.

How to read this note

§11 is the results section and can be read on its own; Sec. 3 of the proton-tagged note gives the proton-tagged subsample, which is secondary scope. Terms of art — smearceptance, A_C , universe, closure, conditioning and the rest — are defined in plain language in the Glossary at the end, in the order a reader meets them. Two companion documents carry material that would otherwise interrupt the argument: the *technical supplement* holds the studies behind the analysis choices — estimator surveys, binning and threshold optimisation, the closure programme, the control-region procedure — and the *change log* holds the history of the analysis, including every correction that changed a result and the values before and after. Neither is needed to follow this note. The additional-smearing and covariance matrices for every extraction are released as text in `report/data_release/`; a prediction cannot be compared with these results without the former, nor used quantitatively without the latter.

2 Analysis status

Approval requested. Approval of the inclusive $CC1\mu1\pi Xp$ analysis strategy, signal definition, event selection, background treatment, unfolding procedure, systematic-uncertainty model and primary result set. Signal-region unblinding is requested only after the absolute flux-normalisation confirmation and the remaining required uncertainty terms are complete and the affected products regenerated. Every FHC and combined number in this version is extracted on the corrected FHC Run-1 exposure.

This section is the one-page version of the analysis, and the status of everything in it.

What is measured. Flux-averaged $\nu_\mu + \bar{\nu}_\mu$ charged-current single-charged-pion production on argon. The signal is defined by what leaves the nucleus — one muon, one charged pion, no neutral pion, any number of nucleons, vertex in the fiducial volume — not by the interaction mechanism, in the phase space $p_\mu > 0.15$ GeV/ c , $p_\pi > 0.175$ GeV/ c and $\theta_{\mu\pi} < 2.6$ rad.

From what. The full NuMI exposure: FHC 7.766×10^{20} POT, RHC 11.082×10^{20} , combined 18.848×10^{20} , each normalised run by run and each with its own integrated flux. The selection reaches $\approx 17\%$ efficiency at $\approx 59\%$ purity; the dominant background is $CC0\pi$ with a proton taken for the pion, then multi-pion events, with beam-off cosmics at $\approx 2\%$ of the prediction.

What comes out. A flux-averaged total from a one-bin extraction, six differential distributions, and a proton-tagged subsample measured as secondary scope; Table 15 is the full inventory. All angles are defined about the neutrino direction, not the detector axis — using the detector axis would move about 65% of signal events to a different $\cos \theta_\mu$ bin.

How to compare a prediction with it. Multiply the prediction by the released additional-smearing matrix A_C , then use the released covariance. A_C is not normalisation-preserving, so the integral of a differential result is not the total cross section and differs between observables; the physical total is the one-bin extraction, where $A_C = 1$.

Status of every product

- **Signal region: blinded.** Every data point in this note is a Poisson throw of the central-value simulation. No statement here is a measurement of beam-on data.
- **Control regions: opened** on beam-on data under the frozen procedure; they indicate no supported class-specific background correction within the transfer uncertainties; the 12–20% excess of data over the pre-fit prediction is within them, and FHC Run 1 is the one period that does not share it (§11.5).
- **Primary:** the inclusive one-bin total and the six inclusive differential distributions, in all three configurations.
- **Proton-tagged subsample:** a separate measurement with its own note; not part of this approval request (Sec. 3 of the proton-tagged note).
- **Withdrawn — do not quote:** the differential $W_{\pi p}$ shape and the five-bin $d\sigma/dp_\pi$ (Appendix C).
- **Validated:** the implementation, the size of the statistical uncertainties, and robustness to a reweighted interaction model (FHC and RHC; no bin beyond 0.8σ , §13.3). **Not yet available:** an independent-generator closure — the one existing NuWro overlay could not be validated against the standalone NuWro prediction and is not used. Not tested for the proton-tagged family.
- **Uncertainties:** the nominal configured covariance. Identified terms that are *not* in it are listed in §15.
- **Before the signal region is opened:** the flux normalisation must be confirmed with the flux contact, and the 13% offset of FHC Run 1 relative to the other five periods in the control regions (§11.5) must be understood. The FHC and combined results are already extracted on the corrected FHC Run-1 exposure (the beam-on sample was recorded without the open trigger, 2.192×10^{20} POT); on beam-on data that correction is worth +14% on the FHC cross section relative to the previous bookkeeping.

3 Physics Context

3.1 Production mechanisms and nuclear effects

At 0.5–5 GeV charged-current single charged-pion production proceeds through baryon resonance production, dominated by the $\Delta(1232)$ and modelled by Rein–Sehgal-type form factors [35, 36, 37, 38, 39]; a non-resonant continuum and deep-inelastic scattering, together $\sim 20\%$ of the rate here and added incoherently by every generator used; and coherent production off the whole nucleus ($\lesssim 10\%$), which satisfies the topological signal definition of §6 and is therefore part of the signal, not a background. On argon, Fermi motion, Pauli blocking and binding modify the kinematics, and final-state interactions (absorption at ~ 25 – 30% , charge exchange, quasi-elastic scattering) mix the $CC\pi^\pm$, $CC\pi^0$ and $CC0\pi$ topologies at the 30% level. That mixing is why the signal is defined by the final-state particle content and not by the production process, and why

$\text{CC}0\pi$ is the largest background: pion absorption and a proton taken for the pion feed it into the selection (Sec. 11 of the technical supplement).

3.2 Generators compared

Four generators are compared with the results; their relevant model choices are summarised in Table 1. The injected model and the source of the response is the MicroBooNE tune of GENIE v3 [16]. In the measured phase space the standalone generators span a factor of ~ 1.6 in the total cross section, GENIE lowest and NEUT highest (Table 16 of the technical supplement), bracketing the tune; because the present results are extracted on central-value fake data, that spread is the context for the model comparisons, not a measured tension.

Table 1: Model content of the generators compared, restricted to what matters for interpreting this measurement.

Generator	Resonance model	Non-resonant / DIS	Final-state interactions
GENIE v3 [9, 20]	KLN–BS, 18 resonances to $W \lesssim 2$ GeV	Bodek–Yang, incoherent; AGKY transition	INTRANUKE hA (one effective interaction)
NuWro [21]	$\Delta(1232)$ only (Paschos–Yu)	separate amplitude; smooth transition to Bodek–Yang	Salcedo–Oset (in-medium Δ , two-body absorption)
NEUT [40, 23]	modified Rein–Sehgal, 18 resonances, Graczyk–Sobczyk Δ form factors	GRV98 with Bodek–Yang above $W \approx 2$ GeV	step-by-step intranuclear cascade
GiBUU [22]	Δ and higher resonances with in-medium spectral functions	PYTHIA fragmentation above $W \approx 1.6$ GeV	transport equations with pion optical potential

3.3 Previous measurements

Table 2 lists the published $\text{CC}\pi^+$ measurements on nuclear targets. The two most directly relevant are MicroBooNE’s own: the BNB measurement [52], on the same detector with an on-axis spectrum peaked at 0.7 GeV and negligible $\bar{\nu}_\mu$ content, gives $\sigma = (3.75 \pm 0.07_{\text{stat}} \pm 0.80_{\text{syst}}) \times 10^{-38} \text{ cm}^2/\text{Ar}$ and finds all tested generators above the data at very forward muon angles; and the NuMI $\nu_e + \bar{\nu}_e$ $\text{CC}\pi^+$ measurement [51], on the same beam and reconstruction, which shares this analysis’s extraction framework and conventions (Sec. 10.1 of the technical supplement). The NuMI off-axis flux used here is softer and broader than the BNB one with a $\sim 35\%$ $\bar{\nu}_\mu$ admixture in FHC, and the flux-averaged total is divided by the whole flux above 60 MeV while pion production opens near 0.3 GeV, so the $\sim 1.0 \times 10^{-38} \text{ cm}^2/\text{Ar}$ found here is not to be compared directly with the BNB value (§4.4). The T2K off-axis measurement on carbon [49] overlaps this spectrum most closely among the other experiments. The extended survey is in the technical supplement (App. 15 of the technical supplement).

Table 2: Selected $\text{CC}\pi^+$ cross-section measurements on nuclear targets. $\langle E_\nu \rangle$ is the approximate mean or peak neutrino energy. “Type” indicates whether a total integrated (T) or differential (D) cross section was reported. “Target” gives the nucleus primarily used for cross-section extraction.

Experiment	Target	$\langle E_\nu \rangle$ (GeV)	Type	Key result / reference
ANL 12-ft [41]	D	1.0	T	Bubble chamber; foundational σ vs E_ν
BNL 7-ft [42]	D	1.5	T	Higher statistics; 15% normalisation tension with ANL
K2K [43]	C	1.3	T	First accelerator $\text{CC}\pi^+$ on carbon near detector
MiniBooNE [44]	CH_2	0.7	D	Q^2 , W , p_μ , θ_μ ; 20% excess over NUANCE
MINERvA [46]	CH	3.6	D	p_π , θ_π , Adler angles; NuMI medium-energy
T2K ND280 [49]	CH	0.85	D	p_μ , $\cos\theta_\mu$; off-axis beam similar to this work
ArgoNEUT [50]	Ar	9.6	T	First $\text{CC}\pi^+$ on Ar; 115 ν + 158 $\bar{\nu}$ events (1.25×10^{20} POT)
MicroBooNE BNB [52]	Ar	0.7	D	Full BNB dataset (1.11×10^{21} POT); $\sigma =$ (3.75 ± 0.80) $\times 10^{-38}$ cm^2/Ar
MicroBooNE NuMI ν_e [51]	Ar	0.73	D	First $\nu_e + \bar{\nu}_e$ $\text{CC}\pi^+$ on Ar; $\sigma = (0.93 \pm 0.28) \times$ 10^{-39} $\text{cm}^2/\text{nucleon}$

4 Beam and Neutrino Flux

4.1 NuMI beam and flux composition

The NuMI beam is produced by 120 GeV protons from the Fermilab Main Injector striking a graphite target. Charged secondary hadrons are focused by two magnetic horns whose polarity selects the sign of the focused mesons. In Forward Horn Current (FHC) mode the horns focus positive pions and kaons, giving a beam with predominantly ν_μ content; in Reverse Horn Current (RHC) mode the polarity is reversed to focus negatives, giving a predominantly $\bar{\nu}_\mu$ beam. MicroBooNE is located 135 mrad (7.71°) off-axis from the NuMI beam axis¹, reducing the peak neutrino energy to the 0.5–2.5 GeV range and narrowing the energy spectrum. Because MicroBooNE sits off-axis and at comparatively low energy, the horn sign-selection is imperfect and both modes carry a sizeable wrong-sign component, most notably RHC, whose wrong-sign ν_μ fraction is large.

The flux composition at the MicroBooNE site is taken from the new-G4 central-value flux histograms, integrated over $E_\nu > 60$ MeV to exclude an unphysical sub-threshold pile-up near $E_\nu = 0$ (removed by the flux group’s low-energy peak removal; below the reaction threshold and irrelevant to the measurement). The muon-flavour split of these central-value histograms matches the author’s authoritative integrated fluxes exactly (Sec. 4.4):

¹From the detector position in beam coordinates, (5502, 7259, 67270) cm: 91.1 m transverse at 672.7 m downstream. Earlier versions of this note quoted 110 mrad. See §4.2.

Table 3: NuMI flux composition at the MicroBooNE site by neutrino type, for the FHC and RHC horn modes (new-G4 central-value flux, integrated over $E_\nu > 60$ MeV, the same threshold as the authoritative integrals of Table 5; above 0.1 GeV the RHC split would read 42.0/55.2%). FHC is ν_μ -dominant; RHC is $\bar{\nu}_\mu$ -dominant with a large wrong-sign ν_μ fraction. The combined row weights each mode by its fluence, $\Phi \times \text{POT}$ (FHC 6.03, RHC 7.14×10^{11} cm⁻²; Tables 4 and 5), giving a near-balanced $\nu_\mu/\bar{\nu}_\mu$ mix. The signal is charge-blind ($|\nu_{\text{pdg}}| = 14$), so both ν_μ and $\bar{\nu}_\mu$ contribute in every mode.

Horn mode	ν_μ	$\bar{\nu}_\mu$	ν_e	$\bar{\nu}_e$
FHC	63.4%	34.0%	1.7%	0.9%
RHC	44.2%	53.3%	1.4%	1.2%
Combined	53.0%	44.4%	1.5%	1.1%

Flux corrections from the Package to Predict the FluX (PPFX) [8] are applied per event via the central-value weight stored in the analysis ntuples. The absolute flux normalisation of Sec. 4.4 uses the same new-G4 histograms.

4.2 Beam direction in detector coordinates

Two different angles describe the MicroBooNE–NuMI geometry, and they must not be confused (Fig. 2). The *off-axis angle* is a property of the beamline: the detector sits 91.1 m off the NuMI axis at 672.7 m downstream of the target, a baseline of 678.8 m and an off-axis angle of 7.71° (134.6 mrad).² That angle says nothing about how the TPC is oriented.

The angle that enters the measurement is a different one: the direction in which the neutrinos travel *relative to the detector z axis*. Averaged over interactions in the active volume, the true neutrino direction in detector coordinates is (0.4624, 0.0489, 0.8853), which lies 28.6° away from z . The detector z axis is therefore a poor measurement coordinate for the beam: it is not the direction the neutrinos came from, nor the NuMI beamline axis (itself 23.2° from z).

²Computed from the detector position in beam coordinates, (5502, 7259, 67270) cm, taken from the same rotation matrix used for the NuMI beamline geometry weights.

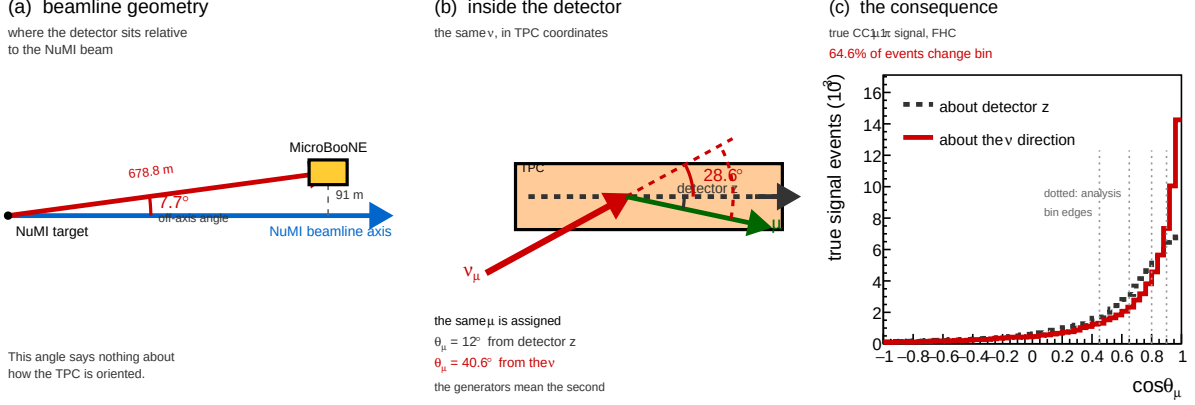


Figure 2: The NuMI geometry seen by MicroBooNE. (a) Beamline geometry, to scale: the detector sits 91.1 m off the NuMI axis at a baseline of 678.8 m, an off-axis angle of 7.7° . (b) The same neutrinos in TPC coordinates, to scale (the x - z projection; the quoted 28.6° is the full three-dimensional angle): the beam arrives 28.6° away from the detector z axis, so one and the same muon is assigned $\theta_\mu = 12^\circ$ about z but $\theta_\mu = 40.6^\circ$ about the neutrino. The generators mean the second. (c) The resulting shift in the true $\cos \theta_\mu$ distribution; dotted lines mark the analysis bin edges. Under the two conventions 64.6% of signal events fall in a different bin (44.6% for $\cos \theta_\pi$).

This matters because every polar angle and every transverse-kinematic-imbalance variable is defined with respect to the neutrino direction. In the generator predictions the neutrino defines $+z$ by construction — the GENIE event records used here have $p_x^\nu = p_y^\nu = 0$ and $p_z^\nu = E_\nu$ exactly — so a measurement that refers its angles to the detector z axis is not being compared with the prediction it appears to be compared with. The effect is not small: 64.6% of signal events change $\cos \theta_\mu$ bin between the two conventions, and 44.6% change $\cos \theta_\pi$ bin. For the TKI variables the axis error exceeds the observable itself, the mean δp_T being 0.742 GeV/ c about z against 0.264 GeV/ c about the beam.

Accordingly, all angular and TKI observables in this analysis are referred to the neutrino direction, following the β convention of the NuMI ν_e CC1 π internal note [1] and of the earlier NuMI ν_e cross-section measurement it derives from [2]. At truth level the exact per-event neutrino direction is used, which reproduces the generator convention exactly. At reconstruction level the neutrino direction is approximated by the vector from the NuMI target — at $(-31388, -3316, -60100)$ cm in detector coordinates, a 684 m baseline to the detector centre (678.8 m to the TPC origin quoted above) — to the reconstructed interaction vertex. Adopting the same β as the ν_e analysis keeps the two NuMI measurements directly comparable.

The choice of reconstruction-level axis is in any case not a sensitive one. A single fixed axis and the per-event β differ by a median of 2.4 mrad (maximum 7.3 mrad), which moves only 0.42% of muons across a $\cos \theta_\mu$ bin edge — to be compared with the 64.6% moved by the detector- z convention. Against the true per-event direction the two perform almost identically, with a mean opening angle of 1.95° for β and 2.00° for a fixed axis. What neither can recover is the physical divergence of the beam: neutrinos reaching the detector come from decays distributed along the decay pipe, not from the target, and over a 684 m baseline a 2.5 m detector subtends only ~ 4 mrad, so the vertex position carries almost no information about the direction. That residual is a resolution effect, absorbed by the response matrix like any other.

Quantities that are invariant under this choice of axis are unaffected: p_μ , p_π , the muon-pion opening angle $\theta_{\mu\pi}$, the invariant mass $W_{\pi p}$, and all integrated cross sections. The event selection is likewise unaffected, as no selection cut and no BDT input variable uses a polar angle.

4.3 Data exposure

The analysis is *blind*: the real beam-on data are not used. Each configuration is extracted on Poisson-thrown central-value fake data, generated and scaled *per run* to that run’s data POT (Table 4). The FHC measurement uses the full FHC exposure (Runs 1, 2, 4 and 5); RHC uses Runs 1, 2, 3, 4a, 4b and 4c; the combined measurement uses all fourteen overlay files. Every run is normalised independently to its own data exposure, so the prediction carries the data-POT mixture of run periods and the framework is ready for real per-run beam-on data (each fake-data sample is a drop-in replacement for its beam-on counterpart).

Table 4: Run-period exposure accounting (final, full FHC+RHC). “MC POT” is each period’s native simulated exposure; “Data POT” is the per-run exposure to which the (blind, per-run) fake data is scaled. *Each run is normalised independently* by its own factor, $\text{Scale} = (\text{run Data POT})/(\text{run MC POT})$, so the prediction is the data-POT-weighted mixture of run periods; the normalisation a real per-run beam-on dataset produces. This replaces an earlier single per-mode pooling factor: because the per-run MC/data POT ratios vary by up to a factor of three (Scale ranges 0.051–0.141 within FHC alone), a single factor would weight the runs by their MC statistics rather than by the data exposure. The RHC Run 4 row sums the 4a/4b/4c sub-files and Run 3 sums aa–ae; sub-files sharing a run’s data POT are scaled by the group MC total so they pool within that run. The combined measurement applies each run’s own factor. Run 5 uses the ν_μ overlay produced with reweighted PPFX flux weights (the only Run-5 overlay available). The fake data are thrown from the same per-run-scaled prediction, so the closure (Table 17) tests the bookkeeping of this scheme, not its correctness against data.

Run period	MC POT ($\times 10^{20}$)	Data POT ($\times 10^{20}$)	Scale
Run 1 (FHC)	23.282	2.192	0.0942
Run 2 (FHC)	24.934	1.268	0.0509
Run 4 (FHC)	28.335	2.075	0.0732
Run 5 (FHC)	19.300	2.231	0.1156
FHC total	95.85	7.766	<i>per run</i>
Run 1 (RHC)	8.997	0.605	0.0673
Run 2 (RHC)	57.865	2.591	0.0448
Run 3 (RHC, aa–ae)	55.185	5.003	0.0907
Run 4 (RHC, 4a–4c)	32.586	2.883	0.0885
RHC total	154.63	11.082	<i>per run</i>
Combined (FHC+RHC)	250.49	19.939	<i>per run</i>

The full exposure substantially reduces the data-statistical term (Table 17 of the technical supplement–Table 19 of the technical supplement), leaving the measurement flux- and detector-systematic limited. When the analysis is unblinded, the real beam-on POT (and gate counts) replace the data-POT values above.

4.4 Integrated flux

The absolute flux normalisation uses the new-G4 GEANT4 NuMI flux integrated over $E_\nu > 60$ MeV (the flux group’s low-energy peak removal, which removes an unphysical ~ 25 – 30 MeV artifact spike). The signal is charge-blind ($|\nu_{\text{pdg}}| = 14$, covering both $\nu_\mu \rightarrow \mu^- \pi^+$ and $\bar{\nu}_\mu \rightarrow \mu^+ \pi^-$), so the normalisation uses the combined $\nu_\mu + \bar{\nu}_\mu$ flux $\Phi = \Phi_{\nu_\mu} + \Phi_{\bar{\nu}_\mu}$. Because RHC is $\bar{\nu}_\mu$ -dominant with a different absolute flux than FHC, each configuration uses its *own* integrated flux:

Table 5: Integrated ν_μ and $\bar{\nu}_\mu$ fluxes per POT used by the analysis (new-G4 flux, $E_\nu > 60$ MeV), and the combined $\nu_\mu + \bar{\nu}_\mu$ value used as the per-configuration normalisation. The combined row is the data-POT-weighted average of the two modes (weights 7.766 : 11.082; Table 4). Each configuration is normalised with its own flux: the RHC flux is 5.4% below the FHC one, so using a single value for both would bias the RHC cross section by that amount.

Config	Φ_{ν_μ} ($10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$)	$\Phi_{\bar{\nu}_\mu}$	$\Phi_{\nu_\mu + \bar{\nu}_\mu}$	$\bar{\nu}_\mu$ frac.
FHC	4.43515	2.37644	6.81159	34.9%
RHC	2.92348	3.52298	6.44646	54.6%
Combined	3.54634	3.05057	6.59691	46.2%

The mean energy of the $\nu_\mu + \bar{\nu}_\mu$ flux above 60 MeV is 0.479 GeV (FHC) and 0.475 GeV (RHC) (ν_μ alone 0.468 and 0.544 GeV; $\bar{\nu}_\mu$ alone 0.500 and 0.417 GeV, the RHC antineutrinos being the focused, softer component at this off-axis angle). Only 40% of that flux lies above 0.3 GeV, 26% above 0.5 GeV and 13% above 1 GeV. Because the flux-averaged cross section divides by the *whole* flux above 60 MeV while single pion production only opens near 0.3 GeV, the total is roughly a factor 2.5 smaller than it would be on the BNB spectrum peaked at 0.7 GeV; this, together with the measured phase space ($p_\pi > 0.175$ GeV/ c , $\theta_{\mu\pi} < 2.6$ rad), is why the $\sim 1.0 \times 10^{-38} \text{ cm}^2/\text{Ar}$ found here is not to be compared directly with the $3.75 \times 10^{-38} \text{ cm}^2/\text{Ar}$ of the BNB measurement of §3.3. These fluxes are set per configuration in the extraction. Because every simulated event carries the PPFX central-value flux correction (§5; mean 0.944 on selected signal), the integrated flux that divides the event rate must be the PPFX-corrected central value, not the raw GEANT4 flux; the central-value histograms of the new-G4 flux file are taken to be that central value, and this should be confirmed with the flux author before unblinding, since a raw-G4 integral would put every absolute cross section $\sim 5\text{--}6\%$ low. **Open normalisation sign-off** (not required for the control-region opening, required before the signal region is opened): written confirmation by the NuMI flux contact that the integrated constants of Table 5 are PPFX-corrected central values consistent with the per-event PPFX central-value weighting. The integrated values were validated four independent ways: standalone NuWro and GENIE event generation on the same flux, the flux author’s numbers, and a GENIE closure.

The number of argon target nuclei in the fiducial volume is computed from first principles as

$$N_{\text{Ar}} = \frac{\rho_{\text{LAr}} \cdot V_{\text{FV}} \cdot N_A}{A_{\text{Ar}}} = 8.710 \times 10^{29}, \quad (2)$$

with $\rho_{\text{LAr}} = 1.3836 \text{ g/cm}^3$, $A_{\text{Ar}} = 39.948 \text{ g/mol}$ and $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$. The fiducial box is $10 < x < 246$, $-101 < y < 101$, $10 < z < 986 \text{ cm}$ ($4.6528 \times 10^7 \text{ cm}^3$), from which the $675.1 < z < 775.1 \text{ cm}$ dead region is subtracted, leaving $V_{\text{FV}} = 4.1761 \times 10^7 \text{ cm}^3$. Without that subtraction the same expression would give 9.705×10^{29} . This is the value the code uses, computed by the extraction from the fiducial box with the dead slab removed, and it is the value entering the conversion factor of §10 and the one-bin total of §11.4. The per-configuration normalisation factor is $N_{\text{norm}} = \Phi_{\text{mode}} \cdot N_{\text{POT,mode}} \cdot N_{\text{Ar}}$, evaluated with each mode’s flux and data POT (Table 4).

5 Monte Carlo Samples

Monte Carlo events are generated with GENIE v3 [9, 20] (the MicroBooNE tune [16]) on argon-40, propagated through the detector with Geant4 [24], and overlaid with real cosmic-ray data (the “overlay” technique) so that simulated events experience in-situ detector noise and cosmogenic activity. Table 6 summarises the input samples.

Table 6: Input samples (final, full FHC+RHC). “ n ” is the number of overlay files; “MC POT” is the summed native simulated exposure ($\times 10^{20}$). The blind Poisson-thrown fake data stands in for the (unused) beam-on data; the detector variations use the native Run-4 FHC and RHC samples. The per-file lists and POT/trigger values are in the configuration tables of the release.

Sample	Composition
ν_μ overlay (FHC)	Runs 1, 2, 4 (pandora) + Run 5 (reweighted-PPFX)
ν_μ overlay (RHC)	Runs 1, 2, 3, 4a, 4b, 4c (pandora)
Dirt MC (interactions outside the cryostat)	GENIE NuMI dirt overlays, per run (Run-1 file standing in for Run 2)
EXT (beam-off)	cosmic data, per run, horn-mode-independent
Detector variations	Run-4 FHC + Run-4 RHC, CV + 8 knobs each
Fake data (blind)	Poisson-thrown central value, per configuration
Beam-on data	blinded, not used until unblinding

Event weights. Each MC event is weighted by

$$w = \text{Scale}[\text{run}] \times w_{\text{PPFX}} \times w_{\text{tune}}, \quad (3)$$

where $\text{Scale}[\text{run}] = (\text{run Data POT})/(\text{run MC POT})$ is the *per-run* POT factor of Table 4: each run period is scaled independently to its own data exposure, so the prediction reproduces the data-POT mixture of runs rather than the MC-statistics mixture (see the caption of Table 4). Operationally this is exactly what the framework does natively each overlay file carries its run’s data POT via a matching per-run beam-on (here, per-run fake-data) sample, and is scaled by `run.bnb_pot` divided by that run’s MC exposure, taken as the sum of the *distinct* exposure values stored by the run’s files (the run total under both processed-file conventions; caption of Table 20). w_{PPFX} is the PPFX central-value flux correction and w_{tune} is the GENIE/GEANT4 reweighting correction. One weight rule is used everywhere:

$$w_{\text{safe}} = \begin{cases} 1, & w \text{ non-finite, } w < 0, \text{ or } w > 30, \\ w, & \text{otherwise (a weight of exactly zero is kept),} \end{cases} \quad (4)$$

applied to the product $w = \text{Scale} \times w_{\text{PPFX}} \times w_{\text{tune}}$ (including the 0.65 normalisation weight the dirt overlay carries, which scales its generated exposure to the out-of-cryostat rate expected at the NuMI site). The universe construction, the covariance, the unfolding inputs, the cut-flow histograms, the count-level tables, the control-region yields, the transfer factors and the beam-on comparison at unblinding all use Eq. (4); one rule everywhere is what makes count-level and covariance-level numbers comparable to the per cent. At unblinding the per-run fake-data samples are replaced by the corresponding per-run beam-on files (same runs, same POT), so the normalisation is already correct.

EXT scaling. Beam-off data are cosmic-only, so a beam-off sample is matched to a beam-on run by *run period* alone, irrespective of the horn polarity it was recorded under. Each run period uses the beam-off sample(s) taken in that period, scaled by that run’s beam-on/beam-off gate ratio (times the 0.98 NuMI beam-occupancy factor); where a run has several beam-off files the framework sums their gate counts and pools them:

$$\text{EXT}_{\text{run}} = 0.98 \frac{\text{bnb_gates}[\text{run}]}{\sum_{\text{files} \in \text{run}} \text{ext_gates}} \sum_{\text{files} \in \text{run}} N_{\text{EXT}}^{\text{sel}}. \quad (5)$$

The gate counts per run are released with the results (`data_release/ext_gates.tsv`). No Run-2 beam-off sample exists and none will be produced, so the Run-1 sample, the nearest pre-CRT period, stands in for Run 2 permanently, scaled by its own gate count. The same beam-off files serve the FHC and RHC beam-on runs of a given period, and the combined configuration

lists each (file, run) pair once. The selected beam-off rate per 10^6 gates is 1.75 (Run 1), 0.43 (Run 3b), 1.15 (Run 4) and 0.52 (Run 5): the pre-CRT Run-1 rate is 1.5–4 times the CRT-era ones (the rates carry 16–35% Poisson uncertainties, from 8, 14, 40 and 10 raw selected beam-off events), which is why period matching rather than a pooled sample is used. At the FHC final cut the beam-off background is 30 events (23 RHC, 53 combined), 1.9% of the selected sample. Figure 3 shows the per-period rates, scale factors and resulting yields; the machine-readable version is `data_release/ext_gates.tsv`, and the normalisation manifest (Sec. 14 of the technical supplement) checks that every (beam-off file, run) pair enters each configuration exactly once and that the gate counts are consistent.

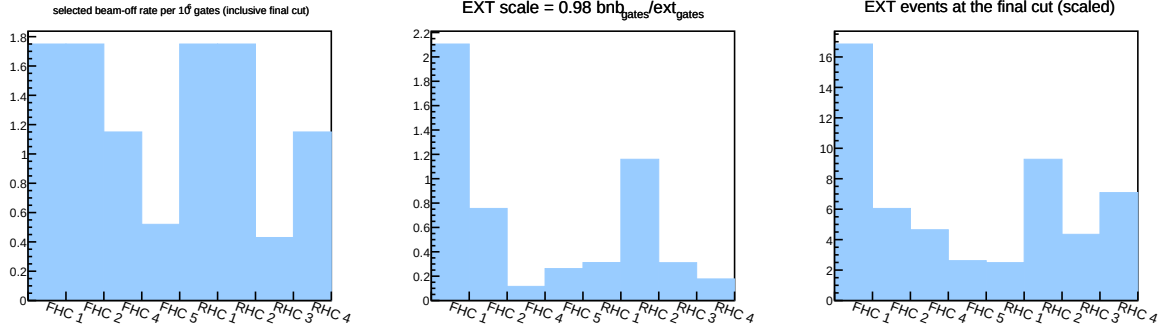


Figure 3: Beam-off validation by run period, from `data_release/ext_gates.tsv`: selected beam-off rate per 10^6 gates at the inclusive final cut (left), the per-run scale factor $0.98 \times (\text{beam-on gates})/(\text{beam-off gates})$ (centre) and the resulting beam-off events at the final cut (right), for each beam-on run period of the FHC and RHC configurations. The Run-1 sample serves Run 2 of both modes as a stand-in (for RHC Run 1 it is the native period sample). The pre-CRT Run-1 rate is 1.5 to 4 times the CRT-era ones.

No further cosmic-rejection cut. Further cosmic-rejection cuts were not adopted because the beam-off contribution is 1.6% of the selected prediction and is removed by an unbiased data-driven subtraction, so a cut can only reduce the subtraction’s statistical uncertainty, about five events on a sample of 1 418; every plausible cut loses more signal than that. The directional and calorimetric variants that were considered are in Sec. 13.2 of the technical supplement.

Software trigger. Data and EXT files are pre-filtered to require the software trigger; the CRT-era beam-off files (Runs 4–5) carry a vestigial trigger flag that is zero throughout, so for those files the pre-CRT trigger flag is used instead. MC events failing the software trigger are removed at the NeutrinoSlice cut stage. In practice this has no measurable effect at the current working point: all MC events passing the topological cut are verified to pass the software trigger.

6 Signal Definition

The signal is defined at generator level and is used only to construct the selection efficiency and response matrix. It is *never* applied as a reconstruction cut. The definition is *charge-blind*: both $\nu_\mu \rightarrow \mu^- \pi^+$ and $\bar{\nu}_\mu \rightarrow \mu^+ \pi^-$ are treated as signal on the same footing.

Neutrino flavour

$$|\nu \text{ PDG}| = 14 \ (\nu_\mu \text{ or } \bar{\nu}_\mu).$$

Interaction type

Charged-current.

True vertex

Inside the fiducial volume (Section 7).

Primary multiplicity

Exactly one muon ($|\text{PDG}| = 13$), exactly one charged pion ($|\text{PDG}| = 211$) *of any momentum* among the primary final-state particles of the interaction (a second charged pion below threshold makes the event background, not signal; pions from the decay of primary hyperons are not primaries and are not counted), zero neutral pions ($|\text{PDG}| = 111$), zero kaons ($K^\pm$, K^0 , K_S , K_L) and no other mesons. Any number of protons (and neutrons) is allowed.

Kinematic thresholds

$p_\mu > 0.15 \text{ GeV}/c$, $p_\pi > 0.175 \text{ GeV}/c$, and $\theta_{\mu\pi} < 2.6 \text{ rad}$. These define the *measured phase space*, chosen from the efficiency turn-on curves (Section 9). The momentum efficiency turns on sharply at $\sim 0.15\text{--}0.20 \text{ GeV}/c$ (set by the 10 cm muon and 20 cm pion track-length cuts); below that it is only $\sim 1\text{--}3\%$ with a strongly model-dependent correction. The opening-angle upper bound (2.6 rad) excludes the large-angle region, which is dominated by cosmic and dirt background ($\sim 7\%$ purity in the $\theta_{\mu\pi} > 2.513 \text{ rad}$ bin). No *lower* opening-angle cut is applied: the small-angle region is high-purity ($\sim 58\%$), so cutting it would only remove signal (it is low-efficiency but not low-purity). The same cut is applied at reco level so the signal and selection acceptances match. The first p_μ/p_π bins and the $\theta_{\mu\pi}$ upper edge start/end at these thresholds.

(Earlier iterations used $p_\mu, p_\pi > 0.1 \text{ GeV}/c$ with the full $0\text{--}\pi$ angle; that left near-dead efficiency bins at low momentum and at the large-angle extreme. The phase space above raises the efficiency from 12.1% to 15.6% at $\sim 55\%$ purity at the beam-off scaling then in use, see Section 9.)

Pion threshold and the sibling ν_e measurement. The companion MicroBooNE NuMI $\nu_e \text{ CC } 1\pi$ analysis defines its pion signal by kinetic energy, $\text{KE}_\pi > 40 \text{ MeV}$, equivalent to $p_\pi > 0.113 \text{ GeV}/c$. Our $p_\pi > 0.175 \text{ GeV}/c$ ($\text{KE}_\pi > 84 \text{ MeV}$) is therefore a higher threshold. Lowering it toward the ν_e value is possible, the signal definition is a truth-level choice and the unfolding maps reco to truth but the pion momentum reconstruction (track range under the muon hypothesis, mass-corrected) degrades rapidly below $0.175 \text{ GeV}/c$. The table below is the study that led to *rejecting* a lower threshold; it is not part of the measurement. The measured phase space is $p_\pi > 0.175 \text{ GeV}/c$ throughout this note, and no cross section is reported below it:

Table 7: Pion momentum reconstruction versus true p_π (correctly reconstructed signal pions). Below the $0.175 \text{ GeV}/c$ threshold the bias and resolution blow up and the reconstructed fraction halves.

True p_π [GeV/c]	Bias $\langle (p_\pi^{\text{reco}} - p_\pi^{\text{true}})/p_\pi^{\text{true}} \rangle$	RMS resolution	Reco fraction
0.080–0.113	+131%	73%	3.4%
0.113–0.150	+66%	56%	3.4%
0.150–0.175	+30%	51%	4.3%
0.175–0.250 (threshold)	−8%	31%	10.0%
0.250–0.400	−30%	21%	10.4%

Throughout this note the pion-momentum bias is the *signed* quantity $b(p_\pi) = \langle (p_\pi^{\text{reco}} - p_\pi^{\text{true}})/p_\pi^{\text{true}} \rangle$, and it has two regimes that must not be conflated. Below threshold, the small subset of pions that reconstructs at all is biased *upward* (+30 to +131% in the table): these are the few tracks long enough to pass, a conditional sample. At the threshold the mass correction removes most of the hypothesis bias (−8%, with 31% RMS resolution). Above about $0.4 \text{ GeV}/c$ the bias turns increasingly *negative*, because pions interact before they stop and the visible range is

truncated; the mean reconstructed momentum saturates near 0.34 GeV/ c , so the bias reaches -65% above 0.8 GeV/ c . In the [0.113, 0.175] GeV/ c region the resolution is 51–56% with only $\sim 3\text{--}4\%$ reconstructability. Extending the measurement to the ν_e threshold would thus add bins carrying large, strongly model-dependent unfolding systematics. For consistency with the ν_e measurement the threshold could be lowered to 0.113 GeV/ c with the [0.113, 0.175] GeV/ c region kept in a single coarse bin to contain those systematics, or both phase spaces reported; this is a systematics-versus-cross-analysis-consistency trade rather than a reconstruction limitation.

Why the threshold was not lowered. Extending to the sibling ν_e threshold was investigated and rejected: the [0.113, 0.175] GeV/ c region has 2.2% selection efficiency and a strongly model-dependent reconstruction correction. The study is recorded in the technical supplement. It is not part of this measurement, and no cross section is reported below $p_\pi > 0.175$ GeV/ c .

7 Fiducial Volume

The fiducial volume (FV) is:

$$x \in [10.0, 246.0] \text{ cm}, \quad y \in [-101.0, +101.0] \text{ cm}, \quad z \in [10.0, 986.0] \text{ cm}, \quad (6)$$

with the dead-wire region $z \in [675.1, 775.1] \text{ cm}$ excluded. The total FV volume is

$$\begin{aligned} V_{\text{FV}} &= (246 - 10) \times 202 \times (976 - 100) \text{ cm}^3 \\ &= 236 \times 202 \times 876 \text{ cm}^3 \approx 4.17 \times 10^7 \text{ cm}^3. \end{aligned} \quad (7)$$

A wider *containment* region (2 cm inset from the active-volume faces, except the top face, which is inset by 10 cm) is used to require that the pion candidate track endpoint is contained:

$$x \in [2.0, 254.35] \text{ cm}, \quad y \in [-113.53, +107.47] \text{ cm}, \quad z \in [2.1, 1034.9] \text{ cm}. \quad (8)$$

7.1 Which volume applies where

Three distinct volumes appear above and it is worth stating plainly which is used for what, since “contained” and “in the fiducial volume” are not the same requirement.

volume	extent	used for
fiducial (FV)	$[10, 246] \times [-101, 101] \times [10, 986] \text{ cm}$ minus $z \in [675.1, 775.1] \text{ cm}$	neutrino vertex, true <i>and</i> reco
containment	$[2, 254.35] \times [-113.53, 107.47] \times [2.1, 1034.9] \text{ cm}$	track endpoints
active TPC	$[0, 256.35] \times [-115.5, 117.5] \times [0, 1036.9] \text{ cm}$	(reference only)

True and reco use identical dimensions. The framework allows the true and reconstructed fiducial volumes to differ, and other selections use that freedom. Here they are defined as the same box. The signal definition tests the *true* neutrino vertex and the selection tests the *reconstructed* vertex, both through the same helper, so both inherit the dead-region veto as well. The truth and reconstructed boundaries are geometrically identical, so there is no *intentional* acceptance mismatch. Vertex-resolution and space-charge effects can nevertheless move a reconstructed vertex across these boundaries, and events that migrate across the FV or dead-region edge are represented in the simulated response (true-in, reco-out as inefficiency; true-out, reco-in as the out-of-FV background category of Table 10) rather than being absent.

The dead-region veto applies to the vertex only. An event whose vertex lies outside the unresponsive-wire slab is kept even if one of its tracks crosses that slab. This is the conventional choice, but it is an assumption rather than a necessity: tracks crossing dead wires have degraded calorimetry and range reconstruction, and the associated inefficiency is absorbed into the response matrix rather than removed by a cut.

The dead slab is also removed from the target count. The argon target number is computed from the FV with the slab subtracted, not from the full box. Retaining the slab would inflate the target count by $100/876 \approx 11\%$ and bias every cross section low by the same factor. Note that this subtraction is applied on the NuMI code path specifically; the generic path computes the target number without it, so the two must not be mixed.

Relation to the active volume. The FV is inset from the active TPC by approximately 10 cm in x , 15 cm in y , 10 cm at the upstream z face and 51 cm at the downstream face. The containment box, by contrast, is inset by only 2 cm (10 cm at the top): track endpoints are required to be inside the detector, not inside the fiducial region, which is the correct requirement for range-based momentum and for the pion-containment cut.

8 Event Selection

The event selection is implemented as a selection class of the extraction framework, which produces the response matrices and spectra for all results in this note; a standalone re-implementation, kept as an independent cross-check, applies the identical cut sequence. The selection comprises ten sequential cuts (Cut 0 is the unselected reference sample).

8.1 BDT-based particle identification

Three gradient-boosted decision trees (TMVA [26]), trained on the MicroBooNE overlay MC, supply the calorimetric particle identification used in the muon- and pion-candidate cuts. All three share a common set of per-track inputs: the three Bragg-peak dE/dx likelihoods under the proton, muon and MIP hypotheses, the log-likelihood-ratio PID score, the Pandora track-vs-shower score, and the space-charge-corrected track end-point coordinates:

- **MIP BDT:** separates minimum-ionising tracks (muons, pions) from stopping protons. Both the muon and the pion candidate are required to have a MIP score above -0.1 (Cuts 4–5).
- **Pion BDT:** a dedicated pion-versus-proton discriminant applied to the pion candidate, pion score above -0.1 (Cut 5).
- **Muon BDT:** the same inputs plus the track length. It applies no threshold: the set of *qualifying* muon-candidate tracks is defined by the cuts of Table 8, and the muon BDT only *orders* that set; the qualifying track with the highest muon-BDT score is chosen as the muon (rather than the longest qualifying track, which the BDT ordering supersedes).

The two candidate-PID BDTs deliberately omit track length so the score does not bias the momentum spectra. A fourth score, a cosmic-rejection BDT, is available for cosmic rejection but is disabled by default (no benefit in an A/B test; the Pandora topological score already provides the cosmic rejection at Cut 3).

8.2 Cut definitions

Cut 0. EventsInTrueFV. All events. Reference denominator for truth-level studies.

- Cut 1. NeutrinoSlice.** ≥ 1 reconstructed track, the software trigger, and a cosmic-BDT requirement that is disabled (no benefit found in an A/B test).
- Cut 2. InFiducialVol.** The SCE-corrected reconstructed neutrino vertex lies inside the FV.
- Cut 3. Topological.** Topological score > 0.67 . This Pandora-produced multivariate score distinguishes cosmic-ray topologies from neutrino-like topologies and provides the primary cosmic rejection: EXT events are reduced by a factor of ~ 28 – 29 between Cut 2 and Cut 3 (Table 10).
- Cut 4. MuonCandidate.** At least one primary PFP track satisfies the muon PID criteria of Table 8. Among the qualifying tracks, the one with the highest muon-BDT score (§8.1) is the muon candidate; track length enters only through the 10 cm qualifying cut and as a BDT input, not as the ordering.
- Cut 5. ContainedPion.** Exactly one primary track (other than the muon candidate) satisfies the pion PID criteria of Table 9 and has its endpoint inside the containment volume. The track with the highest LLR PID score is the pion candidate.
- Cut 6. MuonIn3Planes.** The muon candidate has ≥ 1 hit on each of the U, V, Y planes.
- Cut 7. PionIn3Planes.** The pion candidate has ≥ 1 hit on each of the U, V, Y planes.
- Cut 8. ShowerCut.** Zero primary-level EM showers, unless exactly one shower is present with fewer than 50 Y-plane hits and a vertex distance > 10 cm (consistent with a delta-ray or low-energy nuclear photon).
- Cut 9. OpeningAngle.** The reconstructed μ – π opening angle satisfies $\theta_{\mu\pi} < 2.6$ rad ($\approx 149^\circ$), rejecting the large-angle cosmic/dirt background. No lower bound is applied (the small-angle region is high-purity). The same cut is imposed on the signal definition (Section 6).
- Cut 10.**
Nonprotons (final cut). Fewer than 4 primary tracks have LLR PID > 0.1 (non-proton-like), and the total number of primary reconstructed tracks is less than 5.

Table 8: Muon candidate PID thresholds (applied at Cut 4).

Variable	Threshold	Description
Track score	> 0.80	Track-like PFP
Vertex distance	≤ 4.0 cm	Attached to ν vertex
Track length	> 10.0 cm	Minimum track length
LLR PID	> 0.20	Muon-like
MIP BDT	≥ -0.10	Minimum-ionising-particle

Table 9: Pion candidate PID thresholds (applied at Cut 5). Endpoint containment requires the track endpoint inside the containment volume. The Bragg-pion score is the pion-hypothesis Bragg-peak likelihood score (rejects stopping protons).

Variable	Threshold	Description
Track score	≥ 0.50	Track-like PFP
LLR PID	> 0.10	Pion / non-proton-like
Vertex distance	≤ 4.0 cm	Attached to ν vertex
Track length	> 20.0 cm	Minimum track length
MIP BDT	> -0.10	MIP-like
Pion BDT	> -0.10	Pion-like (vs. proton)
Bragg-pion score	≥ 0.08	Proton rejection
Endpoint contained	yes	Inside containment volume

8.3 Kinematic reconstruction

Muon momentum. A *hybrid* estimator, switched on containment of the muon-candidate track: the range-based momentum is used when the track ends inside the detector, and the multiple-Coulomb-scattering momentum when it exits (range is meaningless for an exiting track). The reported p_μ is the hybrid, not either estimator alone. In the selected FHC sample the split is 37% contained (range) and 63% exiting (MCS [14]), i.e. the measurement is *MCS-dominated*: the NuMI muons are energetic and the MicroBooNE TPC is small, so most of them leave it. This matters for the p_μ resolution and for the detector systematic, since the range and MCS estimators carry different uncertainties. The eight detector-variation knobs do not vary the multiple-Coulomb-scattering momentum scale or resolution directly, and no dedicated MCS systematic is assigned; the MCS estimator’s -10% bias and 30% resolution (Fig. 20) enter through the response matrix only. A dedicated term is listed in §15.

Pion momentum. The starting estimator is the *muon-hypothesis* range momentum. Before a reconstructed p_π bin is assigned it is converted to the pion hypothesis by preserving the inferred kinetic energy, $T = \sqrt{p_{\mu\text{-hyp}}^2 + m_\mu^2} - m_\mu$ and $p_\pi = \sqrt{(T + m_\pi)^2 - m_\pi^2}$; the stored branch is not modified, the conversion is written into the reconstructed-bin definition of the bin configuration (Sec. 2.1 of the technical supplement, where it raises the lowest-bin reconstructed-to-true ratio from 0.888 to 0.976). The muon mass (105.7 MeV) is close to the pion mass (139.6 MeV), so the range-to-momentum conversion is a good proxy for a pion that stops, and, crucially, the result is a *momentum*, directly comparable to the true p_π it is binned against. For a pion that interacts hadronically before stopping the visible range is truncated and the estimator is biased low; that, not the mass hypothesis, is what limits the observable (§6). The proton-hypothesis range *kinetic energy* is not used: it is a different physical quantity from the momentum on the true axis, so binning it with the same edges would shift the migration matrix systematically rather than leave it near-diagonal (a true 0.2 GeV/ c pion gives $KE \approx 0.104$ GeV, two bins low, and a true 0.175 GeV/ c pion falls below the first reco bin entirely). Range momentum is meaningful only for contained tracks, which the containment requirement in the selection guarantees.

Angles. Polar angles are defined with respect to the *neutrino direction* $\hat{u}_\nu$, not the detector z -axis: $\cos\theta_\mu = \hat{u}_\mu \cdot \hat{u}_\nu$, $\cos\theta_\pi = \hat{u}_\pi \cdot \hat{u}_\nu$. The two are far from equivalent — the NuMI beam reaches MicroBooNE 28.6° away from z , and referring the angles to z would place 64.6% of signal events in a different $\cos\theta_\mu$ bin than the generator predictions they are compared with (Sec. 4.2). For reconstructed quantities $\hat{u}_\nu$ is the β direction of Refs. [1, 2], from the NuMI target to the reconstructed vertex; at truth level it is the exact per-event neutrino direction. The opening

angle $\theta_{\mu\pi} = \arccos(\hat{u}_\mu \cdot \hat{u}_\pi)$ is computed from reconstructed track directions and, being an angle between two measured tracks, is independent of this choice.

8.4 Proton-tagged subsample

A subsample requiring, in addition, a contained proton candidate with $p_p > 0.3 \text{ GeV}/c$ is selected by the same chain with one further cut. It gives access to hadronic and transverse-imbalance observables but introduces its own proton-assignment and model-dependence requirements, and it is documented in a separate note (Sec. 2 of the proton-tagged note); it is not part of the approval request made here.

9 Cut-Flow and Selection Performance

Figure 4 shows the full-exposure cut-flow yields for all three configurations as stacked plots, and Table 10 gives the detailed per-category, per-cut breakdown for each.

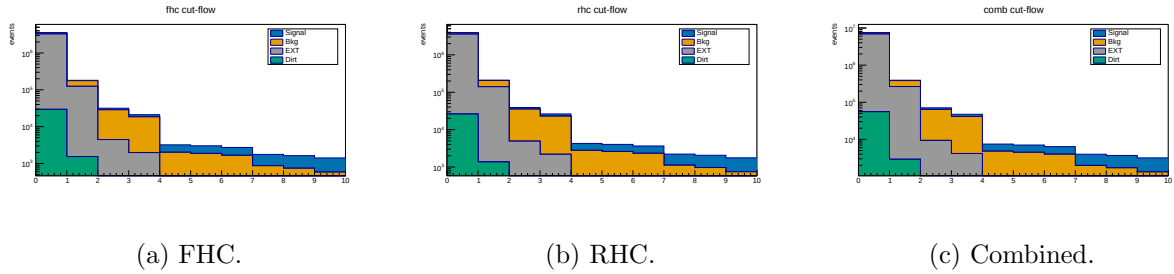


Figure 4: Cut-flow yields at *full exposure* for FHC, RHC and combined (log scale; MC-prediction stack: signal blue, background orange, EXT grey, dirt green; colourblind-safe Okabe-Ito), signal and background POT-scaled per mode. Data is blind, so this shows the MC-prediction stack; the detailed per-category breakdown is in Table 10.

Table 10: Cut-flow yields at *full exposure* for FHC, RHC and combined (MC prediction; data is blind: the fake-data *expectation* equals the central-value prediction by construction, while the realised pseudo-data fluctuate according to the Poisson throw). The histograms behind this table carry the framework’s full central-value weight, $\text{tune} \times \text{PPFX} \times \text{normalisation}$ (§5, with the safe-weight rule of Eq. (4)), the same weight as the fake-data throw and the unfolding, so the final-cut yields agree with the cut-and-count Table 22 to $< 1\%$. Signal and “Bkg” (all non-signal MC: OOFV, NC, $\text{CC}1\pi^0$, $\text{CC}0\pi$, $\text{CCN}\pi$, CCK, CCothers) are from the ν_μ overlays, POT-scaled *per run* (each run by its own $D_{\text{run}}/\text{MC}_{\text{run}}$, Table 4); EXT (beam-off cosmic) is scaled by the (per-run summed) per-run beam-on/beam-off *gate* ratio times the 0.98 NuMI beam-occupancy factor, each run period using the beam-off sample taken in that period (§5); dirt from the GENIE NuMI dirt overlay. Pred. = Signal + Bkg + EXT + Dirt. Eff. is the cumulative signal efficiency relative to the generated signal (the None stage, where all true signal is counted); Pur. = Signal/Pred.

Cut	Signal	Bkg MC	EXT	Dirt	Pred.	Eff. [%]	Pur. [%]
<i>FHC (Runs 1,2,4,5; 7.766×10^{20} POT)</i>							
None	4 753	235 281	3 256 689	29 494	3 526 216	100.0	0.1
InFiducialVol	3 955	52 835	122 860	1 546	181 195	83.2	2.2
Topological	2 995	24 262	4 299	168	31 724	63.0	9.4
MuonCandidate	2 689	16 484	1 870	113	21 156	56.6	12.7
ContainedPion	1 195	1 823	173	21	3 212	25.1	37.2
MuonIn3Planes	1 143	1 722	144	14	3 023	24.1	37.8
PionIn3Planes	1 062	1 549	113	6.0	2 730	22.3	38.9
ShowerCut	896	783	83	5.1	1 768	18.9	50.7
OpeningAngle	890	710	32	1.6	1 634	18.7	54.5
Nonprotons	833	560	23	1.1	1 418	17.5	58.8
<i>RHC (Runs 1,2,3,4a,4b,4c; 1.108×10^{21} POT)</i>							
None	5 778	286 540	3 560 975	26 089	3 879 381	100.0	0.1
InFiducialVol	4 828	65 474	139 122	1 367	210 791	83.6	2.3
Topological	3 743	30 446	4 767	149	39 104	64.8	9.6
MuonCandidate	3 363	20 709	2 090	100	26 262	58.2	12.8
ContainedPion	1 488	2 576	191	19	4 274	25.7	34.8
MuonIn3Planes	1 426	2 430	158	13	4 028	24.7	35.4
PionIn3Planes	1 329	2 194	121	5.3	3 650	23.0	36.4
ShowerCut	1 101	1 026	86	4.5	2 218	19.1	49.7
OpeningAngle	1 093	933	33	1.5	2 060	18.9	53.1
Nonprotons	1 015	720	23	1.0	1 759	17.6	57.7
<i>Combined FHC+RHC (1.885×10^{21} POT)</i>							
None	10 530	521 821	6 817 663	55 583	7 405 598	100.0	0.1
InFiducialVol	8 783	118 309	261 981	2 913	391 986	83.4	2.2
Topological	6 738	54 707	9 066	317	70 828	64.0	9.5
MuonCandidate	6 052	37 193	3 960	213	47 418	57.5	12.8
ContainedPion	2 682	4 399	364	40	7 485	25.5	35.8
MuonIn3Planes	2 570	4 152	302	27	7 051	24.4	36.4
PionIn3Planes	2 391	3 744	234	11	6 380	22.7	37.5
ShowerCut	1 997	1 809	169	9.5	3 985	19.0	50.1
OpeningAngle	1 984	1 643	65	3.1	3 694	18.8	53.7
Nonprotons	1 848	1 280	46	2.1	3 177	17.6	58.2

Why the phase space and the opening-angle cut are what they are. The thresholds were chosen from the efficiency turn-ons and purity studies on the Run-1 FHC sample (Sec. 13.1 of the technical supplement); the measured phase space removes the near-zero-efficiency momentum regions and the low-purity large-opening-angle region. On the full exposure the events the $\theta_{\mu\pi} < 2.6$ rad cut removes are exactly those of the cosmic control region (Table 24): in FHC

6.0 signal, 69 neutrino background, 53 beam-off and 2.2 dirt, a signal purity of 4.6%. Dropping the cut would lower the signal-region purity from 58.5% to 54.5%, raise its beam-off content from 30 to 83 events and recover only 2.7% of the generated signal. No lower opening-angle cut is used because the small-angle region is high-purity ($\sim 58\%$), so cutting it lowers the selected signal without improving purity. The per-run exposure scaling of the yields differs from a single per-mode factor by about 2% in the final-cut signal, a nearly uniform re-weighting that moves efficiency and purity by $< 0.5\%$.

9.1 Key N–1 distributions

Figure 5 shows the N–1 distributions (all cuts applied except the one under study) for the two cuts instrumented per event; the topological score (Cut 3) and the μ – π opening angle (Cut 9), for all three configurations. The shapes are consistent across horn modes, as expected since the selection is identical; RHC and combined simply carry the larger exposure. Signal (blue) and background (orange) are stacked and the dashed line marks the applied threshold. The remaining cut variables are shown at the final cut in Sec. 9.3.

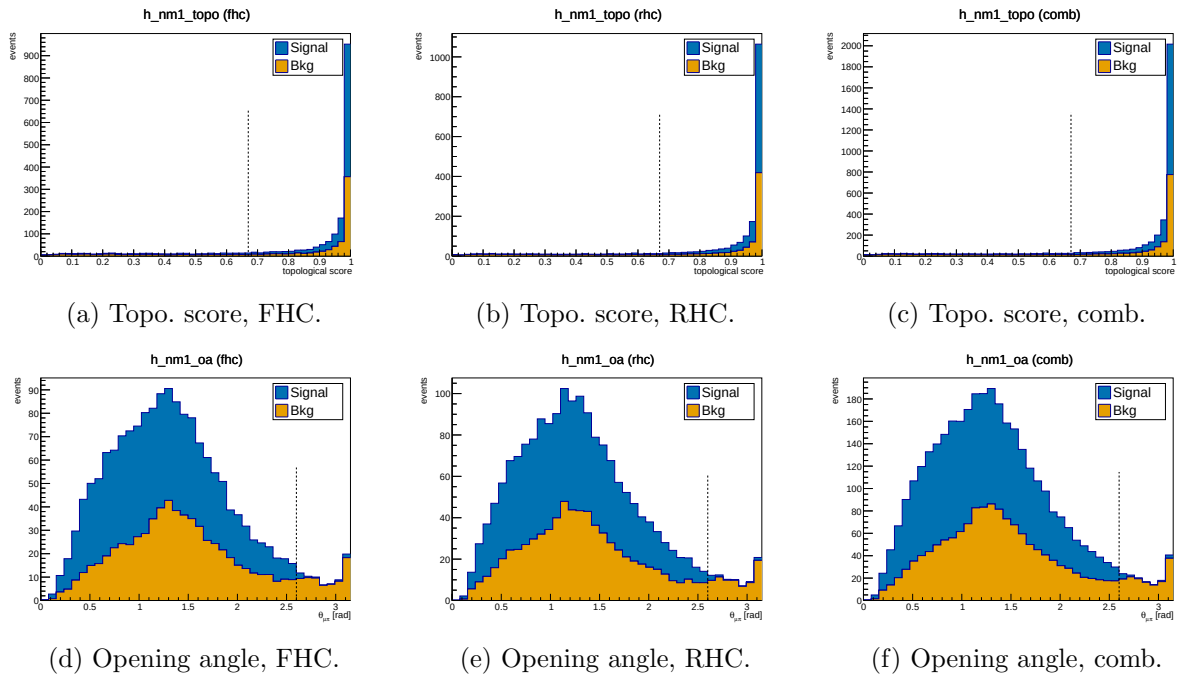
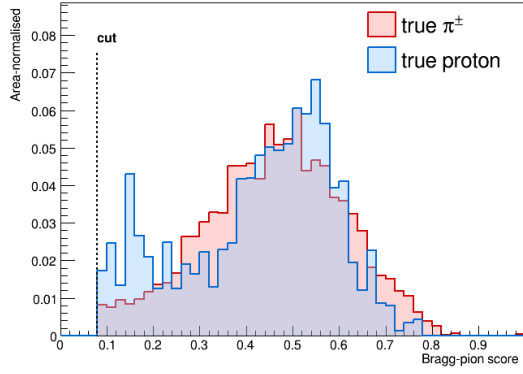


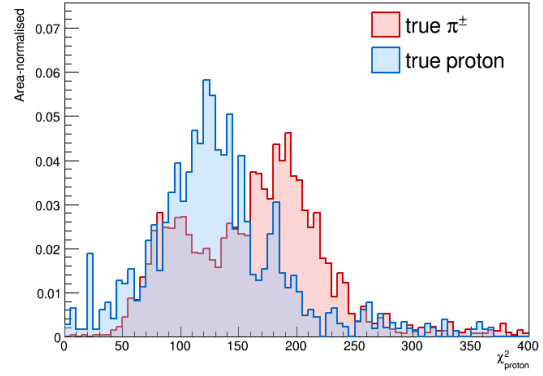
Figure 5: N–1 distributions for the topological score (Cut 3, top) and μ – π opening angle (Cut 9, bottom), for FHC, RHC and combined. Signal (blue) and background (orange) stacked; dashed line = applied threshold. Colourblind-safe Okabe-Ito palette.

9.2 Pion proton-rejection variables

Figure 6 shows the two main proton-rejection variables on the pion candidate, evaluated on MC truth-labelled pions ($\pi^\pm$) and protons at the final cut stage, used to tune the working-point thresholds.



(a) Bragg-pion score. Working-point threshold: ≥ 0.08 .

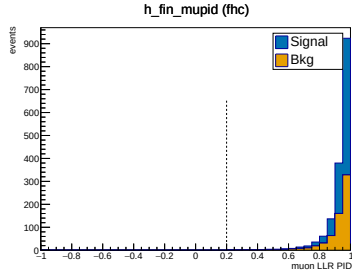


(b) χ^2 proton PID.

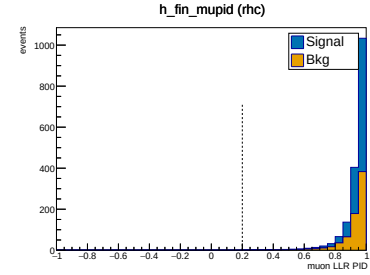
Figure 6: Pion-candidate PID distributions for true $\pi^\pm$ (red) and true protons (blue) at the final cut stage. The Bragg-pion score provides the primary proton rejection; χ_p^2 is studied as cross-check.

9.3 Final-cut data/MC distributions

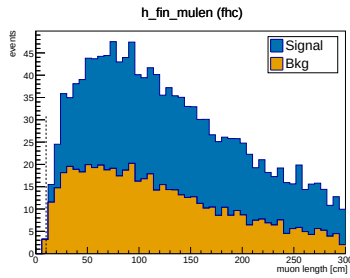
Figures 7 and 8 show the candidate-level distributions at the final selection cut for the four instrumented PID and track-length variables, for FHC and RHC. Signal (blue) and background (orange) are stacked (colourblind-safe Okabe-Ito palette).



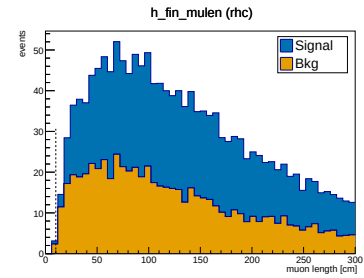
(a) Muon LLR PID, FHC.



(b) Muon LLR PID, RHC.

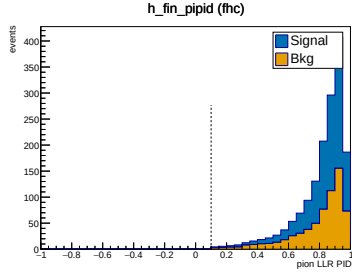


(c) Muon length, FHC.

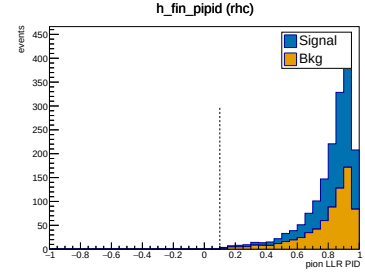


(d) Muon length, RHC.

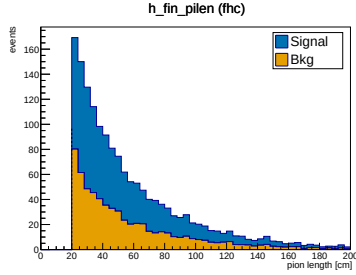
Figure 7: Muon-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC and RHC (the combined configuration is consistent with both and is not shown). Signal (blue) / background (orange) stacked.



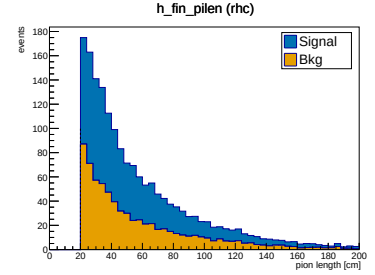
(a) Pion LLR PID, FHC.



(b) Pion LLR PID, RHC.

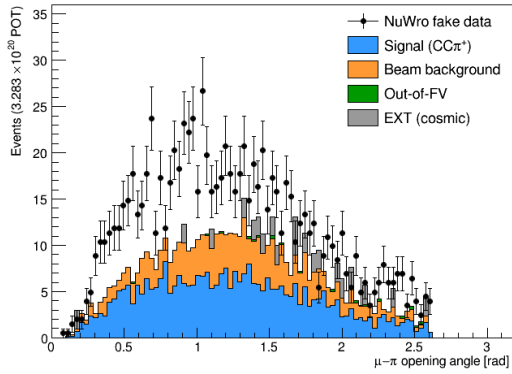


(c) Pion length, FHC.

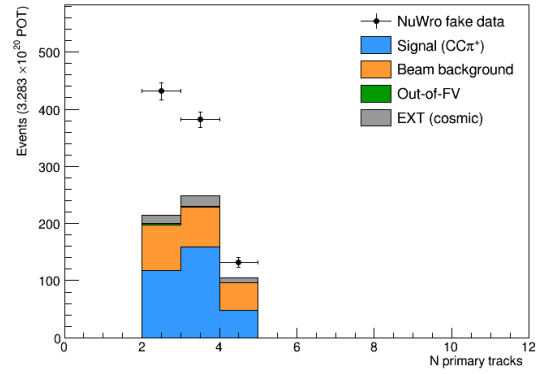


(d) Pion length, RHC.

Figure 8: Pion-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC and RHC (the combined configuration is consistent with both and is not shown). Signal (blue) / background (orange) stacked.



(a) μ - π opening angle.



(b) Number of primary tracks.

Figure 9: Fake-data/MC comparison for topological variables at the final cut.

9.4 Background decomposition

Table 11 shows the truth-level background breakdown at the final cut.

Table 11: Background composition at the final cut, FHC full exposure (truth classes of the selection’s counters, CV-weighted and per-run POT-scaled; the same classes and yields as the signal-region column of Table 22 of the technical supplement). Fractions are of the 637 neutrino-background events; beam-off and dirt are listed for scale.

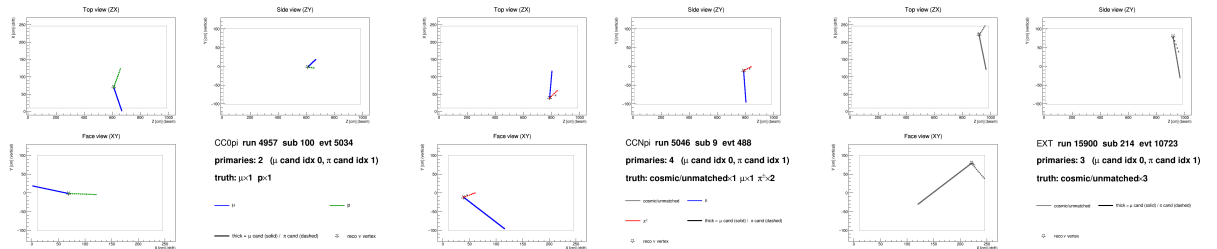
Category	Events	Fraction
ν_μ CC 0π (proton or other track as pion)	244	38%
ν_μ CC $\geq 2\pi^\pm$, no π^0	145	23%
NC and ν_e CC	89.5	14%
ν_μ CC with a π^0	58.0	9%
ν_μ CC $1\pi^\pm$ outside the measured phase space	52.7	8%
Out of fiducial volume	47.4	7%
EXT (cosmic)	30	n/a
Dirt	1.3	n/a

The dominant background (38%) is CC 0π interactions in which a proton is reconstructed as the pion candidate. This mis-identification was verified directly by 2D event displays (truth-labelled by backtracked PDG): the pion candidate track shows a green (proton) truth colour with a stopping Bragg peak, while the genuine muon track is correctly identified.

At the current working point (Bragg-pion score ≥ 0.08), the pion candidate is truth-matched to a proton in $\approx 18\%$ of cases. Tighter Bragg-score thresholds (0.20–0.30) were studied but shown not to improve the figure of merit: the pion-to-proton ratio in CC $\pi^\pm$ signal is $\approx 4 : 1$, so rejecting protons removes genuine pions at nearly the same rate.

9.5 Event displays

Figure 10 shows representative 2D truth-labelled event displays for the two dominant backgrounds and the EXT cosmic sample. Tracks are coloured by backtracked truth PDG: μ (blue), $\pi^\pm$ (red), p (green), cosmic/unmatched (grey). The muon candidate is drawn solid-thick; the pion candidate dashed-thick; the reconstructed ν vertex is shown as a star.



(a) CC 0π background: a proton (green, dashed) is reconstructed as the pion candidate. (b) CC $N\pi$ background: two genuine charged pions present; one is selected as the pion candidate. (c) EXT cosmic event: all tracks are grey (no truth match), faking a neutrino vertex topology.

Figure 10: Truth-labelled 2D event displays (ZX projection) for representative background events at the final cut. Three views (ZX top, ZY side, XY face) are generated per event.

10 Unfolding

10.1 The extraction chain, stage by stage

Before the formalism, it is useful to see the measurement assembled in the order it is actually performed. Each stage is shown for all three beam configurations in the flat reco-bin space in which the unfolding operates (every analysis slice concatenated).

Stage 1, selected events, POT-scaled (Fig. 11). The starting point is the selected reconstructed spectrum, scaled to the full exposure of each configuration with the per-run POT factors of §5. The Monte Carlo is split into signal and background. At the final cut the purity is $\simeq 59\%$ (Table 10), so some 40% of the selected events have to be removed before anything can be interpreted as a signal rate. Because the analysis is blind, the points are the Poisson-thrown fake data rather than beam-on data.

Stage 2, background subtraction (Fig. 12). The estimated background is subtracted bin by bin, leaving the background-subtracted data alongside the central-value MC signal. The neutrino background is taken from the simulation, the cosmic component from beam-off (EXT) data, and the dirt component from the dedicated overlay; the four control regions of §11.6 are what validate this step. This subtracted spectrum, not the raw selection, is the input to the unfolding.

Stage 3, unfolding. The subtracted spectrum still folds in the detector response and the selection efficiency, encoded in the response matrix of §10.3. With 50–99% bin migration a direct inversion is unusable (§10.4), so the measurement is regularised with Wiener–SVD. The price is the additional-smearing matrix A_C : what is recovered is $A_C \times (\text{true signal})$, and *every* model must therefore be multiplied by A_C before being compared with the data.

Stage 4, comparison with models. The unfolded result is then compared with the Micro-BooNE tune and the four external generators, all A_C -smeared into the same space (§11). It is worth separating two effects that are easily conflated. A_C suppresses the integral of each prediction, but it does so by a *common* factor: for FHC p_μ it is 0.78, 0.78, 0.76 and 0.79 for GENIE, GiBUU, NEUT and NuWro respectively (smeared integral over truth-level total, from the released curves). The truth-level totals of those same generators span 0.82 to $1.32 \times 10^{-38} \text{ cm}^2/\text{Ar}$ (Table 16 of the technical supplement), a factor 1.6. A common $\simeq 0.78$ factor cannot produce a $1.6\times$ spread, so the differences between the generators are genuine model differences and not an artefact of the regularisation. The one-bin total of §11.4 provides the independent check, since with a single bin there is no regularisation and $A_C \equiv 1$.

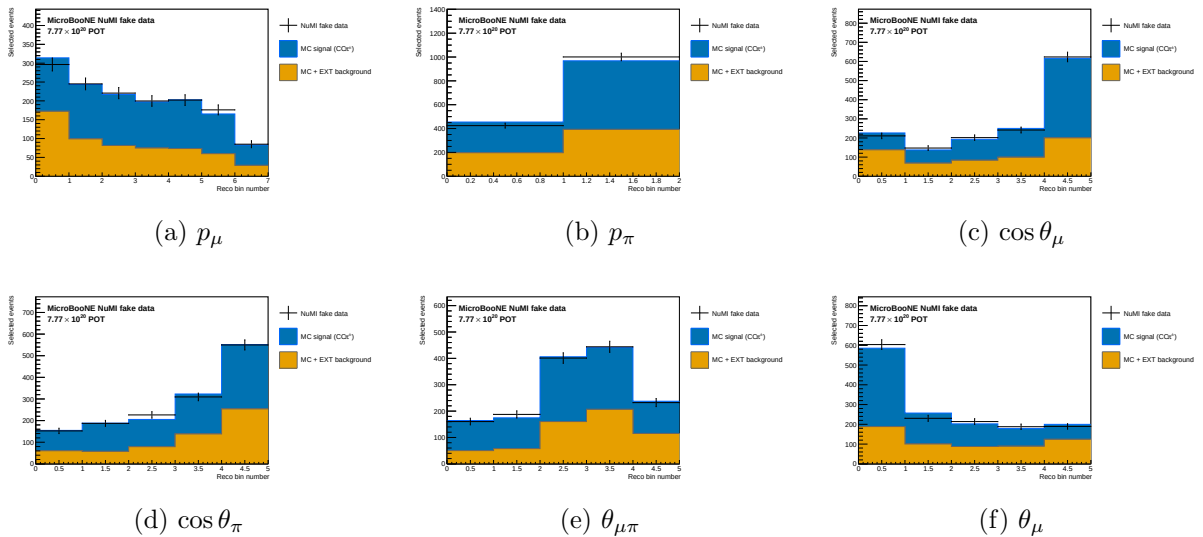


Figure 11: Stage 1, FHC: selected reconstructed spectra scaled to the full exposure, with the MC split into signal and background. Points are the Poisson-thrown fake data. The RHC and combined equivalents are Figs. 13 (the combined configuration is in Fig. 10 of the technical supplement).

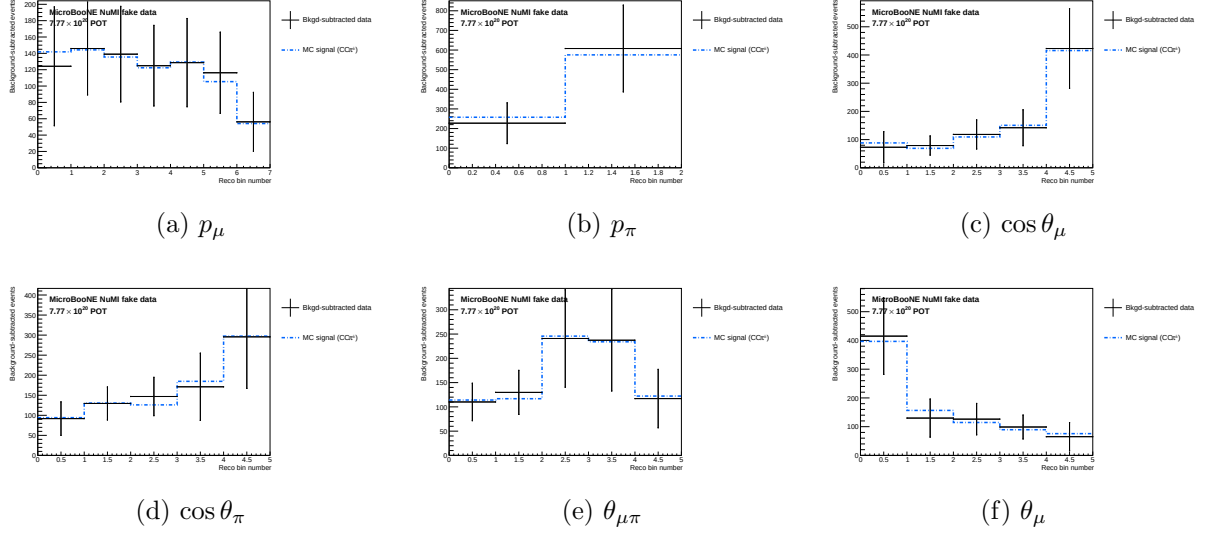


Figure 12: Stage 2, FHC: background-subtracted data compared with the central-value MC signal; the input to the unfolding. RHC and combined: Figs. 14 (combined: Fig. 11 of the technical supplement).

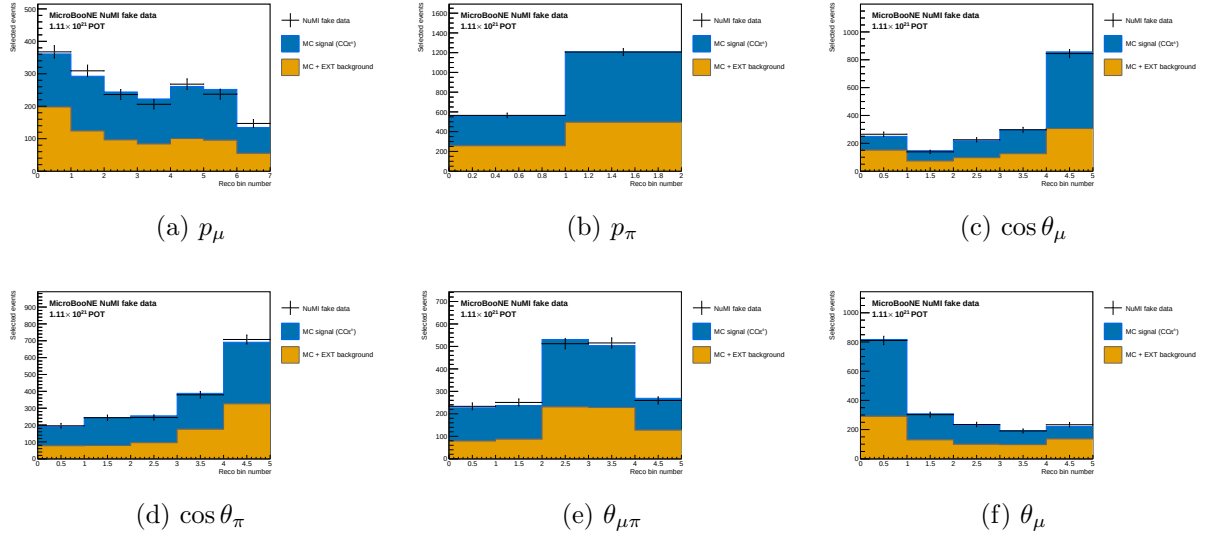


Figure 13: Stage 1, RHC: selected reconstructed spectra scaled to the full RHC exposure, MC split into signal and background.

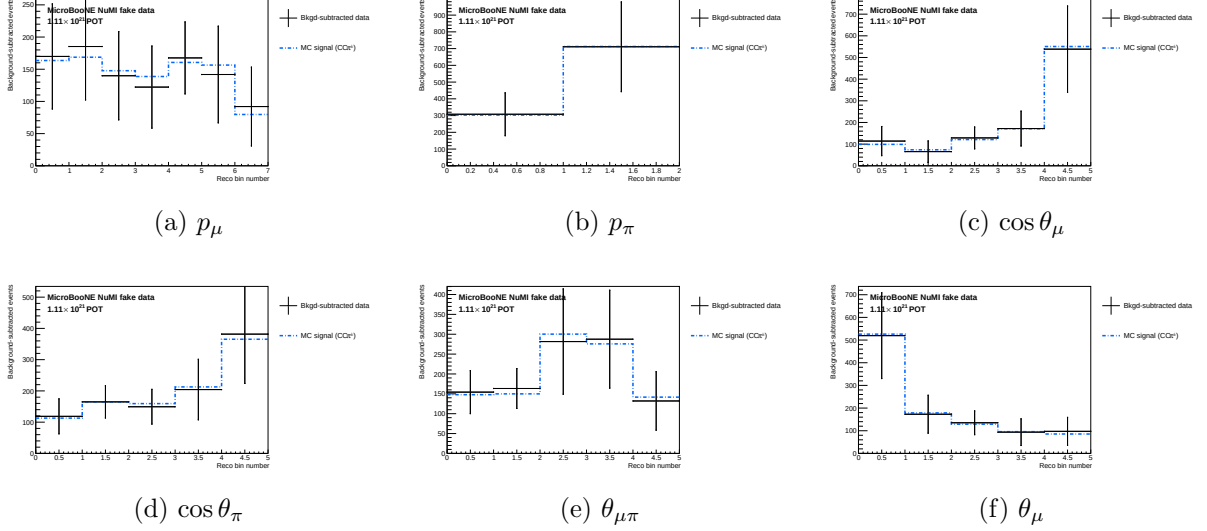


Figure 14: Stage 2, RHC: background-subtracted data compared with the central-value MC signal.

10.2 Signal extraction

After the event selection, the background-subtracted reco-space signal spectrum for observable x is:

$$d_i^{\text{sig}} = d_i^{\text{data}} - b_i^{\nu\text{-bkg}} - b_i^{\text{EXT}} - b_i^{\text{dirt}}, \quad (9)$$

where i runs over reconstructed bins.

10.3 Response matrix

Two distinct matrices are in play and it is worth separating them, since they differ by exactly the efficiency. The *migration* matrix

$$M_{ij} = P(\text{reco in bin } i \mid \text{true in bin } j, \text{ selected}), \quad \sum_i M_{ij} = 1, \quad (10)$$

describes only where selected events move between bins. The *smearceptance* matrix additionally carries the probability of being selected at all,

$$S_{ij} = \varepsilon_j M_{ij}, \quad \sum_i S_{ij} = \varepsilon_j, \quad (11)$$

where ε_j is the ratio of selected to generated signal events in true bin j .

It is S , not M , that enters the unfolding: the unfolder receives the smearceptance matrix, and the efficiency of Fig. 19 is recovered as its column sums. S is built from selected signal MC using the same binning on both axes. The folding relation the unfolder inverts is therefore

$$n_i^{\text{reco}} = \sum_j S_{ij} N_j^{\text{true}} + b_i, \quad S_{ij} = \varepsilon_j M_{ij}, \quad (12)$$

and the unfolded $\hat{N}_j^{\text{true}}$ is already efficiency-corrected: no separate division by ε_j is applied afterwards (§13). Figure 18 plots S itself, exactly the matrix passed to the unfolder (colour scale $P(\text{reco} \mid \text{true}) \times \varepsilon$, so each column sums to the efficiency of that true bin rather than to unity). The migration diagonals quoted in this note, for example 91% and 76% for the two p_π bins, are elements of the column-normalised M , obtained from S by dividing each column by its sum; they are not read off the colour scale of Fig. 18.

10.4 Regularised unfolding: the bias–variance trade-off

Unfolding inverts the folding relation $\vec{m} = A \vec{s} + \vec{b}$ to estimate the true signal spectrum $\vec{s}$ from the measured $\vec{m}$. The naive inverse $A^{-1}(\vec{m} - \vec{b})$ is unbiased but useless: bin migration makes A nearly singular, so its inverse amplifies statistical fluctuations into large, anticorrelated bin-to-bin oscillations. Every practical method therefore *regularises*, trading a small, controlled bias for a large reduction in variance. This analysis uses two complementary methods.

Wiener-SVD (primary). Diagonalising the response with an SVD, each mode k is weighted by the Wiener filter $W_k = \sigma_k^2 / (\sigma_k^2 + N_k)$; the minimum-mean-squared-error weighting, $\simeq 1$ for well-measured modes and $\rightarrow 0$ for noise-dominated ones, so no regularisation strength is tuned by hand. The price is a *known* linear bias: the unfolded data estimates not $\vec{s}$ but the smeared spectrum $A_C \vec{s}$, where the additional-smearing matrix A_C is published so that every theory prediction is smeared by the same A_C before comparison.

D’Agostini iterative Bayesian [27] (cross-check). Bayes’ theorem inverts the migration, $P(C_j|E_i) = R_{ij}\pi_j / \sum_l R_{il}\pi_l$, and the unfolded estimate is $\hat{s}_j = \varepsilon_j^{-1} \sum_i P(C_j|E_i)(m_i - b_i)$. The MC prior π_j is updated with the result and iterated; the *number of iterations* is the regularisation knob (few $\Rightarrow$ prior-biased, many $\Rightarrow$ noisy). It carries no A_C and theory is compared raw. It approximately preserves the total normalisation in the present nominal-model closure; its residual bias is controlled by the iteration count and is biased toward the prior, and can only be evaluated with non-nominal fake data. Its uncertainty here is the data covariance propagated through the iteration at fixed prior.

Table 12: Trade-offs of the two unfolding methods. Neither is “more correct”; they make complementary choices about how to trade bias for variance and how to expose the residual bias.

	Wiener-SVD	D’Agostini (iter. Bayes)
Regularisation	Wiener filter (automatic, from S/N)	Iteration count n (chosen)
Residual bias	Explicit A_C , published	Implicit; set by early stopping
Result integral	Suppressed, observable-dependent	Approximately preserved in nominal closure; prior-biased
Theory comparison	Smear theory by A_C	Raw theory
Positivity	Not guaranteed	Guaranteed for positive input
Uncertainties	Closed-form, linear	Correlated across iters.; needs care

Wiener-SVD is the primary result, for its objective regularisation and its explicit, reportable bias A_C ; D’Agostini serves as the quantitative cross-check (Sec. 5.4 of the technical supplement), where its flat, iteration-stable integral, 23–31% above Wiener-SVD (config-dependent), is consistent with the observable-to-observable spread of the Wiener-SVD integrated cross sections (Table 20) being the A_C smearing rather than a physical or acceptance effect; the row sums of §11.3 establish that directly. That is all it can establish here: with central-value pseudo-data and the same central value as its prior, D’Agostini returns its prior, so its flatness and iteration stability are guaranteed and say nothing about its bias on a different truth. The alternative-generator fake-data pass of §15 would make it a real test.

10.5 Wiener-SVD unfolding

The primary method is Wiener-SVD [7]. Every released result is produced by the Wiener-SVD unfolders of the XSecAnalyzer framework [18, 17]: Cholesky whitening of the data covariance, regularisation inside the SVD of RC^{-1} , and the Wiener filter of Eq. 3.24 of Ref. [7], whose strength is set by the data covariance rather than by hand. The regularisation matrix C is the second-derivative operator on the *density*, corrected for the non-uniform bin widths, with one-sided first-derivative rows at the ends. The width correction matters: on these binnings, whose

widths vary by up to a factor of 14 ($\cos\theta_\mu$; 10 for θ_μ , 6 for p_μ), a uniform-grid stencil penalises the density’s curvature by the wrong weights and drives A_C towards rank one, so that every model is smeared onto one shape and the measurement carries no shape information (Sec. 7.1 of the technical supplement). A standalone replica of the unfold exists for cross-pipeline checks; it imports the framework’s covariance because it throws no universes, and with a statistics-only covariance it under-regularises and over-corrects the integral by $\sim 26\%$, which is why it is not used for any released number.

The differential cross section is:

$$\left. \frac{d\sigma}{dx} \right|_j = \frac{\hat{N}_j^{\text{true}}}{\Delta x_j \cdot \Phi_{\text{total}} \cdot N_{\text{Ar}}}, \quad (13)$$

where $\hat{N}_j^{\text{true}}$ is the Wiener-SVD estimate of the true signal count in true bin j obtained by inverting Eq. (12), $\Phi_{\text{total}} = \Phi_{\text{per POT}} \times T_{\text{POT}}$ is the total integrated flux, and N_{Ar} is the number of argon target nuclei in the FV. Because the response supplied to the unfold is the smearceptance matrix $S = \varepsilon M$, the efficiency correction is part of the inversion and *no explicit* $1/\varepsilon_j$ appears here (one would apply the correction twice). The implementation forms each bin as $\hat{N}_j^{\text{true}}/(\text{conv} \cdot \Delta x_j)$ with the events-to-cross-section conversion $\text{conv} = N_{\text{Ar}} \Phi_{\text{total}}/10^{-38}$ and applies no further efficiency factor. The per-configuration inputs at the final exposures are:

Quantity	FHC	RHC	Combined
N_{Ar} (FV, dead-region excluded)		8.710×10^{29}	
Flux constant ϕ ($E_\nu > 60$ MeV; 10^{-10} cm $^{-2}$ /POT)	6.81159	6.44646	6.59691
Data POT (10^{20})	7.766	11.082	18.848
Integrated flux $\Phi_{\text{total}} = \phi \times \text{POT}$ (10^{11} cm $^{-2}$)	5.290	7.144	12.434
$\text{conv} = N_{\text{Ar}} \Phi_{\text{total}}/10^{-38}$ (events per 10^{-38} cm 2 /Ar)	4608	6223	10830

An integrated cross section is computed as the width-weighted sum,

$$\sigma_{\text{int}} = \sum_j \left. \frac{d\sigma}{dx} \right|_j \Delta x_j. \quad (14)$$

It is *not* required to agree between observables: the Wiener-SVD result estimates $A_C x_{\text{true}}$ and A_C is not norm preserving, so σ_{int} is observable-dependent (§11.3). The observable-independent total is the one-bin total of §11.4.

10.6 Reconstructed spectra and response matrices

Figure 15 shows the background-subtracted reco-space signal spectra for all six observables, overlaid with the MC signal prediction and the individual background components.

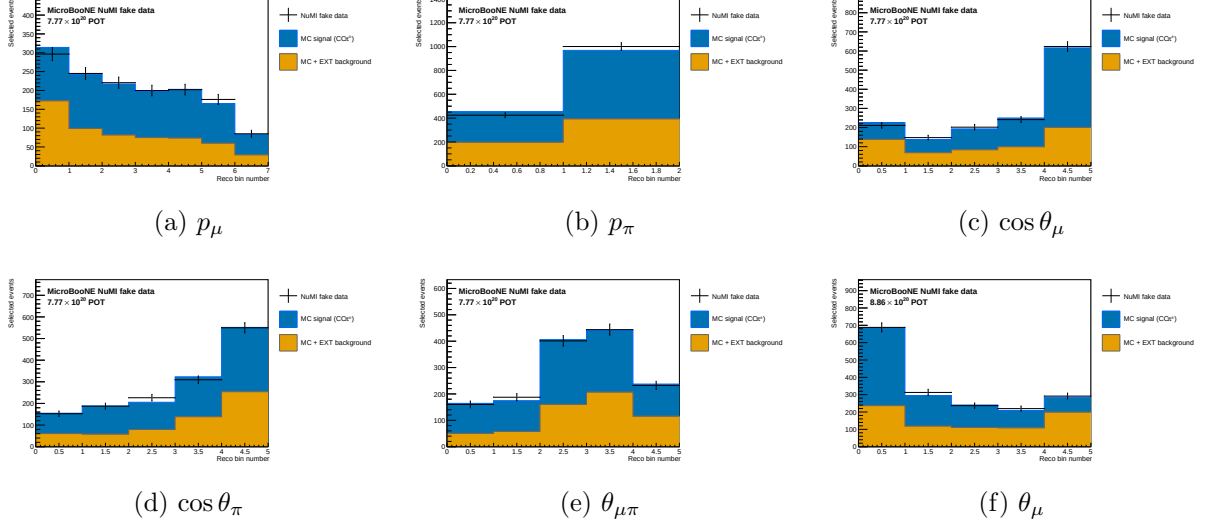


Figure 15: Reconstructed-space spectra at the final cut for all six observables, FHC full exposure (Runs 1+2+4+5): Poisson-thrown central-value fake data (black points), stacked signal (blue), beam background (orange), dirt (green), and EXT (grey). The fake data is drawn from the central-value Monte Carlo and so agrees with the stacked prediction by construction (a reco-level self-closure); the differential results and their closure against the A_C -smeared truth are shown in Fig. 21.

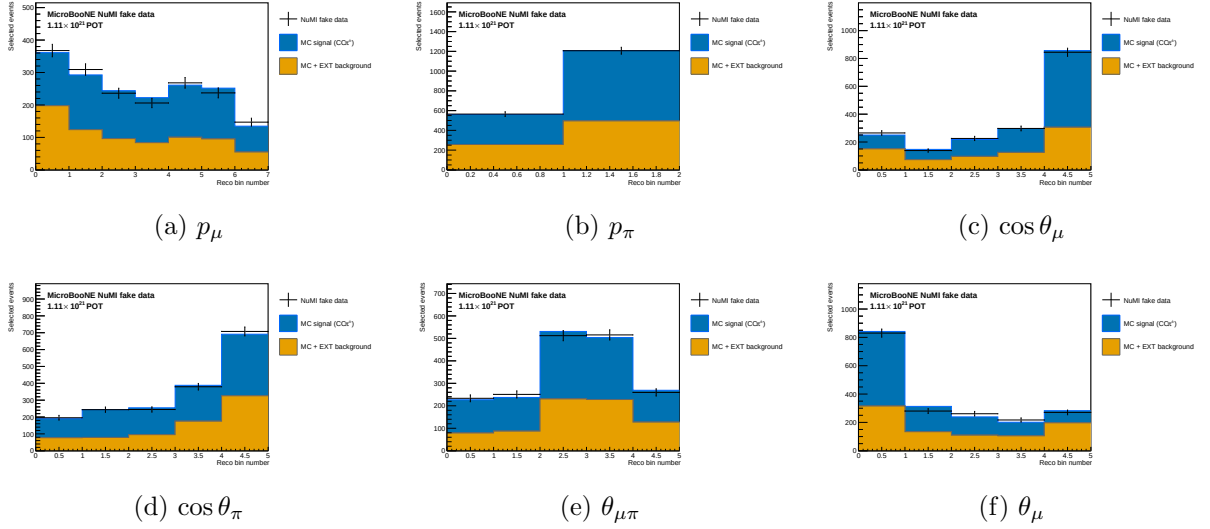


Figure 16: Reconstructed-space spectra, RHC full exposure (Runs 1+2+3+4a+4b+4c); same components as Fig. 15.

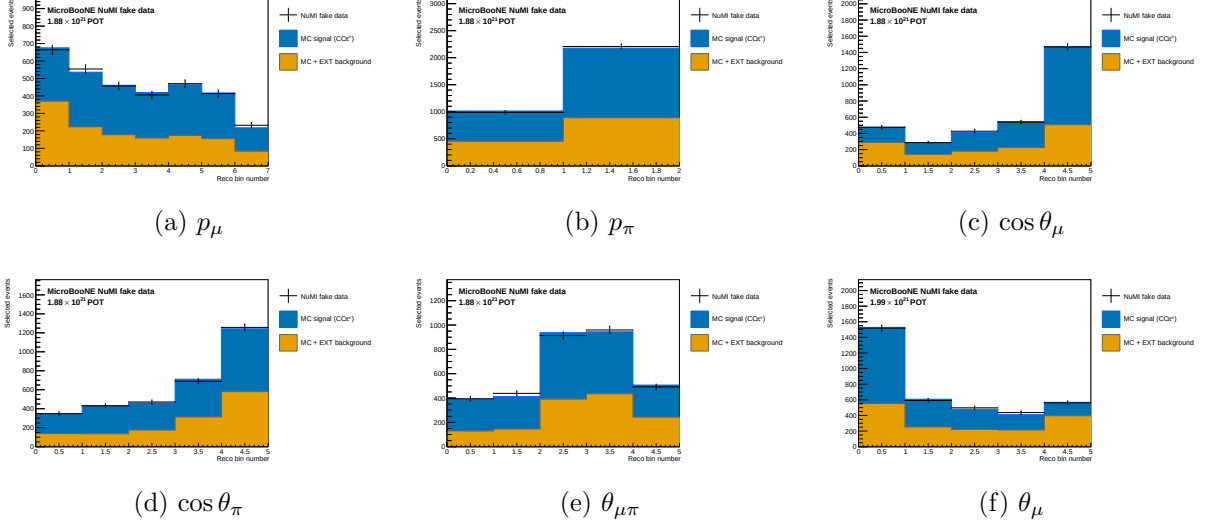


Figure 17: Reconstructed-space spectra, combined FHC+RHC (per-mode normalised); same components as Fig. 15.

Figure 18 shows the smearceptance matrices S (§10.3; colour scale includes the efficiency, so the columns sum to ε_j , not to unity) for the six observables at the released binning. The column-normalised diagonals of the released FHC responses are 0.86–0.95 for the best-resolved bin of each angular observable (the forward bin of $\cos \theta_\mu$, the first bin of the others), 0.67–0.88 elsewhere in $\cos \theta_\mu$, θ_μ , $\cos \theta_\pi$ and $\theta_{\mu\pi}$, and 0.81, 0.65, 0.57, 0.50, 0.45, 0.38, 0.33 for the seven p_μ bins: p_μ is *not* near-diagonal, because its bins (0.2–0.5 GeV/ c wide) are narrower than the 30% resolution of the MCS estimator, which reconstructs 63% of the selected sample and is intrinsically less precise than range. The 0.68 diagonal criterion quoted for p_π below is therefore not a criterion this analysis applies uniformly: for p_μ the Wiener-SVD regularisation absorbs the migration (the released A_C row sums stay at 0.66–0.91, §11.3) and the closure and ensemble tests pass, whereas for p_π the migration is accompanied by a physical saturation of the estimator, which no regularisation recovers. A finer p_μ binning is excluded by the same resolution (Sec. 4 of the technical supplement).

p_π . At any fine binning the p_π response is strongly non-diagonal, because the range estimator saturates near 0.34 GeV/ c once pions interact hadronically before stopping; that is a physical limit rather than an algorithmic one, and the fine-bin study is in the technical supplement (Sec. 2.1 of the technical supplement). The adopted two-bin response shown here is sufficiently diagonal, with column-normalised diagonals of 91% and 76%, but it carries only threshold-versus-above-threshold information: the upper bin is an inclusive region, not a resolved spectral point (§11.1).

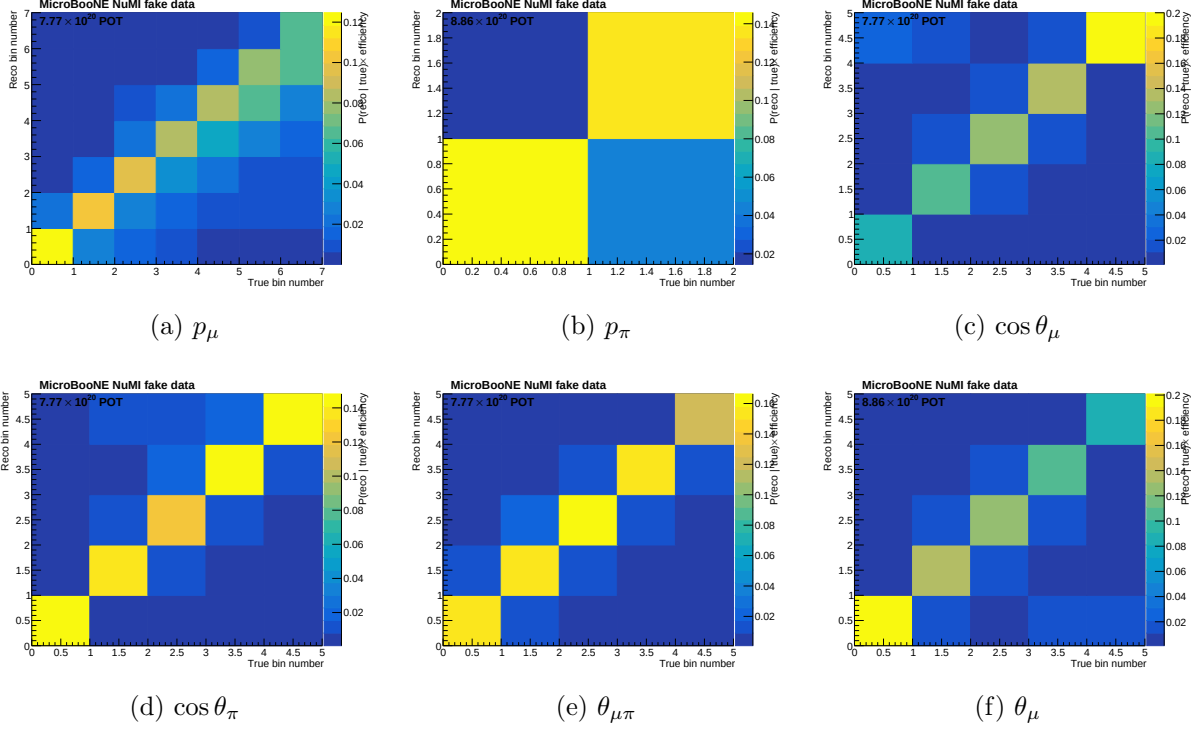


Figure 18: Smearceptance matrices $S_{ij} = \varepsilon_j M_{ij}$, FHC, exactly as passed to the unfoldr. Five panels are at the released binning, including the released two-bin p_π response (91% and 76% column-normalised diagonals); the rejected five-bin p_π response, which shows the reconstructed events piling into the lowest bin from every true bin, is Fig. 12 of the technical supplement.

Figure 19 shows the selection efficiency per true bin for all six observables, computed as the ratio of selected to generated signal MC.

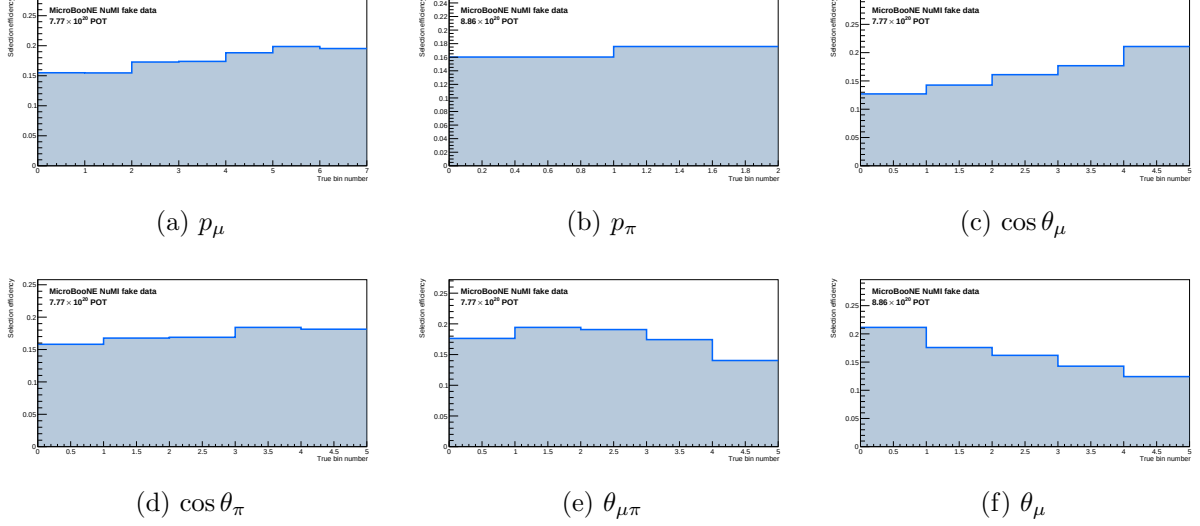


Figure 19: Selection efficiency as a function of true observable value, in the measured phase space ($p_\mu > 0.15$, $p_\pi > 0.175$ GeV/c, $\theta_{\mu\pi} < 2.6$ rad). Efficiency is the ratio of selected to generated signal MC ($\varepsilon_i = \sum_j A_{ij}$, integrated over reco bins). With the momentum thresholds and the 2.6 rad opening-angle cut the efficiency is flat at ~ 13 – 20% across all observables; the former $\sim 5\%$ low-momentum turn-on bins and the cosmic-dominated large-angle region are removed. The lowest point is the smallest opening-angle bin ($\sim 9\%$ efficiency); it is retained because it is high-purity ($\sim 58\%$), not low-purity.

10.7 Reconstruction resolution

Figure 20 shows the reconstruction resolution of each measured observable, evaluated on selected signal events (GENIE ν_μ overlay, measured phase space). For the momenta the relative resolution $(\text{reco} - \text{true})/\text{true}$ is shown; for the angular observables the absolute residual $(\text{reco} - \text{true})$ is used, since the relative metric diverges as $\cos\theta \rightarrow 0$ / $\theta_{\mu\pi} \rightarrow 0$. The red points are the mean residual per true bin; the quoted numbers are the overall mean (bias) and RMS (resolution).

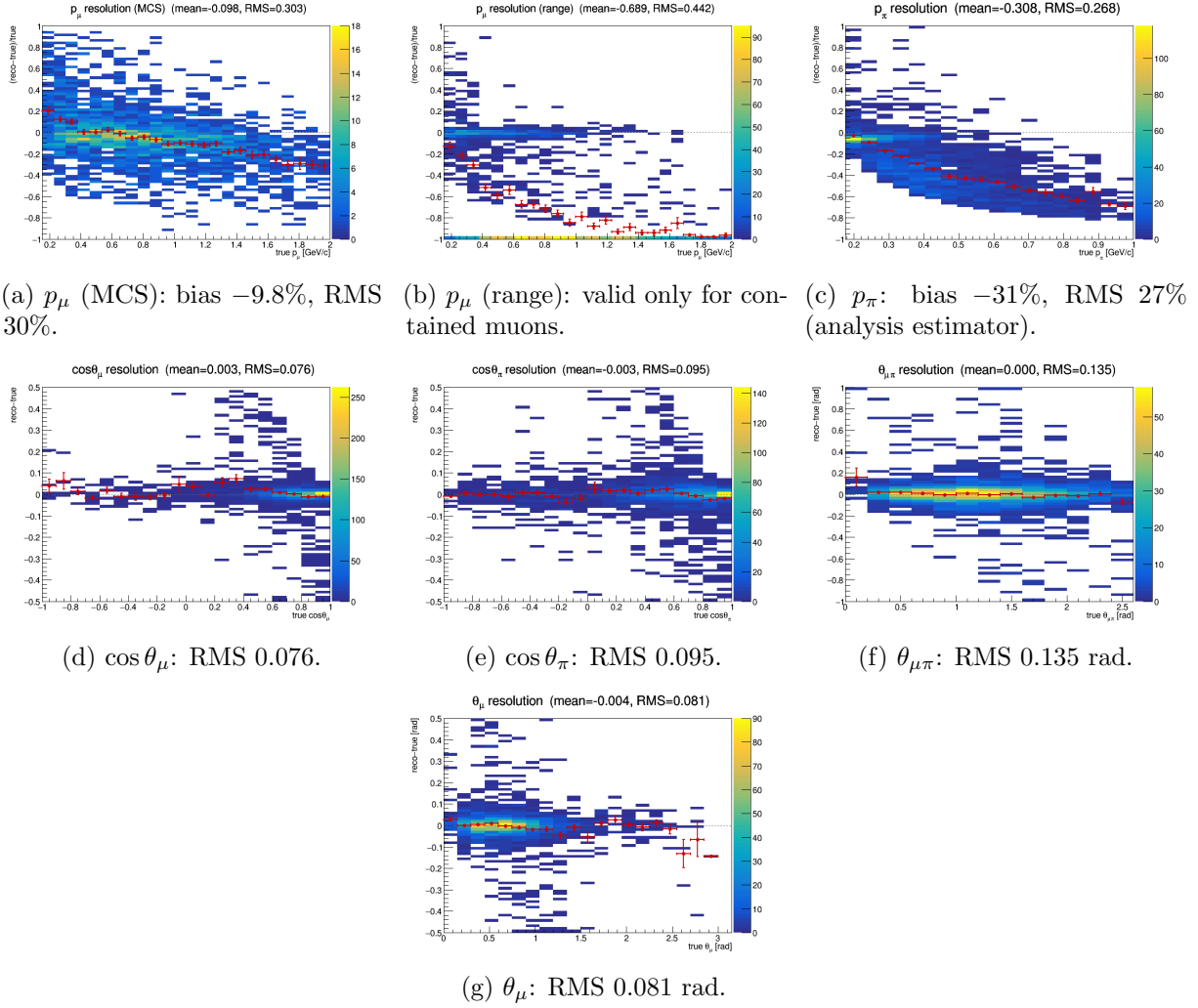


Figure 20: Reconstruction resolution vs. true value, selected signal. Momenta: relative ($\text{reco} - \text{true}$)/ true ; angles: absolute ($\text{reco} - \text{true}$). Red points: mean residual per true bin. θ_μ is derived as $\arccos(\cos \theta_\mu)$, exactly as the bin config does it, so the panel shows the quantity the analysis bins on; it is unbiased (-0.004 rad) with RMS 0.081 rad, the drift below zero above ~ 2.5 rad being a handful of backward-going muons. The muon MCS momentum slightly over-estimates at low p_μ and increasingly under-estimates above ~ 1 GeV/ c ; the range estimator is badly biased for *exiting* muons (range = distance to wall), which is why the analysis uses range only for contained muons (range-or-MCS). The pion panel shows the estimator the analysis actually bins on, not the raw muon-hypothesis range momentum (Sec. 2.1 of the technical supplement): signed mean bias $b = -31\%$, RMS 27% . That average is dominated by high momenta; b is only -2% in the first true bin and reaches -65% above 0.8 GeV/ c , because the reconstructed value saturates near 0.34 GeV/ c once pions interact hadronically before stopping. The angular variables are essentially unbiased with narrow resolution.

10.8 Model comparison: χ^2 against truth, tune and generators

Table 13 collects the bin-by-bin χ^2 of the unfolded fake data against each prediction, for every observable and every beam configuration. All predictions are A_C -smeared, so the comparison is made in the measurement space, and the nominal configured covariance is used. The *truth* column is the closure test: it is small throughout (with the caveat on its p -values in §11.1).

One application of A_C . Every prediction in these tables is smeared by A_C exactly once, by the extractor; the plotting path reads the smeared curves and does not smear again. That is verified externally: the released prescription $\frac{1}{w_i} \sum_j (A_C)_{ij} p_j$ applied to the released truth-level predictions reproduces the released curves to a worst per-bin difference of 1.2×10^{-10} (Sec. 14 of the technical supplement). The point matters because a second application would over-smooth the sharply peaked generators and manufacture an order-of-magnitude χ^2 separation in the muon-angle observables that is not there: NEUT and NuWro in FHC θ_μ give 8.46/5 and 9.10/5, and in $\cos \theta_\mu$ 3.88/5 and 4.04/5 (Table 13), in line with the other observables.

Table 13: χ^2/ndf of the unfolded fake data against each A_C -smeared prediction, for the inclusive selection. “truth” is the fake-data truth (the closure test).

Beam	Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	p_μ	0.47/7	0.71/7	1.71/7	0.81/7	2.00/7	0.68/7
	p_π	1.06/2	1.14/2	1.44/2	3.39/2	1.92/2	1.83/2
	$\cos \theta_\mu$	0.61/5	1.06/5	1.77/5	0.95/5	2.89/5	3.19/5
	$\cos \theta_\pi$	1.21/5	1.21/5	1.92/5	2.56/5	2.26/5	3.09/5
	$\theta_{\mu\pi}$	0.39/5	0.47/5	0.82/5	0.67/5	2.71/5	3.16/5
	θ_μ	1.57/5	2.24/5	4.65/5	4.75/5	7.07/5	7.46/5
RHC	p_μ	0.64/7	2.51/7	3.44/7	2.81/7	3.19/7	3.31/7
	p_π	0.02/2	0.01/2	0.56/2	0.30/2	0.25/2	0.11/2
	$\cos \theta_\mu$	0.35/5	0.37/5	1.11/5	0.24/5	2.71/5	2.67/5
	$\cos \theta_\pi$	0.49/5	0.43/5	1.27/5	2.86/5	1.79/5	1.08/5
	$\theta_{\mu\pi}$	0.19/5	0.34/5	2.52/5	3.54/5	0.46/5	1.08/5
	θ_μ	0.12/5	0.21/5	0.78/5	0.34/5	5.60/5	5.76/5
comb	p_μ	0.34/7	0.61/7	1.71/7	0.62/7	1.22/7	0.92/7
	p_π	0.16/2	0.19/2	0.70/2	1.68/2	0.78/2	0.63/2
	$\cos \theta_\mu$	0.33/5	0.23/5	1.07/5	0.13/5	3.21/5	3.19/5
	$\cos \theta_\pi$	0.58/5	0.52/5	1.61/5	2.74/5	1.68/5	2.10/5
	$\theta_{\mu\pi}$	0.21/5	0.38/5	1.67/5	2.23/5	1.94/5	2.87/5
	θ_μ	0.28/5	0.34/5	1.40/5	2.05/5	7.07/5	7.09/5

Table 14: The same comparison for the muon angle in both parameterisations. The two binnings span the same full range but are different partitions of it: the θ_μ edges (0.43, 0.62, 0.84, 1.17 rad) are not the arccos images of the $\cos \theta_\mu$ edges (0.45, 0.64, 0.86, 1.10 rad). The θ_μ χ^2 are systematically the larger for every column including the truth, a consequence of the different partition and of A_C . Model statements in this note are made on $\cos \theta_\mu$; θ_μ is released as the second parameterisation.

Beam	Variable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	$\cos \theta_\mu$	0.61/5	1.06/5	1.77/5	0.95/5	2.89/5	3.19/5
FHC	θ_μ	1.57/5	2.24/5	4.65/5	4.75/5	7.07/5	7.46/5
RHC	$\cos \theta_\mu$	0.35/5	0.37/5	1.11/5	0.24/5	2.71/5	2.67/5
RHC	θ_μ	0.12/5	0.21/5	0.78/5	0.34/5	5.60/5	5.76/5
Comb	$\cos \theta_\mu$	0.33/5	0.23/5	1.07/5	0.13/5	3.21/5	3.19/5
Comb	θ_μ	0.28/5	0.34/5	1.40/5	2.05/5	7.07/5	7.09/5

11 Full-exposure results: FHC, RHC, and combined

11.1 Scope and status of each result

Table 15 is the inventory of everything this analysis has extracted, generated from the release index so that it cannot drift from what was produced. The fifty-one differential extractions in the release — six inclusive and eleven proton-tagged observables, each in three configurations — carry the systematic covariance of the nominal configured uncertainty model, detector variations included. “Nominal configured” is meant strictly: the identified terms that are *not* in it are listed in §15, and the phrase should not be read as a complete accounting of every uncertainty this measurement has.

Table 15: Every extraction this analysis has produced, generated from the release index (`report/tools/inventory.py`). The extraction count is one per observable and configuration. The withdrawn products are present in the release directory and must not be quoted (Appendix C).

Family	Observables	Extractions	Status
Inclusive differential	6	18	primary
Inclusive one-bin total	1	3	primary
Proton-tagged differential	11	33	secondary
Proton-tagged one-bin total	1	3	secondary
Proton-tagged angular ($\theta_p, \theta_{\pi p}$)	2	6	proposed secondary ¹
Two-dimensional pairs	8	24	two proposed, combined only ¹
<i>Of the above, withdrawn — do not quote</i>			
Differential $W_{\pi p}$	1	3	withdrawn
Five-bin $d\sigma/dp_\pi$	1	—	withdrawn, not in the release

¹ Produced after the release directory was written, so not indexed there.

The proton-tagged detector terms run from 10.5% to 108.5% with a median of 33.0%, against 15.6%–35.0% for the inclusive results; the wide range is the sparse six-bin samples, where the unfolding amplifies them.

The proton-tagged results are secondary scope because the exposure does not support the differential $W_{\pi p}$ shape at any binning, the proton-mistag background lacks a dedicated inverted-proton-PID control region (Sec. 12.6 of the technical supplement), and ensemble coverage has not been demonstrated for this sparser family (§15).

The status of every product — primary, secondary, proposed, validation-only or withdrawn — is given once, in the status box of §2, and the inventory above is generated from the release index. The two are the controlled vocabulary of this note; where a table or caption below says “validation product”, it means a secondary-scope extraction that is complete but not proposed for approval in this review, and “diagnostic” means a distribution shown only to explain why a result is withdrawn.

Table 16: Information content of the primary FHC results. The reported points are the Wiener-SVD estimator $A_C x_{\text{true}}$, not an unsmeared estimate of x_{true} ; an A_C row sum well below one means that bin is set largely by the regularisation. s_{min}/s_1 is the conditioning of A_C ; the uncertainty is the bin-averaged total including data statistics. Migration diagonals are the smallest over bins.

Observable	bins	min. diagonal	A_C row sums	s_{min}/s_1	unc. [%]
p_μ	7	0.33	0.67–0.92	0.006	43.5
p_π (two regions)	2	0.76	0.36–0.54	0.426	53.9
$\cos \theta_\mu$	5	0.68	0.22–0.42	0.093	44.9
θ_μ	5	0.67	0.59–0.80	0.060	32.4
$\cos \theta_\pi$	5	0.74	0.68–0.78	0.430	41.0
$\theta_{\mu\pi}$	5	0.79	0.52–1.04	0.007	37.5

What the reader should know before the tables. Three points condition how the numbers below are read, and are established in Sec. 9 of the technical supplement. *First*, $d\sigma/dp_\pi$ is released in two bins only: charged pions interact before ranging out, so the reconstructed momentum saturates near 0.34 GeV/ c and no finer binning survives the migration criterion; the upper bin is open-ended and its released density represents a bin integral. *Second*, the closure χ^2 quoted with each extraction uses the nominal configured covariance, so its p -values are not expected to be uniform and are not a sensitive test of the extraction — the pseudo-data and the reference share the systematic sources that the covariance includes, so they confirm consistency and nothing more; statistical coverage is evaluated separately with ensembles (§13). *Third*, ensembles of independent Poisson realisations carried through the whole chain — six extractions across FHC, RHC and the combined configuration, 46–100 members each — give per-bin pull widths of 0.90–1.10 and 68% coverage of 61–79%, so the quoted statistical uncertainties are correctly sized in every configuration; the proton-tagged family has not been tested.

This is the primary result of the analysis: the $\text{CC}1\mu1\pi^\pm$ cross section extracted over the full available exposure in forward horn current (FHC), reverse horn current (RHC), and their combination, all on Poisson-thrown central-value fake data. The FHC measurement uses Runs 1, 2, 4 and 5 (data POT 7.766×10^{20}); RHC uses Runs 1, 2, 3, 4a, 4b, 4c (1.108×10^{21}); the combined measurement uses all fourteen overlay files at the joint POT 1.885×10^{21} . All three close cleanly against the A_C -smeared truth for every observable (Tables 17–19); the differential results are shown in Figs. 21–23.

Table 17: FHC (Runs 1, 2, 4, 5; data POT 7.766×10^{20}): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against A_C -smeared truth. All values are read from the released covariance and results table; the anatomy of the flux term is in §14.2.

Observable	σ_{int}	Flux	Detector	Total	Closure χ^2/ndf
p_μ	0.832	28.1%	23.6%	39.9%	0.47/7 ($p = 1.00$)
p_π	0.864	42.3%	41.4%	66.9%	1.06/2 ($p = 0.59$)
$\cos \theta_\mu$	0.746	27.7%	27.1%	41.0%	0.61/5 ($p = 0.99$)
$\cos \theta_\pi$	0.668	27.1%	23.2%	39.0%	1.21/5 ($p = 0.94$)
$\theta_{\mu\pi}$	0.796	28.8%	19.9%	37.1%	0.39/5 ($p = 1.00$)
θ_μ	0.993	23.8%	14.9%	29.7%	1.57/5 ($p = 0.91$)

We extend the measurement to the reverse-horn-current beam as both a standalone measurement and a combined $\nu_\mu + \bar{\nu}_\mu$ result. The RHC *flux* is $\bar{\nu}_\mu$ -dominant (54.6% $\bar{\nu}_\mu$, versus FHC’s 35%), with > 60 MeV integrals $\Phi_{\nu_\mu}^{\text{RHC}} = 2.923 \times 10^{-10}$, $\Phi_{\bar{\nu}_\mu}^{\text{RHC}} = 3.523 \times 10^{-10}$, total $6.446 \times 10^{-10} \text{ cm}^{-2}\text{POT}^{-1}$ (the FHC-convention computation reproduces the authoritative FHC values exactly). The *measured event rate*, however, remains ν_μ -dominant (76% ν_μ , 20% $\bar{\nu}_\mu$)

because the ν_μ cross section is $\sim 2\text{--}3\times$ larger than the $\bar{\nu}_\mu$ one. The RHC overlays (Runs 1, 2, 3, 4a, 4b, 4c) are all verified as genuine ν_μ overlays (flavour composition 100% $\nu_\mu + \bar{\nu}_\mu$ at truth level, events per POT uniform across files) and processed, 4.05×10^6 events, combined MC POT 1.546×10^{22} , mutually consistent (2.62×10^{-16} events/POT) with the PPFX weights, covering the full RHC exposure (Run 3 is the largest period, 5.52×10^{21} MC POT and 1.44×10^6 events). Native Run-4 RHC detector-variation samples (the standard eight-knob set: light-yield down/Rayleigh, recombination, space charge, and the four wire-modification variations) provide the RHC detector systematic, verified as genuine ν_μ overlays with a uniform 3.80×10^{-16} events/POT across all knobs. The RHC dirt prediction uses the FHC dirt overlay, scaled to the RHC exposure, as no RHC dirt sample exists. The standalone result below uses Poisson-thrown central-value fake data at the RHC data POT 1.108×10^{21} . The generator predictions are re-averaged to the RHC flux composition; the combined FHC+RHC predictions are the fluence-weighted combination of the two horn modes (0.458 FHC / 0.542 RHC, each mode weighted by $\Phi_{\text{tot}} \times \text{POT}$).

The combined FHC+RHC measurement uses the summed 14-file response over the full exposure (5.6×10^6 events, combined data POT $D_{\text{comb}} = 1.994 \times 10^{21} = D_{\text{FHC}} + D_{\text{RHC}}$), with each horn mode normalised to its *own* data exposure. Each mode’s simulation and detector-variation samples are scaled to that mode’s own data exposure, so the combined prediction weights the modes by data exposure (0.80:1) rather than by simulated exposure (0.62:1), and the fake data is thrown at the matching rate.

The joint *flux systematic* needs no hand-built cross-horn correlation. All fourteen numuMC files carry the *same* 600 PPFX multisim universes — a universe being one alternative prediction with a systematic parameter varied, a multisim a set of hundreds of them drawn from the parameter uncertainties — which the universe maker accumulates index-matched across both horn modes in a single chain, so universe u is the same hadron-production draw in FHC and RHC. Per-universe summing therefore propagates one parameter draw through both beam configurations and lets each mode’s flux response set the result, capturing the physical (not unit) cross-horn correlation automatically. The detector variations are carried per mode, the Run-4 FHC and Run-4 RHC samples of each knob accumulated under one label, so the combined detector term is exposure-weighted like the central value. The combined closure passes for every observable (Table 19), with an unfolded flux systematic of 24–30% of the same size as the standalone FHC (25–33%) and RHC (25–28%) values (Tables 17–19), as expected for a common flux model propagated through three similarly conditioned responses.

The standalone RHC extraction (native Run-4 RHC detector variations) closes cleanly for every observable (Table 18): the realised A_C -smeared truth is recovered with $\chi^2 \leq 0.7$ on two to seven degrees of freedom and $p \geq 0.98$ throughout. The reconstructed-level flux uncertainty is $\sim 15\%$, below the FHC $\sim 17\%$, because the $\bar{\nu}_\mu$ -dominant flux carries a smaller PPFX hadron-production uncertainty; the unfolded flux term (25–28%, Table 18) is nevertheless comparable to the FHC one, since the background subtraction multiplies both by the same $1 + B/S$ (§14.2).

Table 18: RHC (Runs 1, 2, 3, 4; data POT 1.1082×10^{21}): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against A_C -smeared truth. All values are read from the released covariance and results table; the anatomy of the flux term is in §14.2.

Observable	σ_{int}	Flux	Detector	Total	Closure χ^2/ndf
p_μ	0.690	27.6%	21.5%	37.7%	0.64/7 ($p = 1.00$)
p_π	0.748	28.4%	34.7%	48.6%	0.01/2 ($p = 0.99$)
$\cos \theta_\mu$	0.733	25.8%	35.0%	46.3%	0.35/5 ($p = 1.00$)
$\cos \theta_\pi$	0.615	25.2%	25.5%	38.5%	0.49/5 ($p = 0.99$)
$\theta_{\mu\pi}$	0.831	25.2%	16.6%	32.6%	0.18/5 ($p = 1.00$)
θ_μ	0.805	25.6%	22.4%	36.1%	0.12/5 ($p = 1.00$)

Table 19: Combined (data POT 1.8848×10^{21}): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against A_C -smeared truth. All values are read from the released covariance and results table; the anatomy of the flux term is in §14.2.

Observable	σ_{int}	Flux	Detector	Total	Closure χ^2/ndf
p_μ	0.821	24.5%	21.9%	35.4%	0.34/7 ($p = 1.00$)
p_π	0.791	32.4%	33.8%	52.0%	0.16/2 ($p = 0.93$)
$\cos \theta_\mu$	0.794	25.9%	24.3%	37.8%	0.33/5 ($p = 1.00$)
$\cos \theta_\pi$	0.649	25.7%	23.0%	37.5%	0.58/5 ($p = 0.99$)
$\theta_{\mu\pi}$	0.860	25.8%	18.0%	33.6%	0.21/5 ($p = 1.00$)
θ_μ	0.812	26.2%	21.4%	35.7%	0.28/5 ($p = 1.00$)

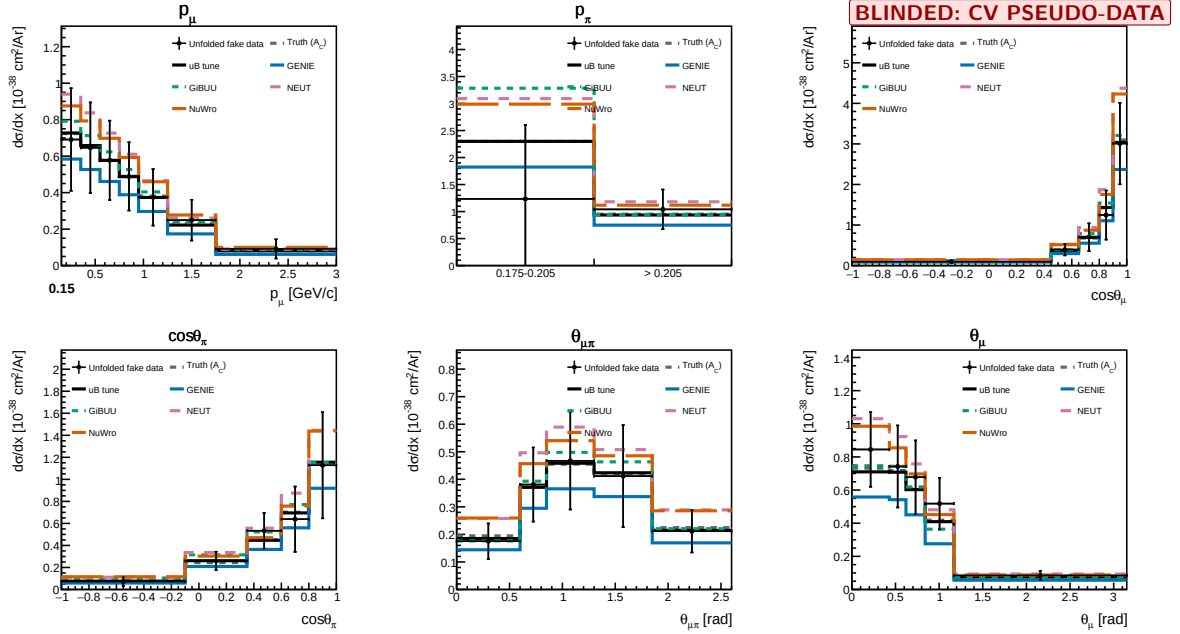


Figure 21: Differential cross sections $d\sigma/dx$ for the six released distributions on the current Poisson-thrown central-value fake data, FHC full exposure: the points are the Wiener-SVD regularised estimator $A_C x_{\text{true}}$, not an unsmeared estimate of x_{true} ; unfolded fake data (black points), A_C -smeared fake-data truth (grey dashed), the MicroBooNE tune (black solid), and the four generator predictions (GENIE, GiBUU, NEUT, NuWro). The unfolded points recover the injected truth for every observable, the same closure summarised in Table 17.

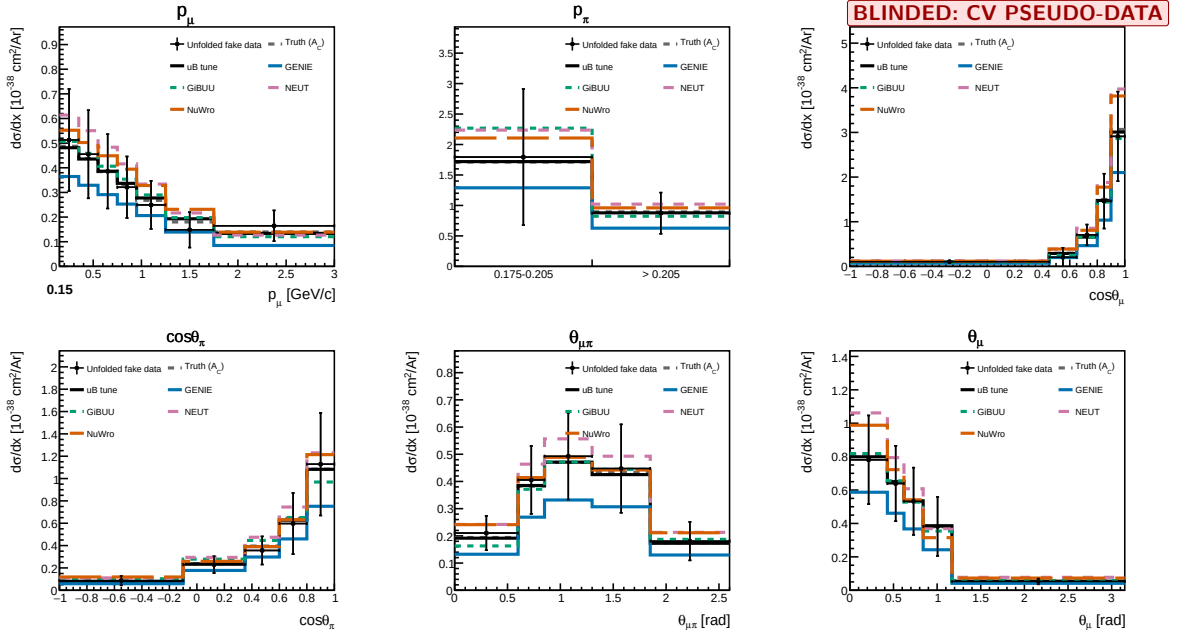


Figure 22: Standalone RHC ($\bar{\nu}_\mu$ -dominant) differential cross sections on the current Poisson-thrown fake data, with the RHC-flux-averaged generator predictions. Same layout as Fig. 21.

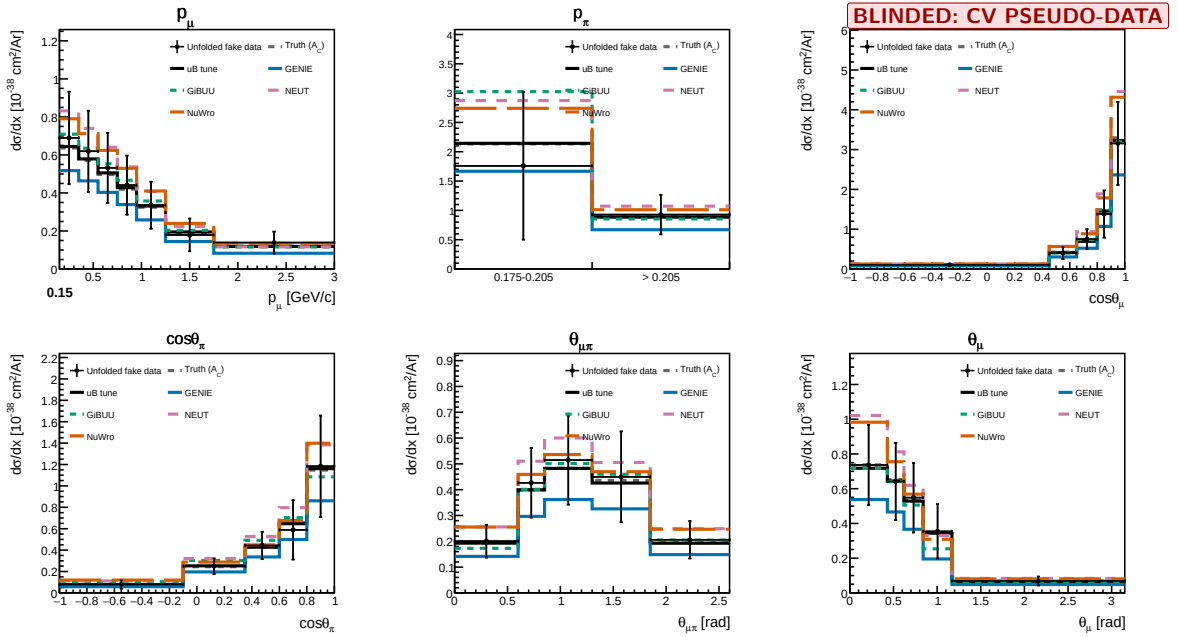


Figure 23: Combined FHC+RHC ($\nu_\mu + \bar{\nu}_\mu$) differential cross sections on the current per-mode-normalised fake data, with the fluence-weighted combined generator predictions.

Table 20: Integrated (unfolded) cross section σ_{int} over the measured phase space, for each observable and configuration (current fake data, five-bin forward-split $\cos\theta_\pi$ and coarsened $\theta_{\mu\pi}$ binning, per-configuration flux; 10^{-38} cm²/Ar). The value differs between observables because the Wiener-SVD additional-smearing A_C is not norm-preserving; the D’Agostini cross-check (Sec. 5.4 of the technical supplement) recovers a flat, 23–31% higher total, consistent with the spread being a smearing effect (the A_C row sums of §11.3 establish it directly). The MC normalisation is evaluated run by run from each run’s exposure. The final row gives the alternative muon-angle parameterisation θ_μ (Sec. 7.1 of the technical supplement), complete for all three configurations; no θ_π parameterisation is extracted. (Values are fake-data closures; the signal is $\nu_\mu + \bar{\nu}_\mu$, so the RHC flux-average is not expected to lie far below FHC.)

Observable	FHC	RHC	Combined
p_μ	0.832	0.690	0.821
p_π	0.864	0.748	0.791
$\cos\theta_\mu$	0.746	0.733	0.794
$\cos\theta_\pi$	0.668	0.615	0.649
$\theta_{\mu\pi}$	0.796	0.831	0.860
θ_μ	0.993	0.805	0.812

Uncertainties shown. Every differential cross section and every integrated value quoted in this note carries *both* statistical and systematic uncertainties. The error bars on the unfolded points are the square root of the diagonal of the *total* covariance matrix, propagated through the unfolding: flux (PPFX multisims), cross-section (GENIE multisims and unisims, a unisim being a single $\pm 1\sigma$ variation of one parameter), detector variations [15], hadron re-interaction [25], POT counting and target number, together with the data, MC and EXT statistical terms. For p_μ in FHC the bin-averaged total is 40.7% (Table 28), dominated by flux (28.0%) and detector (24.7%), with data statistics 14.3% and cross-section 10.7%. The one-bin total cross section of §11.4 carries its own breakdown from the same universes, written there as systematic $\oplus$ data-statistical (FHC: 38.5% $\oplus$ 4.3%); without regularisation its flux and detector terms are not amplified.

11.2 The two-region p_π measurement

Because the upper pion-momentum bin is open-ended, the p_π result is reported as two bin-integrated cross sections rather than as a spectrum. These are the quantities to compare a prediction with; the differential form exists only so that the result can be drawn next to the others, and it carries the reporting width described in Appendix A.

Table 21: The p_π result as bin integrals, 10^{-38} cm²/Ar, on fake data. The upper region is open-ended; no width is assigned to it here and none is needed. Uncertainties are the total of the nominal configured covariance.

Region	FHC	RHC	Combined
$\sigma(0.175 < p_\pi < 0.205 \text{ GeV}/c)$	0.059 ± 0.041	0.054 ± 0.034	0.062 ± 0.038
$\sigma(p_\pi > 0.205 \text{ GeV}/c)$	0.764 ± 0.293	0.694 ± 0.268	0.713 ± 0.270

11.3 How far A_C is from preserving normalisation

Because the Wiener-SVD result estimates $A_C x_{\text{true}}$ rather than x_{true} , and A_C is not norm preserving, summing the differential result over its bins does not give the physical total cross section. This subsection quantifies the effect.

The row sums of A_C are the diagnostic used here: they give the response to a flat bin-integrated truth, for which a row summing to unity passes the normalisation of that bin through

unchanged and a row summing to 0.2 passes one fifth of it. For a non-flat prediction the output of bin i is $\sum_j (A_C)_{ij} t_j$, which depends on the individual elements and on the shape of t ; the row sums describe the dominant behaviour, and the model-specific integral factors of §10.1 (0.76–0.79 for the four generators in FHC p_μ) show how far it carries over.

Observable	FHC	RHC	combined
p_μ	0.66–0.91	0.58–0.99	0.68–1.01
p_π (2-bin)	0.35–0.54	0.40–0.51	0.29–0.54
$\cos \theta_\mu$	0.17–0.39	0.05–0.45	0.09–0.43
$\cos \theta_\pi$	0.68–0.77	0.70–0.84	0.71–0.83
$\theta_{\mu\pi}$	0.52–1.03	0.58–1.15	0.63–1.21
θ_μ	0.58–0.78	0.22–1.16	0.53–0.67

The suppression is large (inclusive $\cos \theta_\mu$ has a row sum as low as 0.05 in RHC) and differs strongly between observables, which is why the integrated cross section obtained by summing a differential result varies from observable to observable while the physical total cannot; the D’Agostini cross-check, whose integral is flat across observables, confirms that the spread is the smearing (Sec. 5.4 of the technical supplement). The proton-tagged row sums, and the two matrices whose least-determined row sums to a small negative value (RHC $W_{\pi p}$ and the first $\cos \theta_\mu$ bin of the proton-tagged RHC sample), are tabulated and discussed in Sec. 13.5 of the technical supplement; a negative row is not forbidden for a regularisation operator, but those two rows should be read with corresponding caution.

Numerical data release. All fifty-one differential extractions are released in `report/data_release/`, indexed by `index_extractions.tsv`: the additional-smearing matrix, the full per-component covariance in cross-section units, and `curves_*.tsv` holding the unfolded data together with every model curve shown here. The six one-bin totals, with their uncertainty breakdown and realised truth, are in `total_xsec.tsv`.

A prediction supplied as a per-bin cross section is compared with these results by

$$\left(\frac{d\sigma}{dx}\right)_i^{\text{smeared}} = \frac{1}{w_i} \sum_j (A_C)_{ij} p_j,$$

with p_j the predicted cross section in true bin j and w_i the width of bin i . This reproduces the published generator curves exactly — the worst per-bin deviation over the released matrices is 1.2×10^{-10} — so an external user can verify their implementation against any `curves_*.tsv` column before applying it to a new model.

The check is part of the release procedure because it is sensitive to the one error an internal closure cannot see, a prediction smeared twice, which would miss it by up to 55%.

The practical consequence is stated in the results section and repeated here because it is easy to get wrong: quote the physical total from the one-bin extraction, which is not subject to A_C , and treat the sum of a Wiener-SVD differential result as a quantity in smeared space rather than as a cross section.

11.4 Total flux-averaged cross section (one-bin extraction)

The total flux-averaged cross section in the measured phase space is extracted with the same machinery as the differential results, as a *one-bin distribution*: a single true bin containing every signal event, defined on $\cos \theta_\mu$ over its full range $[-1, 1]$ so that no overflow bin exists, and a single reconstructed bin over the same range (the proton-tagged version uses the same construction on the proton-tagged selection). With one bin the response $S = \varepsilon M$ reduces to the efficiency, there

is nothing to regularise, and the Wiener filter is switched off, so the “unfolding” is the plain inversion

$$\sigma = \frac{N_{\text{sel}} - N_{\text{bkg}}}{\epsilon \Phi N_{\text{Ar}}}, \quad (15)$$

with $A_C \equiv 1$ exactly (released with the result in `data_release/total_xsec.tsv`). Here N_{sel} is the selected count, N_{bkg} the predicted background (ν -background + EXT + dirt), $\epsilon = N_{\text{sel}}^{\text{sig}}/N_{\text{gen}}^{\text{sig}}$ the selection efficiency in the measured phase space ($\simeq 17.5\%$ for all configurations), Φ the integrated flux ($\Phi = \text{POT} \times \Phi_{\text{POT}}$, Table 5) and $N_{\text{Ar}} = 8.71 \times 10^{29}$ the number of argon nuclei in the fiducial volume (the dead z region removed), the same normalisation $N_{\text{Ar}} \Phi / 10^{-38}$ used for the differential result. Every term of Eq. (15) is evaluated universe by universe, so the covariance contains the flux, cross-section, re-interaction and detector variations, the POT and target-count terms and the MC, EXT and data statistical terms, propagated exactly as for the differential results. Its flux term, 28–30% inclusive and 34–35% proton-tagged, is the flux normalisation times $(1 + B/S)$ with B the neutrino background (§14.2); the detector term is 20–22% (12–27% proton-tagged) and the data-statistical term 3–4% (5–7%). The differential results carry the same terms bin by bin (flux 24–33%, detector 16–35%, statistics 6–18% bin-averaged); the regularisation redistributes them between bins rather than amplifying their sum. Because no additional smearing is applied the total is the physical one: it lies above the A_C -suppressed σ_{int} of Table 20, most visibly for RHC, and agrees with the D’Agostini cross-check (Sec. 5.4 of the technical supplement).

The closed-form cut-and-count evaluation of Eq. (15) on the central-value prediction is retained as the check of the normalisation chain: it reproduces the one-bin extraction of the central-value prediction to four digits (1.0315, 0.9285, 0.9723 inclusive; 0.6269, 0.5506, 0.5831 proton-tagged) and supplies the N_{sel} , N_{bkg} and ϵ columns of Table 22. The MC counts carry the framework’s full central-value weight, $\text{tune} \times \text{PPFX} \times \text{normalisation}$ (§5); the PPFX central-value weight averages 0.944 on selected signal, so a count that omitted it would sit 3–6% above the prediction the unfolding actually uses.

Table 22 is the headline total-cross-section table, for the inclusive selection and for the proton-tagged subsample. The σ column is the one-bin extraction on the Poisson-thrown fake data, so it fluctuates about the injected tune by the throw; the realised truth of the throw is listed alongside, and the closure ratio is 0.952–0.998 with $\chi^2/1 \ll 1$ in all six cases. The uncertainty column is what the measurement will carry at unblinding; it is the one-bin extraction’s own, not a proxy borrowed from a differential extraction, which would carry that extraction’s regularisation and depend on the throw through its unfolding operator. The RHC/FHC ratio of the injected tune is 0.900 (proton-tagged 0.878), the number to be compared with data once unblinded; on this throw the extracted ratio is 0.901 (0.873). Because the extraction is observable-independent, this is also the only proton-tagged total that should be quoted: the integrals of Table 1 of the proton-tagged note–3 are A_C -suppressed and observable-dependent.

Table 22: Headline total flux-averaged cross sections ($10^{-38} \text{ cm}^2/\text{Ar}$) in the measured phase space, from the one-bin extraction (§11.4) on the Poisson-thrown central-value fake data. N_{sel} and N_{bkg} are the predicted selected count and background (ν -background + EXT + dirt) of the central-value prediction and ε the selection efficiency, from the cut-and-count check; σ is the one-bin extraction with its full universe-propagated uncertainty, “stat” its data-statistical term, “syst” the prediction total (flux, cross section, re-interaction, detector, POT, targets, MC and EXT statistics) and “total” their quadrature sum; “truth” is the realised truth of the throw and $\chi^2/1$ the closure against it. MC counts carry the framework’s central-value weight (tune $\times$ PPF $\times$ normalisation), the same as Table 10; the two differ by $\lesssim 1\%$ because the yield comparison uses the Run-1 dirt overlay with per-mode scale factors rather than the per-run dirt files.

Config	N_{sel}	N_{bkg}	ε [%]	σ	stat [%]	syst [%]	total [%]	truth	σ/truth	$\chi^2/1$
<i>Inclusive CC $1\mu 1\pi X p$</i>										
FHC	1 417	584	17.5	1.034 ± 0.396	4.5	38.0	38.3	1.044	0.991	0.001
RHC	1 759	744	17.6	0.932 ± 0.350	4.1	37.4	37.6	0.941	0.990	0.001
Combined	3 176	1 328	17.5	0.975 ± 0.366	3.0	37.5	37.6	0.985	0.990	0.001
<i>Proton-tagged CC $1\mu 1\pi 1p$ ($p_p > 0.3 \text{ GeV}/c$)</i>										
FHC	637	305	11.5	0.608 ± 0.256	7.8	41.3	42.0	0.639	0.952	0.014
RHC	786	395	11.4	0.531 ± 0.251	7.4	46.6	47.2	0.532	0.998	0.000
Combined	1 423	700	11.5	0.564 ± 0.239	5.4	42.1	42.5	0.578	0.976	0.003

Against the standalone flux-averaged generator totals in the inclusive phase space (GENIE 0.82, GiBUU 1.13, NEUT 1.32, NuWro 1.25 for FHC; Table 16 of the technical supplement for all three configurations) the one-bin total sits $0.5\text{--}0.8\sigma$ above GENIE and within 0.8σ of the other three; the generator spread itself, a factor 1.6, is larger than any of these differences. For the proton-tagged phase space the FHC generator totals are GENIE 0.50, NuWro 0.65, GiBUU 0.74 and NEUT 0.78, bracketing the injected 0.63.

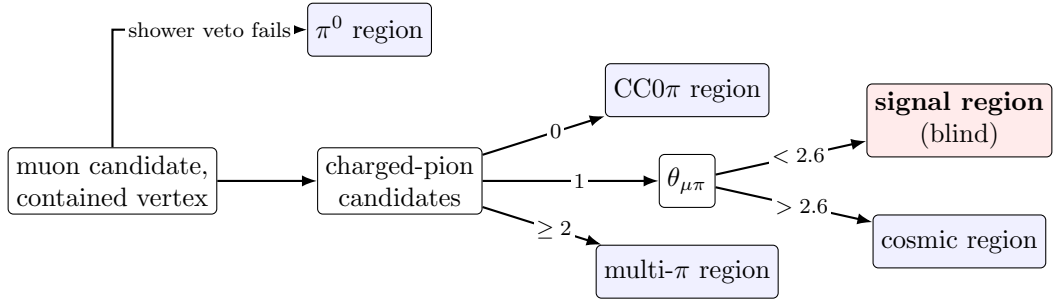


Figure 24: How the four control regions relate to the signal region. Each is one inverted requirement of the signal selection, so no event can be in both: the regions are disjoint from the signal region by construction, which is what allows them to be inspected on beam-on data while the signal region stays blind.

11.5 What the control regions show

The four regions have been opened on beam-on data. They test the background model at reconstruction level, with the signal region untouched, and the procedure they follow — the frozen definitions, the pre-agreed test statistic and the conditions under which a correction may be applied — is given in Sec. 11 of the technical supplement. Three things come out of them.

- **No supported class-specific correction.** A joint FHC+RHC fit of the background class normalisations returns the dominant CC0 π class at 0.949 ± 0.190 over all periods (0.984 ± 0.191 without FHC Run 1), unity within a third of its own uncertainty (Sec. 6 of the control-sample note). The measurement keeps its nominal background prediction. The

fit uses the $\text{CC}0\pi$, π^0 and cosmic-enriched regions only; the multi- π region, which shares 47 FHC and 68 RHC events with the π^0 region, is diagnostic and is not in the fit, so that overlap does not enter the fit covariance.

- **One run’s exposure was wrong; the cause is identified and corrected, and a smaller Run-1 residual remains open.** On the assumed exposure the global test passed in every period except FHC Run 1, where every region sat at 0.60–0.72 and a free exposure factor fitted 0.520 ± 0.020 . Two sets of bookkeeping numbers exist for that run — 2.192×10^{20} POT with 5 748 692 triggers without the open trigger, and 3.283×10^{20} with 9 846 635 with it — and every configuration in this analysis assumes the second. The data say it is the first. The assumption enters the prediction three ways that all scale with it (the neutrino MC through the POT, the beam-off through the gate count, the dirt through the POT), so the four combinations can be told apart: assuming the open-trigger pair gives region ratios of 0.637, 0.715, 0.911 and 0.595, while assuming the non-open pair gives 0.970, 1.090, 1.380 (on 40 events) and 0.948. The two mixed hypotheses leave the beam-off-dominated cosmic region at 0.72–0.75, so both numbers are wrong together rather than one of them.

This does not stay in the control regions. FHC Run 1 was 37% of the assumed FHC exposure, so the correction takes FHC from 8.857 to 7.766×10^{20} POT; on beam-on data it would raise the FHC cross section by 14.0% and the combined one by 5.8% relative to the old bookkeeping, and RHC is untouched. The beam bookkeeping confirms the non-open-trigger pairing, 2.192×10^{20} POT with 5 748 692 triggers; it is adopted in every configuration, FHC Run 1 was re-thrown at it and the FHC and combined extractions were rebuilt on it. Every FHC and combined number in this note is from that rebuild. Because the pseudo-data are thrown at the assumed exposure, the fake-data results do not move by 14%: they change only through the new Run-1 realisation and the run weighting, and the rebuilt integrals are within a few per cent of the previous release. On the corrected exposure the frozen global test passes ($\chi^2 = 8.1/8$, $p = 0.42$). What remains is smaller and unexplained: period by period the data sit 12–25% above the prediction everywhere except in FHC Run 1, which sits at unity, so a Run-1-specific factor of 0.869 ± 0.022 is still preferred ($\Delta\chi^2 = 37$; Table 13 of the control-sample note). The common excess of the other periods is inside the correlated flux and cross-section uncertainties; a 13% offset of one period against the others is not, and its cause (bookkeeping POT, sample completeness, or the Run-1 overlay) is not yet identified. It is listed as an open item for unblinding.

- **The common excess is not signal contamination.** The neutrino-dominated regions sit 12–20% above prediction, inside the correlated flux and cross-section uncertainties. Tightening each region so that it holds less signal — the $\text{CC}0\pi$ region goes from 8.6% to 2.2% signal while keeping 58% of its events — leaves the data-to-prediction ratio where it was, $1.12 \rightarrow 1.13$ (Sec. 7 of the control-sample note), and the cosmic region falls to unity as its beam-off share rises, so the excess sits in the neutrino simulation and does not scale with the signal content of the region.

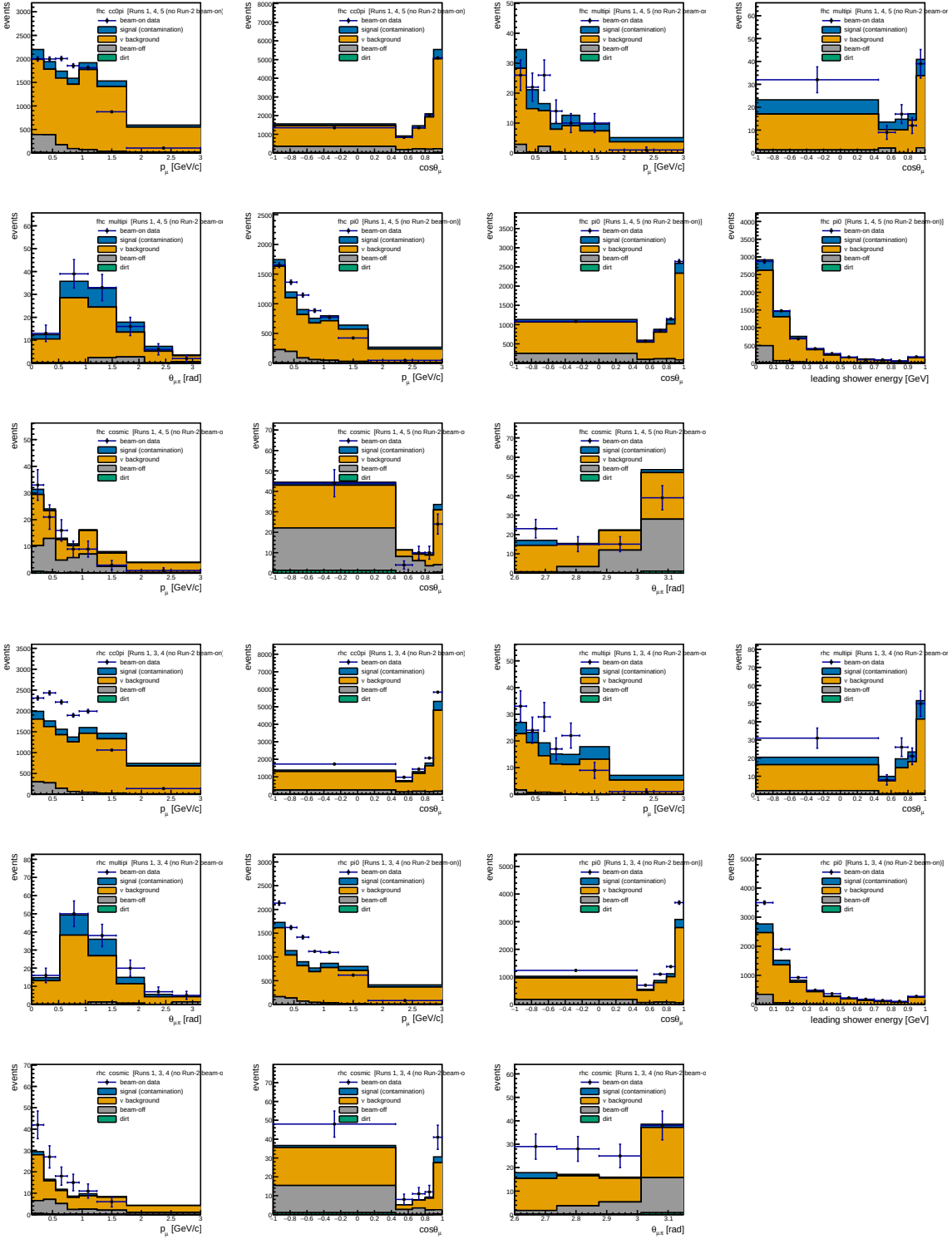


Figure 25: The opened control regions on beam-on data: FHC (top) and RHC (bottom), data against the neutrino-MC + beam-off + dirt prediction in each region. These, not the pseudo-data tests that follow, are the evidence for the background model; the FHC Run-1 deficit visible here is the exposure error of §11.5. Full per-run ratios, the fitted normalisation parameters and their correlations are in the control-sample note.

11.6 Pre-opening validation of the control-region machinery (pseudo-data)

This subsection is the test that was run before the control regions were opened, and it is not a test of the background model. The pseudo-data used here contain only the central-value neutrino-MC component, so what these ratios validate is the implementation — the region definitions, the per-run exposure scaling and the central-value weighting — on a sample where the right answer is known by construction. The full-stack variant of Sec. 12.2 of the technical supplement additionally exercises the bookkeeping path. **The beam-on results are in §11.5**, where the regions are compared with the complete neutrino-MC + beam-off + dirt prediction under the frozen procedure of Sec. 12.5 of the technical supplement.

Because the analysis is blind, the background model cannot yet be confronted with beam-on data in the signal region. To exercise it before unblinding, four signal-depleted control regions (“sidebands”) are defined, each anchored by the inversion of one nominal signal requirement, with the requirements that the inversion leaves undefined dropped explicitly (branch-level definitions in Table 21 of the technical supplement):

- **CC0 π** : the muon selection with no reconstructed charged pion (charged-current quasi-elastic / $2p2h$ -enriched);
- **multi- π** : two or more charged pions (deep-inelastic / resonant multi-pion);
- **π^0** : a reconstructed shower present but the π^0 shower veto failed (CC π^0 / NC π^0);
- **cosmic**: the full signal selection with the muon–pion opening angle inverted ($\theta_{\mu\pi} > 2.6$), enriched in out-of-time cosmic activity.

Each region is evaluated with the same per-run POT scaling and CV event weighting (§5) used for the measurement. Because the current “data” is a Poisson throw of the neutrino Monte Carlo central value only (it contains no EXT or dirt component), the meaningful closure quantity here is the ratio of the pseudo-data to the CV-weighted *neutrino* MC, $N_{\text{data}}/N_{\nu\text{MC}}$; the EXT and dirt yields are shown for scale but enter the background-model test only once real beam-on data are available. Table 23 summarises the four regions for the three beam configurations.

Table 23: Background-control sidebands. “Signal fraction” is the signal contamination of the (CV-weighted neutrino) MC in the region; all four are strongly signal-depleted. $N_{\text{data}}/N_{\nu\text{MC}}$ is the pseudo-data to neutrino-MC ratio: the high-statistics regions (CC0 π , π^0) close to unity across FHC/RHC/combined, validating the per-run weighting and POT-scaling machinery; the low-statistics regions (multi- π , cosmic) scatter within Poisson expectation ($\sqrt{N} \approx 6\text{--}10\%$). The background-model test against the full stack (with EXT and dirt) is performed at control-region unblinding with beam-on data.

Sideband	Signal frac. (of νMC)	$N_{\text{data}}/N_{\nu\text{MC}}$		
		FHC	RHC	Combined
CC0 π	$\sim 10\%$	1.00	0.99	1.00
multi- π	$\sim 23\%$	1.01	1.10	1.06
π^0	$\sim 10\%$	1.00	1.01	1.00
cosmic	$\sim 8\%$	1.09	0.99	1.04

Yields and composition. Table 24 gives the full breakdown for the three configurations. The regions differ enormously in size and in what they are sensitive to, which is what makes them complementary rather than redundant:

- **CC0 π** is by far the largest ($\sim 12\text{k}$ pseudo-data events per beam, $\sim 24\text{k}$ combined). Its neutrino component is dominated by the quasi-elastic and $2p2h$ modes that feed the signal region only through pion *absorption* in the final state. It is therefore the natural handle on the FSI model that converts CC $1\pi^\pm$ into CC0 π and vice versa; the largest single migration path into and out of the signal definition (§6).
- π^0 is the second largest (6.6–14.8k). It is enriched in CC/NC π^0 events, whose showers are precisely what the shower veto (Cut 8) is designed to remove; it therefore tests the veto efficiency and the π^0 production normalisation together. These are the events that leak into the signal region when a photon is missed.
- **multi- π** is small (125–300 events) but has the highest signal contamination of the four ($\sim 23\%$), because a true multi-pion event with one pion lost is exactly the wrong-topology background the selection fights. Its size is what limits its constraining power at present.
- **cosmic** is the region with by far the largest beam-off (EXT) share of the prediction: 53 EXT against 75 neutrino-MC events for FHC, a ratio of 0.7, against 0.1 in CC0 π (with the per-run beam-off treatment of §5). It is the region that tests the beam-off normalisation and shape, and the one whose interpretation changes most at unblinding, since the current pseudo-data contain no EXT component at all. Its beam-off purity is only 41% (FHC; 34% RHC), so under the target-purity requirement of Sec. 12.5 of the technical supplement it is diagnostic: it cannot by itself produce a fitted EXT normalisation correction, only an additional uncertainty.

What the present numbers do and do not show. With Poisson-thrown neutrino-MC pseudo-data, the ratios of Table 23 test the *machinery*, per-run POT scaling, CV weighting, the selection logic in each inverted region, and not the background model. That test passes: the two high-statistics regions sit at $N_{\text{data}}/N_{\nu\text{MC}} = 0.99\text{--}1.01$ in all three configurations, which checks that the per-run scaling is applied identically on the throw and the prediction paths, not the normalisation itself. The apparent deviations in the small regions are statistical: with 82 cosmic-sideband events for FHC, a $\sqrt{N}$ fluctuation is 11%, so the observed 1.09, and the RHC multi- π 1.10 (on 175 events, 8%), are within about one standard deviation.

What the regions could deliver. These yields set the statistical precision that was available when the regions were opened, and the estimates were made beforehand; what they actually showed is §11.5. The combined CC0 π and π^0 samples have 0.6% and 0.8% *raw data statistical* precision. Their effective constraint on the signal-region background is weaker and depends on the correlated transfer uncertainty (Sec. 12.3 of the technical supplement): the transfer factors are small (0.03 and 0.02) but the transfer from a CC0 π event to a selected CC1 π topology is itself model dependent, so the achievable constraint is set by the cross-section and detector terms of that transfer rather than by the sample size. The cosmic region would constrain the EXT normalisation only to $\sim 50\%$ (94 EXT events among ~ 250 combined: $\sqrt{257} = 16$ events of data statistics against the 32% prediction systematic on the 145 neutrino-MC events in the region, ± 46 ; $\sim 17\%$ from data statistics alone); with the beam-off at 1.6% of the signal-region sample that is adequate. No uncertainty is currently assigned to the gate ratios themselves or to the Run-1 sample standing in for Run 2 (14% of the FHC and 23% of the RHC exposure); both are listed in §15. The multi- π region, at ~ 300 combined events, gives $\sim 6\%$, useful as a cross-check but not competitive with the others. The procedure followed was to open the control regions first, use them to test — and where item 4 of Sec. 12.5 of the technical supplement permits, to correct — the normalisation of the dominant background classes, and only then to open the signal region. The signal region remains blinded.

Table 24: Sideband yields at full exposure. “Signal” is the signal contamination of the neutrino MC (events satisfying the inclusive signal definition of §6; all yields and the pseudo-data throw apply the safe-weight rule of Eq. (4)), “ ν bkg” the remaining neutrino MC, EXT the per-run beam-off (cosmic) samples scaled by the run gate ratios (§5), and dirt the out-of-cryostat overlay. The pseudo-data column is a Poisson throw of the neutrino MC only, so it should be compared with Signal + ν bkg; EXT and dirt are listed for scale and enter the comparison only with real beam-on data.

Beam	Sideband	Pseudo-data	Signal	ν bkg	EXT	Dirt	Sig. frac.
FHC	CC0 π	12 181	1 124	11 068	1 358	56	9.2%
FHC	multi- π	125	30	94	7.2	0.1	24.0%
FHC	π^0	6 628	662	5 985	778	12	10.0%
FHC	cosmic	82	6.0	69	53	2.2	7.9%
RHC	CC0 π	12 590	1 206	11 452	1 214	43	9.5%
RHC	multi- π	175	35	124	5.9	0.1	22.0%
RHC	π^0	8 176	780	7 300	678	9.1	9.6%
RHC	cosmic	81	6.4	75	42	1.7	7.9%
Comb	CC0 π	24 771	2 329	22 520	2 572	99	9.4%
Comb	multi- π	300	65	218	13	0.2	22.9%
Comb	π^0	14 804	1 441	13 285	1 455	21	9.8%
Comb	cosmic	163	12	145	94	3.9	7.9%

12 The proton-tagged subsample

A proton-tagged subsample — the inclusive selection with a reconstructed proton candidate above $0.3 \text{ GeV}/c$ — has also been measured: a one-bin total, the calorimetric hadronic-mass proxy, the four transverse-kinematic-imbalance observables, two proton angles, and two-dimensional pairs. It is a different measurement, with its own signal definition, proton-assignment requirements and validation standard, and it is documented in a separate analysis note [53]. Its products are not part of the approval request made here. For cross-reference only, its one-bin total is 0.608 ± 0.256 (FHC), 0.531 ± 0.251 (RHC) and $0.564 \pm 0.239 \times 10^{-38} \text{ cm}^2/\text{Ar}$ (combined), on the same blind fake data and the same provisional exposure as the results above.

13 Validation

Nothing in this section uses beam-on data. Every study here asks whether the machinery does what it claims on simulation, where the right answer is known. The tests that do use beam-on data — the control-region comparison itself — are in the control-sample note.

Validation of an unfolded measurement comes in three layers, and it is worth naming them, because they are not interchangeable and this analysis has completed two of the three.

1. **Implementation closure under the nominal model.** Unfold pseudo-data thrown from the same simulation that built the response, and require the truth back. This tests the implementation — the matrices, the binning, the normalisation, the bookkeeping — and nothing else, because the model is identical on both sides. *Passed*, and reported with every extraction as the closure χ^2 .
2. **Statistical coverage.** Carry many independent Poisson realisations through the whole chain and ask whether the spread of the results matches the uncertainty each one reports. *Passed* for the extractions and configuration stated below; it has not been run for RHC or for the proton-tagged family.

3. **Robustness to model mismatch.** Unfold pseudo-data whose truth differs materially from the model that built the response, and ask how much bias the regularisation and the nominal response then introduce. With *reweighted* GENIE pseudo-data, *passed with a caveat* (§13.3); with an *independent generator, not yet available* — the one existing NuWro overlay could not be validated and is not used.

The third is the one that bounds how much of the result is the data and how much is the model assumed in unfolding it.

13.1 Implementation closure under the nominal model

The first test is the weakest and the most necessary: unfold a spectrum thrown from the same simulation that built the response, and require the truth back. This catches an inverted matrix, a misaligned binning or a normalisation applied twice, and it is reported with every extraction in this note as the closure χ^2 against the A_C -smeared truth. It cannot detect a wrong model, because the model is the same on both sides, and it cannot detect a mis-sized uncertainty, because one throw says nothing about a distribution of throws. Those are the next two subsections.

13.2 Ensemble coverage: are the uncertainties the right size?

A single closure throw cannot say whether the quoted uncertainty is correctly sized, so up to one hundred statistically independent Poisson realisations of the data of each configuration were each put through the whole chain, universes rebuilt from that throw, and the per-bin pull $p_i = (\hat{x}_i - (A_C x_{CV})_i) / \sigma_i$ formed against the fixed central-value truth smeared by that member's own A_C , for six extractions chosen to bracket the conditioning and the configurations: p_μ and $\cos \theta_\mu$ in FHC (the latter's A_C row sums fall to 0.17), $\cos \theta_\mu$ in RHC (0.05), the two-region p_π in the combined configuration, and the one-bin totals of FHC and RHC (no regularisation). Two ensembles are still filling (51 and 46 members); the others have 96–100.

Reference: statistical covariance	pull mean	pull width	68% / 95% coverage
$d\sigma/dp_\mu$, FHC	−0.08	0.96	68.5% / 96.9%
$d\sigma/d \cos \theta_\mu$, FHC	−0.09	1.03	67.6% / 94.6%
$d\sigma/d \cos \theta_\mu$, RHC	−0.07	1.07	66.6% / 93.6%
$d\sigma/dp_\pi$ (two regions), combined	−0.08	0.90	79.4% / 94.1%
one-bin total, FHC	−0.12	0.98	66.7% / 94.8%
one-bin total, RHC	−0.24	1.10	60.9% / 91.3%
expected	0	1	68% / 95%

The widths are consistent with unity, each determined to $\pm 13\%$ with this many members, and the coverage is correct within the same statistics: **the statistical uncertainties are correctly sized in all three configurations, and the strongly suppressed observables behave no differently from the well-conditioned ones.** Against the full covariance the pull widths are 0.21–0.35 (totals: 0.12) with essentially 100% coverage, because the ensemble varies only the data statistics while the quoted uncertainty carries the flux, detector and cross-section terms that are identical in every member; that is the measured form of the closure p -values clustered near unity. The ensemble means sit below the reference by 0.4–1.3%, at 0.5–1.5 standard deviations of the mean, and the per-bin pull means are −0.07 to −0.12 (−0.24 for the RHC total on 46 members): a small common negative offset, consistent with zero at the present ensemble size, and not corrected. The proton-tagged family has not been tested. The reference construction and the full-covariance pulls are discussed in Sec. 13.6 of the technical supplement.

13.3 Robustness to a different interaction model

The closure above cannot see a wrong model, because the model is the same on both sides of it. This test breaks that: the pseudo-data are thrown from a *reweighted* interaction model while the response, the efficiency and the background prediction stay nominal, so whatever bias appears is what the regularisation and a wrong response produce together. The framework’s fake-data universe is filled from the thrown files in truth and reco, so the truth it compares against is the alternative model’s own, and the reweighting is applied to signal and background alike, as a different model would be.

To be precise about what this is and is not: the pseudo-data are reconstructed GENIE events reweighted event by event to an alternative parameter set — signal and background together — with the same detector simulation, unfolded with the nominal response and compared with the nominal covariance. It is a model-distortion closure, not an independent-generator closure; a NuWro or GiBUU detector-level sample would be the stronger test and remains pending. Two variations were used, in FHC and in RHC. The first is GENIE multisim universe 545, a coherent variation of every interaction parameter at once, chosen because it changes *shape* rather than rate: the signal rate moves by 4% while the true-level distributions move by up to 29% in $\cos\theta_\pi$ and 24% in p_π . The second is the $\Delta \rightarrow N\pi$ decay angular unisim, a targeted distortion of the pion angular distribution in resonance events. Both are far larger than the agreement between the tune and the standalone generators, so they are a fair test rather than a token one.

Table 25: Unfolding pseudo-data from an alternative interaction model with the nominal response, FHC (upper block) and RHC (lower). “Ratio” is the mean of unfolded over the alternative model’s own truth, “max dev” the largest per-bin departure from unity, and “bias” the largest per-bin departure expressed in units of the uncertainty that bin carries.

Observable	Alternative model	mean ratio	max dev.	largest bias
p_μ	GENIE multisim u545	0.962	41.8%	-0.68σ
	$\Delta \rightarrow N\pi$ angular	1.145	40.3%	$+0.78\sigma$
$\cos\theta_\mu$	GENIE multisim u545	1.025	11.5%	$+0.23\sigma$
	$\Delta \rightarrow N\pi$ angular	0.955	44.0%	-0.93σ
$\cos\theta_\pi$	GENIE multisim u545	0.980	14.7%	$+0.40\sigma$
	$\Delta \rightarrow N\pi$ angular	0.944	22.6%	-0.59σ
p_π (two regions)	GENIE multisim u545	0.709	60.0%	-0.79σ
	$\Delta \rightarrow N\pi$ angular	1.087	22.2%	$+0.41\sigma$
<i>RHC</i>				
p_μ	GENIE multisim u545	0.843	26.2%	-0.73σ
	$\Delta \rightarrow N\pi$ angular	1.049	14.7%	$+0.34\sigma$
$\cos\theta_\mu$	GENIE multisim u545	0.956	8.7%	-0.22σ
	$\Delta \rightarrow N\pi$ angular	1.013	25.0%	$+0.57\sigma$
$\cos\theta_\pi$	GENIE multisim u545	0.934	26.4%	-0.64σ
	$\Delta \rightarrow N\pi$ angular	1.008	9.3%	$+0.20\sigma$
p_π (two regions)	GENIE multisim u545	0.865	23.2%	-0.32σ
	$\Delta \rightarrow N\pi$ angular	1.036	5.1%	$+0.08\sigma$

No bin anywhere exceeds one standard deviation of its own uncertainty (the largest is 0.93σ), so nothing in this note is invalidated by the interaction model being wrong at the level tested. In the well-populated bins the extraction is genuinely robust: the six populated p_μ bins return 0.92–1.16 of the alternative truth under both variations with biases below 0.5σ , against truth-level distortions of 10–29%.

Three bins are not robust, and all three are sparse. The near-threshold p_π interval, 0.175–0.205 GeV/ c , comes back at 0.40 of the alternative truth under the coherent variation; the top p_μ bin at 0.58 (and its neighbour at 1.4 under the angular variation); the third $\cos\theta_\mu$ bin at 0.56

under the angular variation. These are the bins the note already identifies as the weakest — the first because the range-based pion momentum saturates, the others because they are the sparse tail — and the test says that when the model is wrong, that is where the damage lands.

The honest reading of the σ column is that the bias is sub- σ *partly because the uncertainties are large*: those bins carry 190%, 106% and 85% total uncertainty. At this exposure model bias is a subleading effect, comfortably inside the error bars. It would not stay subleading if the uncertainties shrank, so this test should be repeated before any future exposure is interpreted. In RHC the picture is the same with the roles of the edge bins taken by the sparse sixth p_μ bin (-0.73σ) and the second $\cos\theta_\pi$ bin (-0.64σ): no bin beyond 0.73σ , and the coherent GENIE variation, which lowers the RHC rate, pulls every RHC mean ratio below one (0.84–0.96) where the angular variation does not. The proton-tagged family, tested the same way, shows a larger sensitivity: its low-imbalance δp_T and p_n bins come back 35% below the alternative truth under the coherent GENIE variation, -1.1 to -1.2σ (Sec. 4 of the proton-tagged note).

Independent-generator closure: not yet available. The stronger form of the third test — pseudo-data from a different generator with its own detector simulation — was attempted with the one NuWro overlay of Run-1 FHC that exists. It is not used. The sample’s normalisation does not agree with the standalone NuWro prediction for the same flux (its true signal rate per POT is 1.76 times GENIE’s, against the 1.6 implied by the standalone NuWro cross section and the raw flux), and its provenance could not be established, so a closure against it would test the sample rather than the extraction. A properly produced NuWro or GiBUU overlay remains the outstanding validation item (§15).

13.4 The background fit, validated on pseudo-data

- **Toys**: see §13.5 for ensembles of up to 10^5 toys per model; the 300-toy sets of Fig. 26 give pull means 0.02–0.07, widths 0.89–1.03 and coverage 67–75%.
- **Injections** (Asimov, two-parameter totals fit): class shifts are recovered exactly; a 30% signal excess moves the normalisations by 0.13σ ; an unfitted class at +30% by -0.07σ ; a detector knob by at most 0.14σ ; flux and GENIE universes by -0.97 to $+0.84\sigma$, i.e. within the uncertainty they are part of.
- **FHC versus RHC**: a likelihood-ratio test of shared against beam-specific normalisations inside the joint covariance (a parameter goodness-of-fit test is not valid here, the modes share universes). A +15% shift of the RHC neutrino Monte Carlo alone gives $\Delta\chi^2 = 12.9/2$ ($p = 0.002$, totals) and $20.3/2$ ($p = 4 \times 10^{-5}$, 2D).
- **MCMC** (adaptive Metropolis, four chains, flat priors): posterior means within 0.02 and widths within 2% of MIGRAD/MINOS on the Asimov dataset and on a $CC0\pi \times 1.25$ injection, $\hat{R} \leq 1.004$. With a few parameters and a near-quadratic likelihood the profile likelihood is sufficient; the sampler is kept as a cross-check.

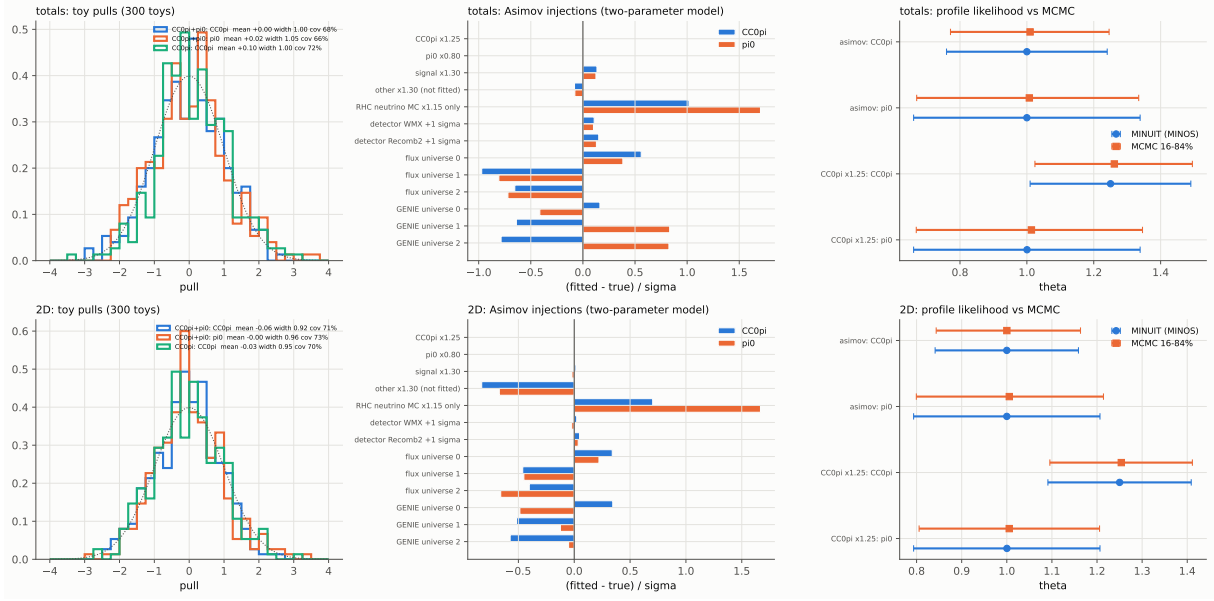


Figure 26: Validation of the background fit on simulation, for the totals-only fit (top) and the two-dimensional fit (bottom): pull distributions from toy ensembles, recovery of injected normalisation shifts, and the profile-likelihood interval against the Markov-chain posterior.

13.5 Pull-mean convergence of the background fit

With 300 toys the error on a pull mean is 0.058, so the $+0.07$ of the first validation could not be separated from zero. Ensembles of 10^5 toys per model (error 0.003) with three diagnostic variants, each of 5×10^4 , settle it (Table 26, Fig. 27). With the combined Neyman–Pearson data variance the mean does *not* converge to zero in the 2D fits: a residual $+0.019$ ($CC0\pi$) and $+0.034$, $+0.031$ ($CC0\pi+\pi^0$) remains at six to ten standard deviations of the ensemble. It persists when the Poisson fluctuation is replaced by a Gaussian one ($+0.031$), so it is not Poisson counting in low bins; it disappears when the data variance is the prediction at the nominal parameters ($+0.000$, -0.004 , -0.001). The CNP variance depends on the fluctuating data, which correlates the weight of a bin with its own fluctuation. The bias is small — 0.3–0.7% on θ against a 16–21% uncertainty — but it is avoidable, so the Pearson form is the default of the fit and every data result below uses it. Pull widths are 1.00 and coverage 68% in every variant.

Table 26: Pull means $\pm$ their ensemble error. “standard”: systematic throw from the covariance, Poisson, CNP data variance (10^5 toys); “Gaussian”: Gaussian instead of Poisson fluctuation; “Pearson”: standard toys fitted with the prediction as data variance (5×10^4 each).

Bins	Model (parameter)	standard, CNP	Gaussian, CNP	standard, Pearson
totals	$CC0\pi$	$+0.003 \pm 0.003$	$+0.005 \pm 0.005$	-0.011 ± 0.005
totals	$CC0\pi+\pi^0$ ($CC0\pi$)	$+0.012 \pm 0.003$	$+0.014 \pm 0.005$	$+0.003 \pm 0.005$
totals	$CC0\pi+\pi^0$ (π^0)	$+0.016 \pm 0.003$	$+0.014 \pm 0.005$	$+0.008 \pm 0.005$
2D	$CC0\pi$	$+0.019 \pm 0.003$	$+0.031 \pm 0.005$	$+0.000 \pm 0.005$
2D	$CC0\pi+\pi^0$ ($CC0\pi$)	$+0.034 \pm 0.003$	$+0.031 \pm 0.005$	-0.004 ± 0.005
2D	$CC0\pi+\pi^0$ (π^0)	$+0.031 \pm 0.003$	$+0.029 \pm 0.005$	-0.001 ± 0.005

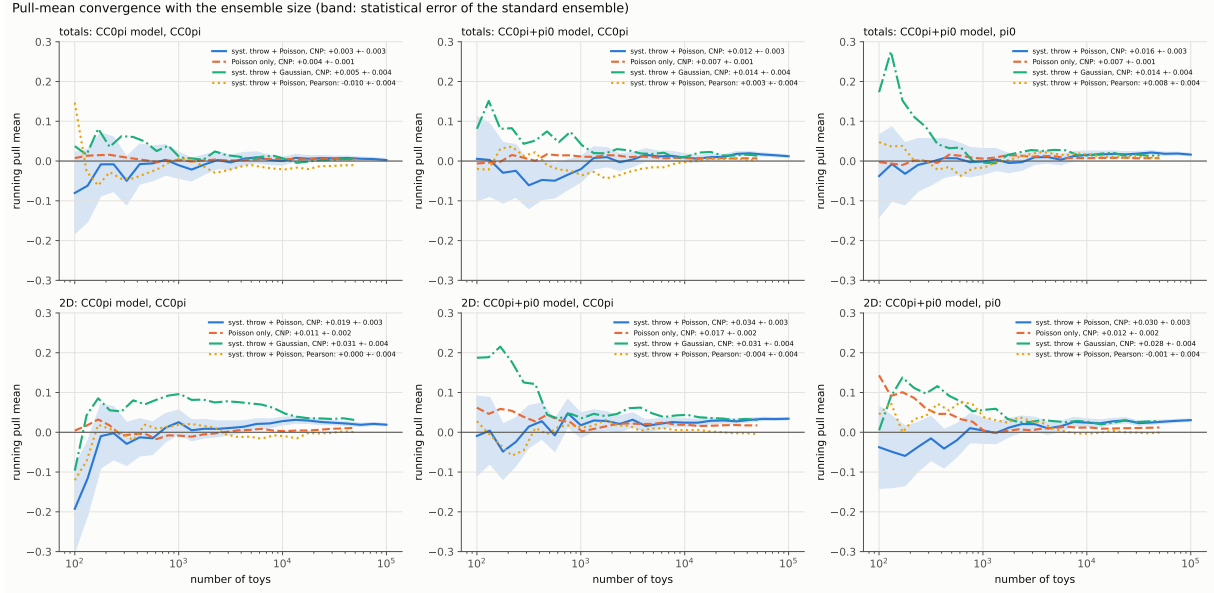


Figure 27: Running pull mean against the ensemble size for each model and parameter; band: statistical error of the standard ensemble. The ensemble thrown with statistical fluctuations only, and no systematic variation, has narrower pulls as expected.

14 Systematic Uncertainties

The configured systematic covariance is propagated for all fifty-one extractions, complete with respect to the configured model (the identified unpropagated terms, the MCS momentum scale (evaluated at $\pm 5\%$ on the data side and released as a separate term for FHC p_μ , §15), the beamline geometry (evaluated at 1.6–3.3% and bounded there, §14), the beam-off gate ratios and the Run-2 beam-off stand-in, are listed in §15). Every result carries flux, cross-section-model, hadron-reinteraction, detector-variation, POT, target-count and statistical terms, propagated through the unfolding by the error propagation matrix and released per component (Sec. 11.3). Table 27 lists the sources.

Table 27: Systematic sources and their bin-averaged fractional contribution, computed from the released per-component covariance rather than estimated. Ranges span observables and configurations; the two families are given separately because the proton-tagged sample is both statistically sparser and more detector-sensitive.

Source	Branch / method	Inclusive (%)	Proton-tagged (%)
Flux (PPFX multisims)	weightsFlux	23.8–42.3	17.6–74.6
Detector response	Dedicated samples	14.9–41.4	9.9–142.5
Cross-section model	weightsGenie	7.8–22.5	6.7–71.3
Hadron re-interaction	weightsReint	3.4–16.5	3.6–33.5
POT counting	Beam toroids	2.6–5.0	2.1–8.7
Target count	FV geometry	1.3–2.5	1.1–4.4
MC statistics	universe spread	1.9–8.9	2.2–26.3
EXT statistics	beam-off sample	0.5–3.4	0.5–5.4
Data statistics	thrown fake data	6.6–27.9	7.3–85.3
Prediction total	quadrature sum	29.7–66.9	22.0–173.9
Total	incl. data stats	31.1–73.0	23.8–194.2

Flux and detector response are the two largest terms in both families; which of the two leads depends on the observable and configuration (Table 28; Table 18 of the technical supplement–Table 19 of the technical supplement). Every entry is propagated into the released covariance; no source declared in the release systematics configuration is left unpropagated. That is a statement about the configured model, not a completeness claim: the beamline-geometry flux weights exist in the ntuples but are not in the release configuration, and the other identified omissions (the MCS momentum scale, evaluated at $\pm 5\%$ and released as a separate term for FHC p_μ but not propagated inside the framework; the beam-off gate ratio; the Run-2 beam-off stand-in) are listed in §15.

14.1 Systematic breakdown by source

The framework builds a covariance matrix for every systematic. The bin-averaged fractional uncertainty from each source group for the released FHC extractions is given below; the RHC and combined tables, generated from the same release dumps, are Table 18 of the technical supplement and Table 19 of the technical supplement; the numbers are those of Table 17.

Table 28: FHC (Runs 1, 2, 4, 5): bin-averaged fractional uncertainty (%) by source group for the six released extractions, read from the release systematics dumps “Prediction total” excludes the data statistics; the flux row is the PPFX multisim term. Every row is a bin average of the per-bin fractional uncertainties, so the rows do not combine in quadrature to the totals (the quadrature is taken per bin before averaging).

Source	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$	θ_μ
Prediction total	39.9	66.9	41.0	39.0	37.1	29.7
Cross section (GENIE)	10.4	22.5	10.5	11.4	10.6	7.8
Flux (PPFX)	28.1	42.3	27.7	27.1	28.8	23.8
Detector	23.6	41.4	27.1	23.2	19.9	14.9
Reinteraction	4.4	16.5	3.7	5.1	4.1	3.4
MC stat	4.5	8.9	4.7	4.2	2.8	2.7
EXT stat	0.7	3.4	1.6	1.2	1.0	1.0
Data stat	14.5	27.9	15.1	13.6	8.9	8.7
POT + targets	3.4	5.5	3.6	3.4	3.6	2.9
Total (incl. data stat)	42.7	73.0	43.8	41.5	38.3	31.1

The NuMI flux uncertainty has two parts: the PPFX hadron-production term (the dominant systematic) and a beamline-geometry term. The latter is built from the 12 geometry-variation weights (horn current, horn positions, beam spot size and position, horn water layers and target position) injected into the numuMC overlay, summed in quadrature as two-sided unisim variations. Evaluated in a dedicated pass it is 1.6–3.3% across the observables and sub-dominant; it is *not* part of the released covariance, whose configuration omits the beamline block. Added in quadrature to the $\sim 28\%$ PPFX term it would raise the flux row by $\lesssim 0.2\%$ absolute, so the omission is immaterial to the budget, but the term is listed as absent rather than propagated. The per-bin breakdowns are shown in Fig. 28.

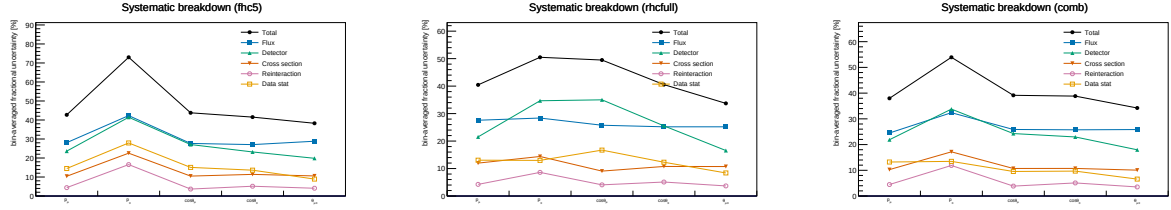


Figure 28: Bin-averaged fractional systematic uncertainty by source vs. observable, for FHC (left), RHC (centre) and combined (right); the figure form of Table 17 of the technical supplement–Table 19 of the technical supplement. Colours use the Okabe-Ito colourblind-safe palette with distinct markers per source: total (black), flux (blue), detector (green), cross section (vermillion), reinteraction (purple), data stat (orange).

14.2 Anatomy of the flux term: why 17% becomes 30%

Why a 17% flux uncertainty becomes 28–30% on the total. A flux universe scales signal and neutrino background together, and the extracted signal is $(N_{\text{pred}} - B)/(\epsilon\Phi N_{\text{Ar}})$, so its fractional flux variation is $\delta\Phi/\Phi \times (1 + B/S)$: $17.7\% \times (1 + 637/943) = 29.7\%$ for FHC against the one-bin value 30.25%. The one-bin total has no regularisation, so this is background subtraction, not unfolding amplification. The lever is purity or a data-driven background constraint (§11.6).

The unfolded PPFX term is 24–33% (Tables 17–19) and the one-bin total carries 28–30% (Table 22), against the $\sim 17\%$ normalisation uncertainty of the NuMI flux itself. The difference is not amplification by the unfolding operator: the one-bin total, which has no regularisation, carries it too. The factor is the background subtraction. A flux universe scales the whole reconstructed prediction, signal and neutrino background alike, and the framework’s covariance is that of the prediction of the measured quantity, $(N_{\text{pred}} - B)/(\epsilon\Phi N_{\text{Ar}})$ with B and Φ moving together in every universe; relative to the signal that is a fractional variation of $\delta\Phi/\Phi \times (1 + B/S)$, with B the neutrino background (the beam-off and dirt carry no flux weight). With $B/S = 637/943$ (FHC, Table 10) and a reconstructed-level flux term of 17.7% this gives 29.7% against the one-bin

FHC value of 30.25%; RHC ($15\% \times (1 + 720/1015) = 25.6\%$ against 27.7%) and the proton-tagged FHC total ($17.7\% \times (1 + 338/376) = 33.6\%$ against 34.9%, a sample with a different purity) follow the same arithmetic. The differential flux terms of 24–33% are the same effect bin by bin, with the response conditioning adding a few per cent in the worst-conditioned bins (p_π , $\cos\theta_\mu$) and nothing in the best ($\theta_{\mu\pi}$).

Two consequences follow. First, the flux term is fully correlated between the flux normalisation and the subtracted background, which is the correct treatment when both come from the same flux prediction; a reader who wants the “pure” normalisation term should read 17%, and the remainder is the flux dependence of the background model. Second, the lever for reducing it is the purity of the selection (or a data-driven background constraint from the control regions of §11.6), not the regularisation. The reconstructed-level flux term is 16.4–17.7% bin-averaged and flat across observables, as expected of a normalisation term (the 600 PPFX universes reweight the same events, so it carries no Monte Carlo statistics); that is the 17% referred to throughout.

15 Limitations

This section states what limits the measurement as it stands. The studies behind the scope decisions are in the technical supplement; the history of the analysis is in the change log.

The analysis is blind, and that is the dominant limitation. No beam-on signal region has been opened. Every “data” point is Poisson-thrown from the central-value Monte Carlo, so everything demonstrated here concerns the extraction machinery rather than nature. In particular, no statement in this note tests the background model against data. At $\approx 59\%$ purity, with proton-to-pion misidentification the largest neutrino background, the measurement depends on that model substantially, and fake-data closure cannot probe it: the same simulation supplies both the pseudo-data and the background estimate. This is the single largest untested assumption in the analysis.

Pion momentum is not recoverable, and that sets the scope. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near $0.34\text{ GeV}/c$ and carries little information above roughly $0.4\text{ GeV}/c$. Every alternative estimator was tried — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, the proton-tagged selection — and a truth-level golden selection performs no better than the trained classifier, which places the limit in the physics rather than the estimator. This is why p_π is reported in two coarse bins and why the differential $W_{\pi p}$ shape is withdrawn: at this exposure no binning satisfies both the migration and the statistics requirement.

The extraction may carry a small negative offset. The ensemble of thirty-two throws on the current release puts the mean below the reference by 0.5% (p_μ), 1.0% (one-bin total) and 2.7% ($\cos\theta_\mu$), at 0.4, 1.3 and 1.9 standard deviations of the mean, all from the same throws. It is not significant individually, but it has the same sign everywhere and is largest in the most regularised observable. No correction or uncertainty is assigned; a larger ensemble, and one on RHC, would resolve it.

The W_{had} binning has not been edge-scanned. The migration-level edge scan that fixed the p_π boundary (Sec. 2.1 of the technical supplement) was not run for W_{had} , whose upper bins are narrower than its resolution half-width; the six-bin W_{had} result is released with its closure and generator overlays, and its shape should be read with that caveat until the scan is made (not gating).

No alternative-generator fake-data test exists for the released chain. Every closure in this note is a same-sample self-closure: the fake data are thrown from the central-value tune that also supplies the response. The model dependence of the efficiency is therefore untested where it matters most: the one-bin total (a single efficiency averaged over the whole phase space) and the open-ended $p_\pi > 0.205$ GeV/ c bin, inside which the efficiency varies strongly with the true momentum. A NuWro or GiBUU fake-data pass through the released configurations is required before the signal region is opened; it is not required for the control-region opening, which compares reconstruction-level yields and does not involve the unfolding.

Coverage is demonstrated for three extractions, in one configuration. The pull and coverage study covers FHC p_μ , $\cos\theta_\mu$ and the one-bin total. All give widths consistent with unity, and $\cos\theta_\mu$ was chosen because its A_C suppression is the most severe in the analysis, but RHC and the proton-tagged family — where the statistics are sparser — have not been tested.

The proton-tagged results are secondary, and their model comparisons are the weakest part. Their covariance is complete with respect to the configured model and their central values current, but the exposure does not support a differential $W_{\pi p}$ shape, and the sample has no dedicated control region for the $\approx 13\%$ proton mis-tag: every existing sideband applies the proton requirement rather than inverting it, so nothing directly constrains a wrong-track proton assignment, which is exactly what distorts the hadronic observables.

The TKI prescription is borrowed from the single-proton case. δp_T , $\delta\alpha_T$ and $\delta\phi_T$ are exact for the $\mu + (\pi p)$ system, but p_n and W_{had} use the single-nucleon residual-mass and removal-energy assumptions of that tool with the summed hadronic system in the proton slot (Sec. 6.1 of the technical supplement). Data and generators are treated identically, so the comparison is consistent, but no truth-level closure of this prescription against a pion-production-specific definition has been made and the bias from the residual-mass assumption is not quantified; p_n should be read as a proxy.

No dedicated systematic covers the MCS muon momentum. Sixty-three per cent of the selected muons exit the detector and take their momentum from multiple Coulomb scattering, whose scale and resolution are not varied by the eight detector-variation knobs; the estimator enters only through the response matrix. A data-driven MCS-versus-range comparison on contained muons is the standard way to bound it and has not been done here. Before the signal region is opened one of three things is required: that comparison, on contained muons in the beam-off and overlay samples; a propagated MCS scale and resolution variation through the response; or a quantitative argument that the existing detector variations, several of which change the MCS estimate through the space-charge and recombination models, already cover it. The second of these has been done for $d\sigma/dp_\mu$ (FHC). The momentum of every selected muon is exactly the range value when it is contained and exactly the MCS value when it exits (63% of the selected sample), so the MCS component can be scaled alone. A momentum-scale error is a mismatch between data and simulation, so the scale is applied to the *data* side only: the fake data are re-thrown with every MCS-measured muon momentum multiplied by 1 ± 0.05 while the simulation, the response and the selection flag stay nominal (the flag is precomputed, so the threshold does not move, which is the limit of this bound), and the spectrum is re-extracted with the nominal response. The half-difference between the two variations is 8–13% of the bin value below 0.95 GeV/ c , where the regularised spectrum is steep, 2% in the fifth bin and 8–10% (opposite sign) in the two highest bins: $0.05\text{--}0.34\sigma$ of the quoted per-bin uncertainty, at most 0.34σ . The integral moves by -6.8% / $+2.1\%$ (half-difference 4.5%). It is released as a fully correlated covariance term for FHC p_μ (`cov_MCSscale.txt`, with `cov_total_plusMCS.txt`); adding it raises the per-bin total uncertainty from 38–58% to 39–58%. It is not yet propagated

inside the framework, and the RHC and combined p_μ extractions and the other observables (where it enters only through the p_μ acceptance) do not carry it. (An earlier version scaled the MCS momentum through the bin configuration, i.e. in the simulation and the fake data alike; a common scale cancels in the extraction, and the 0.07σ that test returned measured only the regulariser’s response to a remapped variable. It is withdrawn.) The control-region comparison does not depend on it, since it is made at reconstruction level with the same estimator on both sides.

Two beam-off terms are not in the covariance. The released covariance carries the beam-off statistical term but no uncertainty on the per-run gate ratios or on the use of the Run-1 beam-off sample for Run 2, which has no sample of its own (14% of the FHC and 23% of the RHC exposure). With the beam-off at 1.9% of the selected sample a 20% uncertainty on either would add 0.4% to the total; they are omitted for that reason and stated here. The beamline-geometry flux term (1.6–3.3%) is likewise absent (§14).

A_C is not norm preserving, so two different totals exist. The Wiener-SVD differential result estimates $A_C x_{\text{true}}$, and the row sums fall as low as 0.050. Summing it over bins therefore does not give a physical total cross section; the physical total is the one-bin extraction, quoted separately. Two matrices have negative row sums in their least-determined bin.

Actions required before unblinding. The prerequisites, split by the decision they gate, are the checklist of Sec. 11 of the technical supplement. In brief: the four inclusive control regions are to be opened first, under the frozen procedure of Sec. 12.5 of the technical supplement once the conveners have ratified it; the signal region only after any control-region discrepancy has been treated under that procedure, the flux constants have been confirmed as PPFx-corrected central values (§4.4), the real beam-on gate counts have replaced the fake-data ones (the Run-1 beam-off sample stands in for Run 2 permanently, §5), and a final clean-cache reproduction has been made; and any proton-tagged result only after the inverted-proton-PID control region exists (Sec. 12.6 of the technical supplement). Nothing has to be switched for the beam-off treatment: the framework detects fake data by the presence of MC truth and selects the treatment itself, and refuses a real beam-on file unless an explicit unblinding switch is set.

The five automated guards that protect against regression — the normalisation manifest, the systematics/file-list coupling assertion, the staleness check, the table-consistency check and the blinding guard — are described in Sec. 14 of the technical supplement.

16 Summary

We have developed a measurement of $\nu_\mu + \bar{\nu}_\mu$ charged-current single charged-pion production on argon using the full MicroBooNE NuMI exposure: forward horn current (7.766×10^{20} POT), reverse horn current (1.1082×10^{21} POT), and their combination (1.8848×10^{21} POT).

The analysis is blind. No beam-on signal region has been opened. Every “data” point in this note is Poisson-thrown from the central-value Monte Carlo, so the results demonstrate that the extraction works; they are not measurements, and no statement here about a generator is a statement about nature.

Measured phase space and selection. $|\text{PDG}_\nu| = 14$, charged current, true vertex in the fiducial volume, exactly one muon and one charged pion, no neutral pions or kaons, any number of protons, with $p_\mu > 0.15 \text{ GeV}/c$, $p_\pi > 0.175 \text{ GeV}/c$ and $\theta_{\mu\pi} < 2.6 \text{ rad}$. The selection reaches $\approx 17.5\%$ efficiency at $\approx 59\%$ purity. Angles and transverse observables are defined about the per-event neutrino direction, not the detector z axis.

What is measured. Fifty-one differential extractions, eighteen inclusive and thirty-three proton-tagged, across the three configurations, plus six one-bin totals. The primary scope is the total flux-averaged cross section, $d\sigma/dp_\mu$, the muon and pion angular cross sections, $d\sigma/d\theta_{\mu\pi}$, and a deliberately coarse two-bin $d\sigma/dp_\pi$. Inclusive differential total uncertainties run 32.4–53.9%, with the detector term 15.6–35.0% and flux the largest single contribution; the one-bin totals carry 37.6–38.7%.

The proton-tagged subsample is secondary scope, with differential total uncertainties 25.1–147.7% (the largest for the six-bin combined W_{had}) and one-bin totals of 41–47%. Its covariance is complete with respect to the configured model, but the exposure does not support a differential $W_{\pi p}$ shape at any binning, so that result is withdrawn and only the integrated proton-tagged cross section is offered.

What limits the measurement. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/ c . A survey of alternatives — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, and the proton-tagged selection — recovers none of it, and a truth-level golden selection performs no better than the classifier, which places the limit in the physics rather than the estimator. This is what reduces p_π to two bins and $W_{\pi p}$ to an integrated quantity, and it is the single most consequential limitation of the analysis.

Validation. The extraction reproduces its own input: the mean closure ratio over the eighteen inclusive extractions is 1.010 with second-derivative regularisation. The algebraic identity $UR = A_C$ holds to 4.4×10^{-6} (App. 8 of the technical supplement), which tests the unfolding construction rather than only its behaviour on one sample. The released A_C row sums establish that the observable-to-observable spread of the Wiener-SVD integrals is the additional smearing rather than physics or acceptance; D’Agostini iterative unfolding, whose integral is flat across observables, is consistent with that (it returns its prior on central-value input, so it cannot test more).

The closure χ^2 values should not be read as agreement: their p -values cluster near unity because the measurement and the truth it is compared against share their systematics, which cancel in the residual while still inflating the covariance used to judge it.

Reproducibility from raw inputs. The release is built from the processed ntuples with every cached universe deleted, in a single pass, so no preserved intermediate exists. An independent rebuild of the same kind, into a separate tree so that a failure could not damage the release it was verifying, reproduced an earlier release *bin by bin*, with no bin differing by more than 10^{-9} in any of the fifty-one extractions; repeating that comparison on the release to be unblinded is a prerequisite (Sec. 14 of the technical supplement). This is a different question from the closure and ensemble tests: those ask whether the extraction is right, this asks whether what is published is what the code produces. The external-reproduction check of the released A_C prescription (§11.3) belongs to the same class: a user must be able to regenerate every published generator curve from the released matrix, and can.

History. The analysis reached this state through a series of corrections, each recorded in the change log with the values before and after; the automated guards of Sec. 14 of the technical supplement exist because of them. This note describes the analysis as it stands and does not repeat that history.

Before unblinding. Control regions must be opened in data first, with the permitted response to any discrepancy fixed beforehand, and a clean rebuild from empty caches must reproduce the release. The configuration-to-systematics coupling is asserted by the framework (Sec. 14 of

the technical supplement). The ensemble of throws covers three FHC extractions; extending it to RHC and to the proton-tagged family would turn the closure caveat above into a coverage statement for every result.

Glossary

Signal definition The set of particles leaving the nucleus that counts as the process being measured, with momentum thresholds and a fiducial requirement. It is deliberately *not* a statement about the interaction mechanism, because final-state interactions mix mechanisms and only the final state is observable.

Fake data / pseudo-data A simulated dataset used in place of beam-on data while the analysis is blind. Here it is a Poisson throw: each simulated event is replicated a Poisson-distributed number of times with mean equal to its weight at data exposure, so the sample has the statistical fluctuations of a real dataset.

Blind The signal region has not been looked at in beam-on data. Every plotted “data” point in these documents is fake data.

Beam-on, beam-off (EXT), dirt Beam-on data are recorded in the beam spill; beam-off data are recorded between spills and measure cosmic-ray activity; “dirt” is simulated neutrino interactions outside the cryostat whose products enter the detector. The prediction is neutrino simulation + beam-off + dirt.

Gate ratio The factor scaling beam-off data to the beam-on exposure: the ratio of the number of beam-on to beam-off trigger gates in a running period.

Tune, central value (CV) The MicroBooNE tune is the collaboration’s adjusted GENIE model; the central value is the nominal prediction from it. “CV weight” is the per-event weight that turns the raw simulation into the tuned prediction.

Safe-weight rule Any event weight that is non-finite, negative or above 30 is replaced by 1. One rule, applied everywhere, so that the prediction and every pseudo-data throw agree.

Universe One alternative version of the prediction with a systematic parameter varied. A *multisim* is a set of hundreds of universes drawn randomly from the parameter uncertainties; a *unisim* is a single $\pm 1\sigma$ variation of one parameter. PPFX universes vary the flux; GENIE universes the interaction model; reinteraction universes the hadron transport.

Detector variations (detVar) Dedicated simulation samples with one aspect of the detector response changed — light yield, recombination, space-charge, wire response — used to estimate the detector uncertainty.

Migration matrix, efficiency, smearceptance The migration matrix M_{ij} is the probability that an event in true bin j is reconstructed in bin i ; the efficiency ε_j is the fraction of true bin j that is selected at all. Their product $S_{ij} = \varepsilon_j M_{ij}$ is the response the unfolding inverts; we call it the smearceptance matrix.

Migration diagonal The fraction of events reconstructed in a bin that were truly in that bin (the column-normalised diagonal of M). A binning is accepted only if every bin’s diagonal exceeds 0.68.

Unfolding, Wiener-SVD Solving for the true spectrum given the reconstructed one and the response. A plain inversion amplifies statistical noise; Wiener-SVD regularises it with a filter built from the simulation, trading a small bias for a large reduction in variance.

Additional-smearing matrix A_C The price of regularisation, made explicit: the unfolded result estimates $A_C T$ rather than the truth T . A prediction must be multiplied by A_C before it is compared with the result. A_C is released with every extraction.

Not normalisation-preserving The rows of A_C do not sum to one, so the integral of an unfolded differential result is not the total cross section and differs between observables of the same sample. The total is measured separately.

One-bin extraction The total cross section, extracted with a single true bin covering the whole phase space. Nothing is regularised, $A_C = 1$, and the result is the physical total.

Conditioning, $s_{\min}/s_1$, rank-deficient The ratio of the smallest to the largest singular value of the regularised response. Near zero, some direction in the true spectrum cannot be recovered from the data at all (rank-deficient); the result in that direction is the prior. A binning can pass the migration-diagonal criterion and still be unusable here.

Closure Unfolding pseudo-data and checking that the truth comes back. Closure against the A_C -smeared truth tests the implementation; it cannot test whether the model is right, because the same model is on both sides.

Covariance The matrix of uncertainties and their correlations between bins, one per systematic source, propagated through the unfolding. “Nominal configured covariance” means the sum over the sources implemented in the analysis; the sources known to be missing are listed in the limitations.

Flux-averaged cross section The cross section averaged over the neutrino energy spectrum of the beam, per argon nucleus, in the stated phase space. It depends on the flux, which is why each horn configuration is a separate measurement.

FHC, RHC Forward and reverse horn current: the two polarities of the NuMI focusing horns, giving a neutrino-dominated and an antineutrino-enriched beam.

Beam frame All angles are measured about the neutrino direction, which at MicroBooNE’s off-axis position is 28° from the detector z axis.

Control region, sideband A sample selected by inverting one requirement of the signal selection, so that it is disjoint from the signal region and enriched in one background. It can be inspected on beam-on data while the signal region stays blind.

Transverse kinematic imbalance (TKI) Variables built from the momenta transverse to the neutrino direction of the lepton and the hadronic system: δp_T , $\delta \alpha_T$, $\delta \phi_T$ and the inferred nucleon momentum p_n . They isolate nuclear effects from the unknown neutrino energy.

LLR particle identification The log-likelihood-ratio score built from track dE/dx that separates protons from muons and pions.

A Observable Binning

Generated directly from the released `curves_*.tsv` files, so these are by construction the bin edges of the published results rather than a transcription. The binnings are identical across the three beam configurations; the inclusive $\cos \theta_\pi$ uses the forward-split five bins below, the proton-tagged $\cos \theta_\pi$ four (Sec. 4 of the technical supplement).

Table 29: Bin edges of every released observable, read from the data release.

Observable	Bins	Edges
<i>Inclusive</i>		
p_μ (GeV/ c)	7	0.15, 0.35, 0.55, 0.75, 0.95, 1.25, 1.75, 3
p_π (GeV/ c)	2	0.175, 0.205, 1*
$\cos \theta_\mu$	5	-1, 0.45, 0.65, 0.8, 0.9, 1
$\cos \theta_\pi$	5	-1, -0.1, 0.35, 0.6, 0.8, 1
$\theta_{\mu\pi}$ (rad)	5	0, 0.6, 0.85, 1.3, 1.85, 2.6
θ_μ (rad)	5	0, 0.43, 0.62, 0.84, 1.17, 3.15
<i>Proton-tagged</i>		
p_μ (GeV/ c)	7	0.15, 0.35, 0.55, 0.75, 0.95, 1.25, 1.75, 3
p_π (GeV/ c)	2	0.175, 0.205, 1*
$\cos \theta_\mu$	5	-1, 0.45, 0.65, 0.8, 0.9, 1
$\cos \theta_\pi$	4	-1, -0.1, 0.35, 0.75, 1
$\theta_{\mu\pi}$ (rad)	5	0, 0.6, 0.85, 1.3, 1.85, 2.6
$W_{\pi p}$ (GeV/ c^2)	6	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.9
W_{had} (GeV/ c^2)	6	0, 0.78, 1.08, 1.21, 1.31, 1.44, 2.74
δp_T (GeV/ c)	2	0, 0.3, 2.5
$\delta \phi_T$ (deg)	2	0, 50, 180
$\delta \alpha_T$ (deg)	2	0, 120, 180
p_n (GeV/ c)	2	0, 0.375, 2
θ_p (rad) [†]	4	0, 0.659, 1.068, 1.539, π
$\theta_{\pi p}$ (rad) [†]	3	0, 1.193, 2.010, π

[†] Proposed secondary scope: the proton polar angle and the pion–proton opening angle (Sec. 3.3 of the proton-tagged note). θ_p uses edges derived on the RHC chain, which condition three times better than the FHC-derived ones at the same bin count.

* Numerical plotting boundary only; the upper truth bin is open-ended ($p_\pi > 0.205$ GeV/ c), and the released density over the conventional width 0.795 GeV/ c represents the bin integral (Table 20 of the technical supplement).

The migration-diagonal criterion itself was also relaxed, from 0.68 to 0.50 admits twenty-six new candidate binnings, and *every one of them* fails the secondary gate on the conditioning of A_C , which is not the diagonal. The only healthy candidate is a two-bin W_{had} , and the six-bin scheme is kept, since two bins buy conditioning by discarding the shape the observable is measured for. No binning in this note changes as a result.

Every released binning was tested for whether finer or more bins are admissible (Sec. 4.3 of the technical supplement): p_μ , $\cos \theta_\mu$, θ_μ and p_π cannot go finer, $\theta_{\mu\pi}$ gains bins but no resolved information, and $\cos \theta_\pi$ admits a fifth bin in the forward hemisphere (edges $-1, -0.1, 0.35, 0.6, 0.8, 1$) with better-conditioned A_C and unchanged per-bin uncertainties, which is the released binning; the generator predictions are computed at these edges from the event-level files.

The top p_π bin is defined at truth level as $p_\pi > 0.205$ GeV/ c with *no upper edge*. An open-ended bin has no width, so it has no differential density: the 1 GeV/ c listed above is a reporting convention that assigns the bin a finite width of 0.795 GeV/ c so that it can be plotted on the same axis as the first, and it is not a kinematic limit of the sample. A height read off that plot is that convention, not a measured $d\sigma/dp_\pi$. The quantities that are unambiguous are the two bin integrals, which is how the result should be quoted and compared (Table 21).

B Alternative binnings considered

Every released binning is the coarsest that satisfies the migration-diagonal criterion, and finer schemes were tested rather than assumed unusable. The criterion itself was also relaxed, from the 0.68 column-normalised diagonal to 0.50, to see what that would admit: twenty-six further

candidates pass the relaxed diagonal, and all but one fail the secondary requirement, which is the conditioning of the additional-smearing matrix A_C . A binning that migrates acceptably can still give a response that cannot be inverted, and that — not the diagonal — is what limits this measurement. Table 30 lists the alternatives and their fate.

Table 30: Alternative binnings examined. $s_{\min}/s_1$ is the conditioning of A_C (FHC/RHC unless stated); a value near zero means the regularised response cannot be inverted, whatever the migration diagonal says. Only the two-bin W_{had} is healthy, and it was not adopted.

Observable	Alternative	$s_{\min}/s_1$	Status
$\cos \theta_\mu$	6 bins	0.11/0.09	not adopted: conditioning
	7 bins	0.02/0.05	not adopted: conditioning
θ_μ	7, 8 bins	0.00/0.00	not adopted: rank-deficient
p_π	3 bins	0.05/0.01	not adopted: conditioning
δp_T	3 bins	0.00/0.00	not adopted: rank-deficient
$\delta \alpha_T$	3 bins	0.03/0.01	not adopted: conditioning
$W_{\pi p}$	2 bins	0.26/0.15	sound response, no shape (App. C)
W_{had}	2 bins	0.58/0.54	healthy; <i>not</i> adopted, six bins keep the shape
$\cos \theta_p$	4 bins	0.45/0.02	superseded by θ_p
	6 bins	0.00/0.00	not adopted: rank-deficient
θ_p	5, 7 bins	≤ 0.03	not adopted: conditioning
	3 bins (RHC edges)	0.12/0.05; COMB 0.49	alternative to the released four bins
$\cos \theta_\pi$	4 bins	0.27/0.24	superseded by the five-bin scheme

Two entries deserve a sentence. The two-bin W_{had} is the only candidate whose response is healthier than the released one, and it was still not adopted: two bins buy that conditioning by discarding almost all of the shape, and the shape is what W_{had} is measured for. The three-bin θ_p reaches the best combined conditioning of anything tried (0.51), but the released four-bin scheme is better balanced across the two configurations in which θ_p is quoted, and keeps a bin.

C Withdrawn results

Two results were extracted, examined and withdrawn. They are recorded here rather than in the body because the reasons are properties of the response, not of the measurement being reported, and because their files remain in the data release, where they could otherwise be mistaken for results. **Neither is to be quoted.**

C.1 The differential $W_{\pi p}$ shape

The directly measured pion–proton invariant mass is the observable that would show the $\Delta(1232)$ directly, and it is the reason the proton-tagged subsample was built. At this exposure its shape is not measurable at any binning tried. The six-bin scheme that resolves the resonance region has a top bin only 2% diagonal: essentially every event reconstructed there comes from somewhere else. Coarsening to two bins restores the response ($s_{\min}/s_1 = 0.26$ FHC, 0.15 RHC) but leaves an observable with no shape to speak of, and the two-bin scheme was not adopted for that reason. Closure against the A_C -smeared truth does not rescue the six-bin result: closure tests whether the extraction inverted the response it was given, and cannot restore information the response does not contain.

What the proton-tagged sample does support is the *integrated* cross section, which is released, and the calorimetric hadronic mass W_{had} , whose response is sound at six bins.

C.2 The five-bin $d\sigma/dp_\pi$

Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/c and the migration matrix loses its diagonal above it: four of the five bins fall below the 68% criterion, the top bin reaching 6%. The released $d\sigma/dp_\pi$ is therefore a two-bin result — a near-threshold interval and one broad inclusive interval — and is a two-region partition rather than a spectrum. The limitation is the estimator, not the binning: range-based momentum is only valid for a pion that stops, and no rebinning of a saturated variable recovers the information.

D Selection Cut Parameter Summary

Table 31: All selection threshold constants.

Parameter	Value	Ntuple variable
Topological score	> 0.67	Pandora score
Muon track score	> 0.80	Pandora track score
Muon vertex distance	≤ 4.0 cm	
Muon track length	> 10.0 cm	
Muon LLR PID	> 0.20	LLR PID
Muon MIP BDT	≥ -0.10	TMVA BDT
Pion track length	> 20.0 cm	
Pion vertex distance	< 4.0 cm	
Pion LLR PID	> 0.10	LLR PID
Pion MIP BDT	> -0.10	TMVA BDT
Pion pion-BDT	> -0.10	TMVA BDT
Pion Bragg score	≥ 0.08	Bragg likelihood
Max. opening angle	< 2.6 rad	
Shower Y-plane hits	< 50 (if 1 shower)	
Shower vertex distance	> 10 cm (if 1 shower)	
Non-proton multiplicity	< 4	LLR PID > 0.1
Primary track multiplicity	< 5	
Cosmic BDT	> 0.0 (disabled)	TMVA BDT
Software trigger	$= 1$	
<i>Proton-tagged selection only (Sec. 2 of the proton-tagged note)</i>		
Leading-proton LLR PID	< 0.05	most proton-like non- μ/π primary
Proton end point	in FV	
Proton momentum	≥ 0.3 GeV/c	proton-hypothesis calorimetric KE

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E The MicroBooNE detector

E.1 Operating Principle

A liquid argon time projection chamber records three-dimensional ionisation images of charged-particle tracks by exploiting two properties of ultra-pure liquid argon:

1. Ionisation electrons produced by a traversing charged particle are not captured by argon atoms and can drift centimetres or metres under an applied electric field without significant recombination, provided the purity is sufficient (O_2 -equivalent $\lesssim 100$ ppt).
2. Argon scintillates at 127 nm (vacuum ultraviolet), producing $\sim 24\,000$ photons per MeV of deposited energy, enabling nanosecond-scale timing.

An applied electric field drifts the ionisation electrons to a set of sense-wire planes at the anode, which record three two-dimensional projections of each track. The third coordinate (along the drift direction) is reconstructed from the known drift velocity and the timing of the signals relative to the beam-gate start, combining the two-dimensional wire images into a full 3D reconstruction.

MicroBooNE operated from October 2015 to July 2021 at the Fermilab Accelerator Laboratory, collecting data in both the Booster Neutrino Beam (BNB) and, as a parasitic experiment, the NuMI beam [19]. It was the world’s largest operating LArTPC until the start of SBND and ICARUS data-taking.

E.2 Active Volume and Electric Field

The MicroBooNE active volume has rectangular geometry: 256 cm (drift, x) $\times$ 233 cm (height, y) $\times$ 1037 cm (beam, z), containing ≈ 87 tonnes of liquid argon at ≈ 87 K. The cathode is a stainless-steel plane at $x = 256$ cm held at -70 kV; the anode wire planes are at $x = 0$. The resulting 273 V/cm drift field gives an electron drift velocity of ≈ 1.11 mm/ μs , so the maximum drift time across the full gap is approximately 2.3 ms.

Liquid argon purity is maintained by continuous recirculation through oxygen and water absorbers, keeping the electron lifetime well above 5 ms (corresponding to O_2 -equivalent contamination below ~ 100 ppt). At this purity level, charge attenuation over the full 2.56 m drift is less than 10%.

E.3 Wire Readout and Signal Formation

Three wire planes at the anode sample the drifting charge:

- **U plane** (first induction): 2400 wires at $+60^\circ$ from the vertical, 3 mm pitch.
- **V plane** (second induction): 2400 wires at -60° from the vertical, 3 mm pitch.
- **Y plane** (collection): 3456 vertical wires, 3 mm pitch.

Induction-plane wires are biased to be transparent to the drifting electrons, producing bipolar (derivative-like) induced signals; collection-plane wires absorb the charge and produce unipolar signals. The combination of two stereo induction views at $\pm 60^\circ$ and a vertical collection view provides sufficient 2D information in each plane to resolve ambiguities during 3D reconstruction.

Front-end electronics (application-specific integrated circuits, ASICs) are mounted inside the cryostat on the wire planes and cold-multiply each channel at ≈ 87 K, reducing thermal noise. The analogue signals are digitised at 16 MHz inside the cryostat by 12-bit ADCs and transmitted over ~ 110 m of cold cables to warm readout electronics. The data stream is sampled into a circular buffer; beam triggers and cosmic triggers select 4.8 ms readout windows (corresponding to $\approx$ two full drift windows) for permanent storage.

A dead-wire region at $z \in [675.1, 775.1]$ cm interrupts the collection plane over a 100 cm span due to a shorted wire bundle. Tracks crossing this region have degraded hit-counting and range-based momentum estimation; the fiducial volume (Section 7) excludes this region.

E.4 Photon Detection System

Thirty-two 8-inch Hamamatsu R5912-MOD photomultiplier tubes (PMTs) are installed behind the collection-plane wire frame facing the active volume. The PMTs detect the 127 nm argon scintillation light via a tetraphenyl-butadiene (TPB) wavelength-shifting coating that converts VUV photons to visible (~ 430 nm) light. The system serves two purposes in this analysis:

1. **Beam-gate synchronisation:** the PMT flash time identifies which $1.6 \mu\text{s}$ NuMI spill window produced the neutrino interaction.
2. **Flash-matching for cosmic rejection:** the Pandora reconstruction associates PFParticle hypotheses with the PMT flash closest in time and position, suppressing out-of-time cosmic-ray muons that overlap with beam-induced events in the long 2.3 ms drift window.

E.5 Calibration and Corrections

Three independent calibration inputs are used:

- **UV laser system** [11]: two pulsed laser beams produce straight ionisation tracks at precisely known locations. Deviations of the reconstructed track positions from the known trajectories are used to build a three-dimensional space-charge distortion map [10], which is applied to all reconstructed hits as a position correction. Distortions reach ~ 10 cm near the anode and cathode edges.
- **Stopping cosmic-ray muons:** the Bragg peak of cosmic muons stopping inside the active volume calibrates the absolute energy scale and the effective recombination factor as a function of local electric field and drift distance. Through-going muons calibrate relative wire-by-wire gains and the electron lifetime.
- **Michel electrons:** $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$ decay products from stopped muons provide a check of the calorimetric reconstruction at $E_e \lesssim 55$ MeV.

E.6 Cosmic Ray Environment and Reconstruction

Operating at the surface with minimal overburden, MicroBooNE is exposed to a cosmic-ray muon rate of ≈ 5 kHz through the active volume. Each 4.8 ms readout window contains on average ~ 10 cosmic-ray tracks unrelated to the beam interaction. The $1.6 \mu\text{s}$ NuMI spill itself contains $\lesssim 0.1$ expected cosmics on average, but the drift-window pile-up is the dominant background at early selection stages (Section 8).

Cosmic rejection proceeds in stages:

1. Flash-matching (PMT timing): candidate neutrino slices must have a consistent optical flash within the beam window.
2. Topological score: the Pandora-based topological neutrino score discriminates between neutrino-like vertex structures and through-going/stopping cosmic topologies.
3. Dedicated BDT scores: the MIP and cosmic scores trained on track-level and event-level features further suppress residual cosmics that passed the topological cut.

The CRT (Cosmic Ray Tagger) sub-detector [34], consisting of scintillator panels surrounding the cryostat, provides an additional cosmic veto but is not used as a cut in the present analysis: the CRT branches are carried through the event processing as pass-through quantities only, they are populated only for the CRT-era data (Runs 3–5) and zero for the pre-CRT Run-1 exposure, and a cut would require a data/MC tagging systematic; the residual beam-off background is subtracted instead (§5).

Reconstruction uses the Pandora [6, 12] multi-algorithm pattern recognition framework, which groups hits into PFParticles organised in a hierarchy with the neutrino interaction as the top-level particle. Track candidates are fitted with a Kalman-filter-based track fitter to obtain

momentum from track curvature (negligible for <300 MeV/ c muons at 273 V/cm) and from range for contained tracks. Log-likelihood ratio particle identification (LLR PID) [13] using the dE/dx profile along the track is the primary tool for muon/pion/proton separation (Section 8).